

Guidelines for the Management of Congenital Heart Diseases in Childhood and Adolescence

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Management of a child with a cardiac murmur

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Abstract Cardiac murmurs are frequent findings in congenital heart defects; therefore diagnostic evaluation is mandatory especially in neonates or, when a new murmur is diagnosed or a known murmur has changed its quality or intensity. Echocardiography is a key diagnostic tool. Typical innocent heart murmurs do not require immediate evaluation.

Keywords: cardiac murmur, children, guideline, congenital heart disease

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Heart murmurs are graded according to their timing in the cardiac cycle, intensity, frequency, tone quality, point of maximum intensity, and sound radiation. They represent turbulent blood flow.

Types of murmurs

Innocent heart murmurs

No pathology of the cardiovascular system, never louder than 3/6, considered harmless (see table 1).

Functional heart murmurs

Based on flow phenomena, occur in increased blood flow across valves and vessels or within the heart with increased flow velocity (i.e. fever, anemia, or hyperthyroidism).

Organic heart murmurs

These are based upon a pathology in the cardiovascular system. Stenosis of a valve or vessel, valve regurgitation, or a pathological shunt.

Intensity

Heart murmurs are graded on a 6-point scale:

- 1/6: faint murmur, not audible over breathing
- 2/6: soft, readily audible murmur
- 3/6: loud murmur not associated with a thrill
- 4/6: loud murmur with a thrill
- 5/6: loud murmur audible through the palpating finger
- 6/6: loud murmur audible without a stethoscope

Table 1. Innocent and functional murmurs

Heart murmur	Age peak	Point of maximum intensity	Timing in cardiac cycle, character	Remarks
Still murmur	Toddlers	Left 3rd–5th ICS parasternal	Systolic, musical tone	The sound is typically loudest when the patient is lying, becoming softer in upright position.
Turbulence at the bifurcation of the pulmonary artery	Neonates, young infants	Left and right 2nd ICS parasternal, extending to the back	Systolic, harsh	In fetuses, the stem of the main pulmonary artery is large, but the branches are smaller, as they receive very little intrauterine blood flow. After the closure of the ductus, the entire cardiac output suddenly flows through the relatively hypoplastic branches of the pulmonary arteries
Pulmonary artery flow murmur	Toddlers, school-age children, adolescents	Left 2nd ICS parasternal	Systolic, harsh	Typical in fever
Supraclavicular arterial murmur (carotid bruit)	School-age children, adolescents	Left 2nd ICS parasternal	Systolic, harsh	
Venous hum	Toddlers	Supraclavicular, more pronounced on the right, extending in infraclavicular direction	Continuous systolic-diastolic sound (louder diastolic segment), soft humming tone	Venous turbulence that typically disappears when the head is turned

Timing

Systolic heart murmurs

Occur during systole, can be heard between the first and second heart sounds.

- **Systolic ejection murmurs** arise from obstructions between the ventricles and the great vessels. The maximum intensity is mid-systolic.
- **Systolic regurgitant murmurs** are a result of AV valve regurgitation (including murmurs caused by a VSD). They begin with the first heart sound and are holosystolic.

Diastolic heart murmurs

are usually pathological.

- **Diastolic regurgitant murmurs** are caused by regurgitation of the semilunar valves. They begin with the second heart sound, decrease in intensity (decrecendo), and last almost until the first heart sound.
- **Diastolic filling murmurs** are caused by increased blood flows through an AV valve. This usually involves a relative AV valve stenosis (i.e. ASD and tricuspid valve). “True” AV valve stenosis is in Europe less frequent (i.e. rheumatic mitral stenosis).

Continuous heart murmurs

can be heard during both systole and diastole. They usually reach maximum intensity around the second heart sound. Examples are patent ductus arteriosus (PDA), aortopulmonary window (APW), ruptured aneurysm of the sinus of Valsalva or AV fistulas.

Extracardiac Murmurs

– are not caused by intracardiac structures.

Pericardial friction rub occurs in pericarditis or after surgery, it is best heard during inspiration and when the patient is sitting and leaning forward.

Diagnostic work up

Patients with a newly diagnosed murmur or those with a change of the intensity or quality of a known murmur do require a diagnostic work up.

Physical examination

Auscultation, blood pressure at all 4 extremities, 4x SO₂, non-specific (Typical locations of murmurs → figure 1).

Pulse oximetry

Severity of cyanosis, differential cyanosis ? (see guideline cyanosis)

Echocardiography

Main diagnostic tool. Confirms the diagnosis of the cardiac anatomy, quantification of the cardiac function, proof or exclusion of cardiac defects.

ECG

Non-specific for innocent cardiac murmurs.

Indications for treatment

Depends on the underlying findings (see corresponding guideline)

Recommendations for follow-up

Depends on the underlying disease

Table 2. Typical auscultation and other indicative findings of common congenital heart defects in children

Diagnosis	Heart murmur	Point of maximum intensity/radiation	Remarks
Small VSD	Grade 2/6–3/6 high-frequency systolic murmur	Left 3rd–4th ICS parasternal	Asymptomatic children (“much ado about nothing”)
Moderate VSD without pulmonary hypertension	Harsh, grade 3/6–4/6 systolic murmur	Left 3rd–4th ICS parasternal	Possibly mild signs of heart failure, normal or slightly increased pulmonary vascular markings
Large VSD with pulmonary hypertension	Grade 1/6–3/6 systolic murmur, pronounced 2nd heart sound	Left 3rd ICS parasternal	Usually clear signs of heart failure
Large AVSD	Grade 2/6–3/6 systolic murmur, pronounced 2nd heart sound	Left 3rd ICS parasternal	Left heart axis in ECG, association with trisomy 21
Small PDA	Grade 2/6–3/6 systolic murmur	Left 2nd ICS parasternal	Frequent in premature infants
Moderate PDA	Grade 2/6–4/6 1 systolic-diastolic continuous murmur	Left 2nd ICS parasternal	Strong pulse, frequent in premature infants
ASD with relevant left-to-right shunt	Grade 2/6–3/6 systolic ejection murmur (relative pulmonary stenosis murmur), fixed split 2nd heart sound	Left 2nd ICS parasternal	Usually asymptomatic, possible frequent respiratory infections
Pulmonary stenosis	Harsh grade 2/6–4/6 systolic ejection murmur, possibly ejection click with valvular stenosis	Left 2nd ICS parasternal, radiation to the back and both lungs	Usually asymptomatic
Aortic stenosis	Harsh grade 2/6–5/6 systolic ejection murmur, possibly ejection click with valvular stenosis	Right 2nd ICS parasternal, radiation to carotids	Usually asymptomatic, chest pain or syncope under stress if stenosis is pronounced
Aortic regurgitation	High-frequency, grade 2/6–3/6 diastolic murmur	Left 3rd–4th ICS	Strong pulses, often associated with aortic stenosis
Coarctation of the aorta	Grade 1/6–3/6 systolic ejection murmur	Interscapular	Blood pressure difference between upper and lower limbs
Tetralogy of Fallot	Grade 2/6–3/6 systolic ejection murmur	Left 2nd–3rd ICS parasternal, radiation to the back	Reduced pulmonary vascular marking and normal-sized heart in chest X-ray

Conflict of interests for this guideline:

Nikolaus A. Haas: None

Karl R. Schirmer: None

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Management of a child with cyanosis

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Abstract An oxygen saturation above 95% measured by pulse-oximetry is normal. Cyanosis is a symptom of inadequate oxygen saturation that requires immediate investigation. Echocardiography is the key diagnostic modality.

Keywords: Cyanosis, children, guideline, congenital heart disease

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Cyanosis is a symptom of inadequate oxygen saturation below 95%, that requires immediate investigation. Cyanosis is visible if the level of deoxygenated hemoglobin in the blood exceeds 5 g/dL. Anemia can mask a lower oxygen saturation. An oxygen saturation above 95% measured by pulse-oximetry is normal.

Forms

- Central cyanosis: R/L-shunt, intracardiac or intrapulmonary
- Peripheral cyanosis: peripheral hypoperfusion, eg. due to a decreased cardiac output

- Acrocyanosis: Normal cardiac output, poor peripheral perfusion of extremities; for example in the cold; functional and harmless
- Differential cyanosis: Saturation differs between upper and lower body (i.e. arm 98%, foot 80%)

Aetiology

- In term and preterm infants: Sign of an impending life-threatening disorder that needs immediate investigation → See algorithm below (figure 1)
- In infants: pulmonary hypoxia, seizures, sepsis or congenital heart disease
- In children and adolescents: Usually the result of known cardiac or pulmonary disease → See algorithm below (figure 2)

Differential diagnosis

- CHD and diseases of the cardiovascular system
- Diseases of the respiratory system
- CNS disorders
- Metabolic/hematologic disorders
- Infections

Clinical presentation

depends on the underlying disease

Diagnostic work-up

Physical examination

Auscultation, blood pressure and O₂-saturation at all four extremities, nonspecific

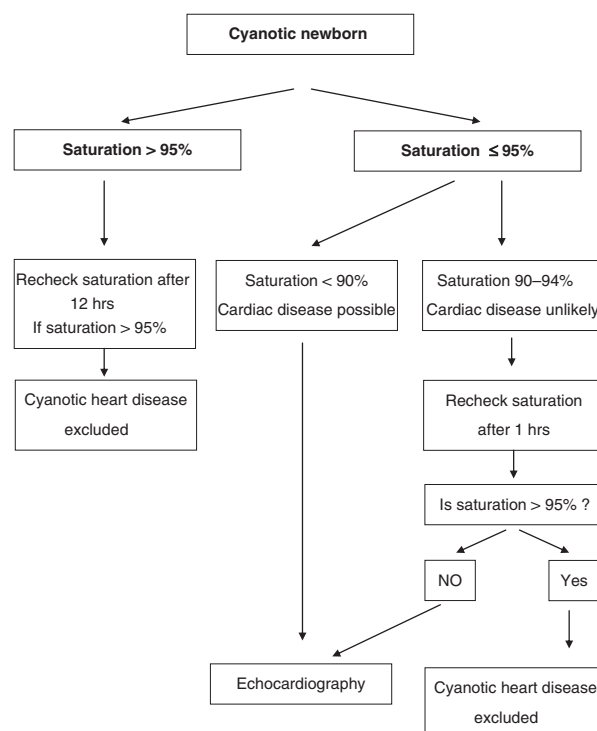


Figure 1.
Flow-chart of the management of a newborn with cyanosis.

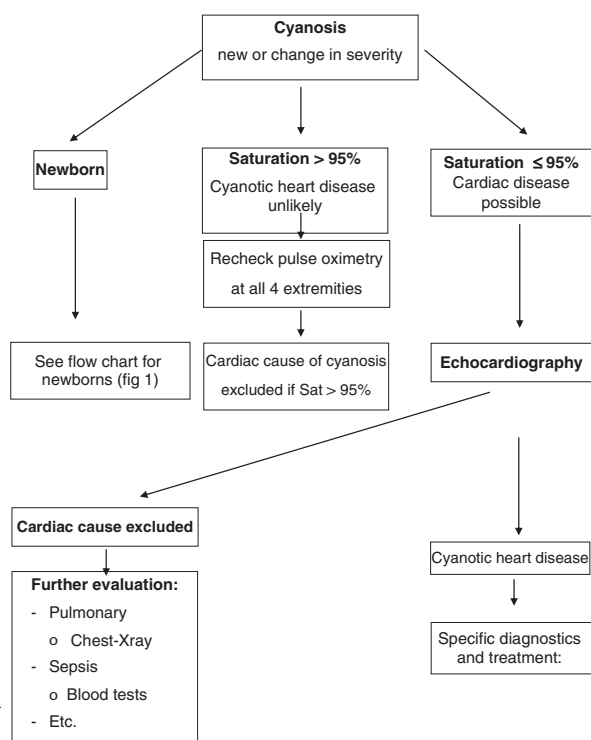


Figure 2.
Flow-chart of the management of a patient with cyanosis.

Pulse oximetry

Note: all neonatal units should routinely perform a general pulse oxymetry screening in all newborns.
Degree of cyanosis, differential cyanosis?

Hyperoxia test

Differentiation between cardiac or pulmonary aetiology; the results are however uncertain (historical approach, should not postpone echocardiography).

Laboratory tests

Blood gas analysis including lactate = gold standard
Blood count: Hb/Hct (polycythemia in chronic cyanosis), Met-Hb

Echocardiography

Mandatory investigation to clarify severe or persistent cyanosis

Electrocardiogram

nonspecific

Chest X-ray

May show underlying pulmonary diseases. Otherwise nonspecific

Cardiac catheterization

Not required if echocardiogram is conclusive

MRI/CT

Provides additional information in selected patients.

Indications for treatment

Depend on the underlying disease (see corresponding guideline)

Conflict of interests for this guideline

Nikolaus A. Haas: None

Jörg Franke: None

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Chest pain

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Abstract: Chest pain is a common complaint in healthy children and adolescents. Usually a cardiac cause can be ruled out by simple diagnostic tests. In patients with a non-organic cause for the pain reassurance of the patient and his family is important to prevent persistence. Warning signs for a possible organic cause for the complaints requiring further diagnostic tests are: syncope, palpitations, dyspnoea, recent onset, prior febrile infection, suspected drug abuse and a positive family history of sudden cardiac death.

Keywords: *chest pain, children, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- Chest pain – usually left sided without radiation – is a symptom and has no organic cause in the vast majority of patients.
- Prevalence is about 8–10% of healthy children and adolescents
- Cardiac aetiology is present in approximately 1% in a non-selected patient population
- Life threatening underlying diseases are extremely rare.

Medical history and clinical presentation

Table 1. History

History of pain	Clinical information
Onset and duration	• acute onset and persistent chest pain, duration ≤ 48 hours: acute heart disease possible • duration 2–7 days: may be less serious heart disease • duration > 6 month: mostly harmless • lasting for seconds: mostly harmless
Intensity	• high in myocardial ischemia, aortic dissection, pneumothorax, peri-/myocarditis, • otherwise not specific
Localization and character	• often diffuse with cardiac etiology
Radiation	• coronary ischemia: left arm and neck, possibly to the teeth • aortic dissection: to the back
Pain during exercise	• majority of children with chest pain during exercise are without any cardiac disease • may occur in coronary anomalies • painful crying and cold sweating in infants with an anomalous left coronary artery from pulmonary artery (Bland-White-Garland-Syndrome)
Awakening from pain	• unspecific
respiration-dependent	• pneumonia, musculoskeletal cause
Concomitant symptoms	
Dyspnea	• probably of organic reason (myocarditis, exercise induced asthma, pneumothorax, pulmonary embolism, pneumonia/pleuritis)
(Pre-)Syncope	• cardiac origin possible

Table 1. Continued

History of pain	Clinical information
Palpitations	• possible underlying cardiac diseases: arrhythmias: primary or in relation to myocarditis, myocardial infarction, transplant rejection
Additional pain in the legs	• vitamin D deficiency? • pulmonary embolism?
Hyperventilation	• anxiety disorders
Concomitant and underlying diseases, family history	
Preceding febrile illness	• peri-/myocarditis • Kawasaki syndrome
Underlying (cardiac) disease	• cardiac surgery • connective tissue disease (Marfan-syndrome) • hypertrophic cardiomyopathy
Vasoactive substances	• stimulants for attention-deficit/hyperactivity disorder • triptans • cocaine and other vasoactive drugs • abuse of energy drinks
Family history	• Marfan syndrome • hypertrophic cardiomyopathy • hereditary arrhythmia syndromes • sudden unexplained death < 40 years of age • psychosocial stress

Clinical presentation

The following findings are suspicious of an organic cause for chest pain:

- A pale, dyspnoeic, cold-sweating patient in reduced general condition complaining of chest pain needs urgent diagnostic work-up.
- A newly discovered heart murmur has to be clarified.
- Pathological pulmonary auscultation findings have to be clarified.

Diagnostic evaluation

Basic diagnostics (in general sufficient):

- History
- Physical examination
- Blood pressure measurement

Advanced diagnostics in suspected cardiac disease

- If necessary, laboratory tests (Troponin)
- Electrocardiogram
- Echocardiography
- Oxygen saturation
- If necessary Holter-monitoring, event-recorder
- If necessary stress test, cardiopulmonary exercise test

Specialized diagnostics are rarely needed to rule out coronary anomalies, myocarditis or cardiomyopathy:

- CT
- MRI
- Cardiac catheterization

Differential diagnosis

Table 2. Cardiac causes of chest pain

Underlying disease	Diagnostics	Comments
Myocarditis	ECG echocardiography laboratory tests	chest pain can mimic acute coronary syndrome,
Pericarditis	ECG echocardiography	usually perimyocarditis
Aortic dissection	imaging has top priority: echocardiography, CT, MRI	- extremely rare - predisposition in Marfan- and Ullrich-Turner syndrome - triggered by trauma or vigorous physical activity
Acute myocardial ischemia	ECG	- very rare - typical pectangina - in strong clinical concern: - work-up without delay - first try sublingual nitroglycerin spray - 1 application = 0.4 mg - administration in adolescents with stable blood pressure
- congenital:	troponin echocardiography lipid analysis if necessary: additional coronary imaging	
- coronary anomalies (see below) - familial hypercholesterolemia		
- acquired:		
- coronary spasm - triggered by drugs (cocaine, synthetic cannabinoid), therapeutic drugs (methylphenidate, triptans) - Kawasaki-syndrome - post cardiac surgery		
Coronary anomalies	ECG stress test imaging	- rare - chest pain mainly during exercise - syncope during exercise - Caveat: low sensitivity of stress test
- Bland-White-Garland syndrome - large coronary fistulas - intramural course of coronary arteries - origin from the opposite sinus with an interarterial course	- echocardiography - heart catheterization - MRI - multidetector CT	

Further rare differential diagnoses in chest pain:

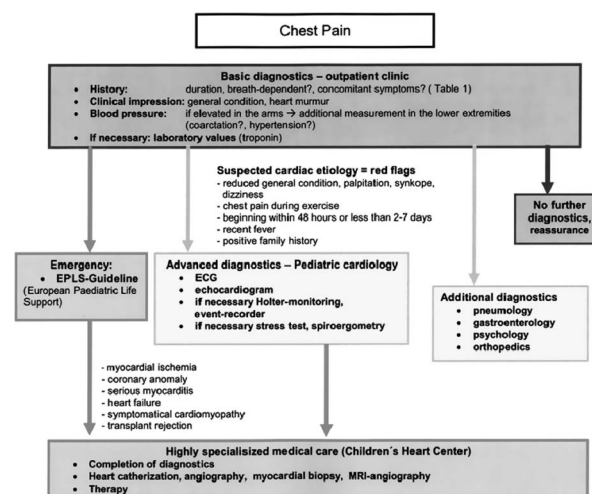
hypertrophic cardiomyopathy, Takotsubo cardiomyopathy, arrhythmias, mitral valve prolapse, outflow tract obstruction, heart transplantation, vertically acquired HIV infection, pulmonary hypertension, hypertensive crisis, pulmonary embolism, HIV – human immunodeficiency virus

Table 3. Non cardiac causes of chest pain

Underlying disease	Diagnostics, comments
Pneumonia, bronchitis, pleuritis, tracheitis, tuberculosis, (exercise induced) asthma, pneumothorax	- correlation with mycoplasma pneumonia - oxygen saturation - chest X-ray - selective laboratory tests - cardiological work-up rarely necessary
Coxsackie type B infection	- leading symptom: severe breathing dependent chest pain - pathogenesis: pleuritis sicca, myositis - chest X-ray - serology, PCR - if necessary: echocardiogram, ECG
Acute chest syndrome in sickle cell disease	- severe chest pain (need for opiates), fever, tachypnea - oxygen saturation - chest x-ray - laboratory work-up - serum vitamin D level
Vitamin D-deficiency	
Musculoskeletal	
costochondritis	- none-specific pain - painful clicking sensation of anterior ends of lower costal cartilages upon activity, with and without scoliosis - rare
slipping rib-syndrome	

Table 3. Continued

Underlying disease	Diagnostics, comments
Tietze-syndrome	- painful swelling of second and third costochondral junctions - very rare
Gastrointestinal: reflux, esophagitis, ulcer, foreign body, pancreatitis, tumor	- related to food intake? - radiation? - specialised diagnostic work-up
herpes zoster	visual diagnosis
traumatic injury, muscle ache	- history - rule out heart contusion if necessary - echocardiogram, further imaging
Tumor: mediastinum, pleura, bony thorax (ribs), abdomen adjacent to diaphragm/thorax	
Drugs: cocaine, opiats, synthetic cannabinoids, amphetamines triptans, methylphenidate	- history - drug screening, troponin - pathogenesis: vasoconstriction, myocardial ischemia
Psychosomatic origin	chest pain can occur in panic disorder, elevated anxiety sensitivity



Therapy

Therapy depends upon the underlying disease. Ruling out an organic origin of the chest pain will reassure the patient and his family.

Prognosis and Follow-up

Non-cardiac chest pain is often a recurrent problem. Therefore psychological work-up should be offered to prevent persistence of the symptoms. An active, healthy life style and sufficient outdoor activity may help to prevent unspecific musculoskeletal discomfort.

Conflict of interests for this guideline

Liane Kändler: none

Marc Schlez: none

Jochen Weil: none

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Secundum atrial septal defect

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Abstract Secundum atrial septal defect is the second most common congenital heart defect. Hemodynamically significant secundum atrial septal defects should be treated by transcatheter device or surgical closure in pre-school age. If untreated or inadequately treated there is increased morbidity, the life expectancy is reduced.

Keywords: atrial septal defect, guideline, congenital heart disease

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- Tissue defect within the atrial septum allowing for shunt at the atrial level
- Prevalence: 17/10.000 live births (corresponding to 15–20% of all congenital heart defects).

Relationships to consider

Associated lesions

Due to the relatively high prevalence, an atrial septal defect can occur in association with virtually all congenital heart defects.

Associated syndromes

E.g. Holt–Oram, Noonan, Alagille syndrome

Differential diagnosis

Patent foramen ovale, common atrium, partial atrioventricular defect (primum atrial septal defect), anomalous pulmonary venous drainage

Haemodynamics

- left-to-right shunting at atrial level due to better diastolic compliance of the right ventricle in comparison to the left ventricle
- volume overload of right-sided heart structures with subsequent dilatation
- in adulthood (if untreated): elevation of pulmonary resistance and pressure, reduction of right ventricular function and diminished left ventricular functional reserve.

Clinical presentation

- systolic heart murmur (flow mediated pulmonary stenosis) and widely-fixed second heart sound
- only with high shunt volume: soft mid-diastolic murmur (flow mediated tricuspid stenosis)
- occasionally history of frequent chest infections
- dyspnoea at exercise
- failure to thrive.

Diagnostic work-up

Clinical examination

Auscultation; pulse oximetry excludes right-to-left shunting in the older patient

ECG

May be unspecific, often with signs of right ventricular volume overload, (incomplete) right bundle branch block, right atrial enlargement

ECHO

Key modality for diagnosis, must include atrial septal defect size and location, exclusion of anomalous pulmonary venous drainage, right ventricular size, function and estimate of right ventricular pressure. Doppler determines shunt direction and allows for non-invasive shunt quantification. Transesophageal echocardiography may be necessary if transthoracic echocardiography windows are inadequate.

MRI

Not necessary if echo is adequate. Good visualization of anatomy (especially pulmonary venous drainage) in older and obese patients, allows for quantification of shunt.

Cardiac catheterization

Diagnostic use only if pulmonary arterial hypertension is suspected or testing for acute pulmonary vasoreactivity is needed, otherwise only if intervention/treatment is planned.

Indications for treatment

- Treatment is indicated if there is right ventricular volume overload (right ventricular enlargement) attributable to the defect.

- Treatment is contraindicated if there is severe pulmonary arterial hypertension and no longer left-to-right shunting.

Management

Treatment aim is complete closure of the defect in preschool age.

Medical therapy

Limited role of medical therapy in selected patients (e.g. anti-congestive medication in premature infants with concomitant lung disease)

Catheter interventions

Catheter interventional closure of ASD II is the therapy of choice with high efficacy and low complication rates. Device closure is not feasible in

- ostium primum and sinus venosus defects,
- secundum defects with inadequate septal rim for safe device anchorage
- inadequate distance to adjacent structures (caval and pulmonary veins, atrioventricular valves, coronary sinus)
- patients with multiple defects that cannot be closed by one or two devices.

Risks: Mortality <0.1%, morbidity 5–10%: e.g. vascular injury at the access site (bleeding, occlusion), device embolization, perforation, thrombo-/embolism, rhythm disturbances. [Morbidity according to the annual report of the German quality assurance for congenital heart disease 2015: 6.0%, n = 29/480; mortality 0%, n = 0/480]

Surgery

Alternative treatment and in patients where interventional treatment is not possible, minimally invasive thoracotomy often possible, requires extracorporeal circulation, patch or direct closure (depending on defect size and location).

Risks: Mortality <0.1%, morbidity 5–10%: e.g. pleural effusion requiring drainage, infection, bleeding requiring reoperation, pneumothorax. [Morbidity according to the annual report of the German quality assurance for congenital heart disease 2015: 8.7%, n = 28/322; mortality 0%, n = 0/322].

Prognosis

Excellent, with normal life expectancy if treated in childhood. Untreated patients with hemodynamically significant ASD statistically have a reduced life expectancy. Starting in their third decade of life, these patients have a higher prevalence of pulmonary arterial hypertension and/or supraventricular rhythm disturbances (atrial fibrillation or flutter, sinus node dysfunction). Further risks are development of exercise intolerance, right heart dysfunction, tricuspid insufficiency and paradoxical embolism.

Recommendations for follow-up

- Regular short-term follow-up after defect closure, lifelong follow-up examinations especially after late defect closure, Echo, ECG at regular follow up; Holter, at least every five years.
- Sport: Usually no exercise restrictions required
- Endocarditis prophylaxis for 6 months after surgical or device closure, aspirin (2–5 mg/kg) or clopidogrel (0.2–0.5 mg/kg) for the first 6 months after device closure.

Conflict of interests for this guideline

Christian Jux: proctor and advisory board St. Jude Medical Inc.

Michael Vogel: stockholder CellAegis Inc.

Jens Scheewe: None

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Isolated Ventricular Septal Defect (VSD) in children and adolescents

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Abstract Ventricular septal defects (VSDs) are the most common form of congenital heart disease. Direct surgical repair should be performed in the first 6 months of life if pulmonary arterial hypertension or congestive heart failure is present. The periventricular hybrid technique is an option to close perimembranous and mid-muscular VSDs in a beating heart technique in selected cases. Large defects with pulmonary hypertension beyond six month of life are recommended to receive pulmonary vascular reactivity testing and anti-PAH-medication in individual cases prior to closure. Elective repair beyond infancy is recommended in patients with left ventricular volume overload, aortic

regurgitation or clinical symptoms. Transcatheter closure is an alternative to surgery in selected patients beyond infancy.

Keywords: *ventricular septal defect (VSD), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Communication between the left and the right ventricle

- The defect can be located in the perimembranous, the outlet, the inlet and the muscular part of the ventricular septum.
- Prevalence of the isolated VSD: 53/10.000 live births (corresponding to 49% of all congenital heart diseases; of notice: 2/3 of this number are small or muscular defects)

Associated lesions, syndromes, differential diagnoses

- Additional cardiac lesions are present in 22% of all patients with VSD

Haemodynamics

- Left-to-right shunt which is related to:
 - size of VSD
 - the ratio of pulmonary and aortic vascular resistance
- A moderate to large VSD initially causes left atrial and left ventricular volume overload with symptoms of congestive heart failure and / or an increase in pulmonary pressure and resistance.
- Long lasting increase of pulmonary hypertension and pulmonary vascular remodelling leads to Eisenmenger's complex and right-to-left shunting across the VSD.

Clinical presentation

Clinical presentation and haemodynamics are linked to management (table 1)

Diagnostic work up

Echocardiography

Delineation of VSD size, localisation and relation to membranous and muscular parts of septum, to the AV-valves and to aortic and pulmonary valve. Left atrial and left ventricular dimensions, systolic and diastolic pressure gradient, assessment of pulmonary artery pressure.

ECG

Not indicative in neonates; LV hypertrophy in volume overload, RV hypertrophy in pulmonary hypertension

Cardiac catheterization

Not required for diagnostic purposes; indicated in elevated pulmonary artery pressure in patients >6 months of life. Testing of pulmonary vascular reactivity if pulmonary vascular resistance is elevated above $R_p:R_s > 0.3:1$ or if transcatheter treatment is intended.

Cardiac MRI

Provides calculation of the left to right shunt for planning of treatment in patients beyond infancy.

Table 1. Clinical presentation, haemodynamics and management of VSD

Size of VSD	Mean PA-Pressure	PA-Resistance	Physical Examination and Symptoms	Management
large, non-restrictive flow	elevated (>25 mmHg)	normal – lightly elevated ($R_p:R_s < 0.2$) elevated ($R_p:R_s = 0.2-0.3$) highly elevated ($R_p:R_s > 0.3$)	noisy systolic murmur, prominent 2. sound, signs of congestive heart failure maybe present systolic heart murmur, less noisy prominent 2. sound, possibly mild cyanosis	<6 months of life: direct repair; in older patients testing of pulmonary vascular reactivity may be required >6 months of life testing of pulmonary vascular reactivity is recommended. An anti-PAH-medication maybe considered >6 months of life treatment belongs to individual findings, anti-PAH-medication is recommended >12 months of life closure of the defect if left ventricular volume overload is present without tendency of spontaneous closure of the defect Echocardiographic follow up; surgical repair if the aortic valve is involved in the defect and aortic regurgitation develops
medium size	normal-slightly elevated	normal-slightly elevated	noisy systolic murmur 2. sound normal	
small, restrictive flow	normal (<20 mmHg)	normal	noisy systolic murmur, typically palpable thrill	

PA – pulmonary artery; PAH – pulmonary artery hypertension;
 $R_p:R_s$ – relation of pulmonary to systemic vascular resistance;
 VSD – ventricular septal defect;

Indications for treatment

- **Small VSD's:** No treatment needed due to excellent long term prognosis. Indication for repair if aortic regurgitation or other symptoms (rare) develop.
- **Medium size VSD's:** Closure should be performed beyond infancy in presence of left atrial and left ventricular volume overload.
- **Large VSD's with pulmonary hypertension:** Closure is indicated in the first 6 months of life. Older patients should undergo investigation of pulmonary vascular resistance before closure.

Management

Surgery

Patch repair is the standard therapy in most patients. Rare complications are: AV-block, residual shunt, lesion of the tricuspid or aortic valve. Temporary pulmonary banding is a palliative option in patients with multiple VSD, or a VSD which is difficult to access or in patients with contraindications for cardio-pulmonary bypass.

Catheter interventions

Device closure of the VSD is feasible in selected patients. Complication risks are comparable to surgery. A possibly elevated risk of AV blocks compared to surgery is under debate if stenting device is used. Particular risk factors for AV-block are direct neighborhood to the aortic valve and to the AV-node.

Hybrid technique

Large muscular and perimembranous defects in infancy are feasible to be closed in hybrid technique with right ventricular puncture.

Medical therapy

Anti-congestive drugs to treat congestive heart failure are only recommended as short term interventions to arrange casual therapy until closure of the defect. Medication to lower pulmonary vascular resistance maybe an option in borderline patients to achieve criterias to close the defect. In patients with Eisenmenger's complex anti PAH drugs have been shown to improve physical capacity and quality of life and there are some notes that the life prognosis maybe extended by this drugs.

Prognosis

- Natural history is unimpaired in almost all patients with small VSD, which are haemodynamically not significant.
- Prognosis is excellent after timely closure of the defect.
- Life expectancy and physical capacity is probably impaired in patients with pulmonary arterial hypertension after defect closure.
- Patients with Eisenmenger's complex are not suitable for defect closure and suffer from reduced physical capacity and life expectancy.

Recommendations for follow-up

- After closure of VSD regular follow up should be performed until adulthood.
- In patients with residual lesions and complications after surgical or interventional closure lifelong follow up is recommended.

Conflict of interest for this guideline

Sven Dittrich has participated at Siemens and Berlin Heart advisory board meetings. The Department received fundings from Actelion, Medtronic, Orion Pharma, St. Jude Medical. The author declares no conflict of interest.

Peter Ewert is proctor for Medtronic, Edwards and St. Jude Medical. Trong-Phi Lê is proctor and advisory board member of PFM Medical Karl R. Schirmer: No conflict of interest
Jürgen Hörer: The Department received funding from corlife, The author declares no conflict of interest.

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Atrioventricular Septal Defect (AVSD)

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Abstract Atrioventricular septal defects are classified as complete, partial or intermediate. Hemodynamics is based on the degree of left-to-right shunting via a primum atrial septal defect and an inlet ventricular septal defect as well as the incompetence of the left sided atrioventricular valve.

Treatment consists in timely closure of the defects and reconstruction of the atrioventricular valves.

Genetic anomalies are present in more than 60% of the patients.

Keywords: *Atrioventricular Septal Defect (AVSD), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition/morphology

Anatomic lesions shared by all forms of AVSD

- Deficiency of the atrioventricular septum at atrial and/or ventricular level
- Anomalies of the atrioventricular valves:
 - inferior displacement and septal insertion at the same level
 - cleft of the left atrioventricular valve (also considered as trileaflet atrioventricular valve), directed towards the interventricular septum
- Elongation and narrowing of the left ventricular outflow tract by anterior and cranial displacement of the aortic valve, fusion of the anterior bridging leaflet to the septal crest and anomalous insertion of the septal papillary muscles of the left atrioventricular valve (“gooseneck” malformation)

Characteristics of the different types of AVSD

- Partial AVSD (pAVSD)
 - separate right and left atrioventricular valve annuli
 - frequently large septum primum defect
 - cleft of the anterior leaflet of the left atrioventricular valve with regurgitation of variable degree
- Complete AVSD (cAVSD)
 - large septal defect with an atrial component (septum primum defect) and a ventricular defect (inlet portion)
 - common atrioventricular valve ring and common atrioventricular valve with five leaflets: one posterior and one anterior bridging leaflet; two lateral leaflets, one right anterior leaflet (type A, B and C of the Rastelli classification, fig. 1)
 - incompetence of the right- and left-sided parts of the AV valve
- Intermediate AVSD
 - septum primum defect, small restrictive inlet ventricular septal defect
 - two separated atrioventricular valves
- Unbalanced AVSD
 - imbalance of ventricular sizes:
 - right ventricular dominance in left ventricular hypoplasia and vice versa; often in association with complex cardiac anomalies (heterotaxy, in particular asplenia)

Prevalence

About 2.7/10.000 live births (corresponding to 3–5% of all congenital heart diseases)

Relationships to consider*Associated lesions*

- Secundum atrial septal defect (20%)
- Persistent arterial duct (10%)
- Tetralogy of Fallot (0.4%)

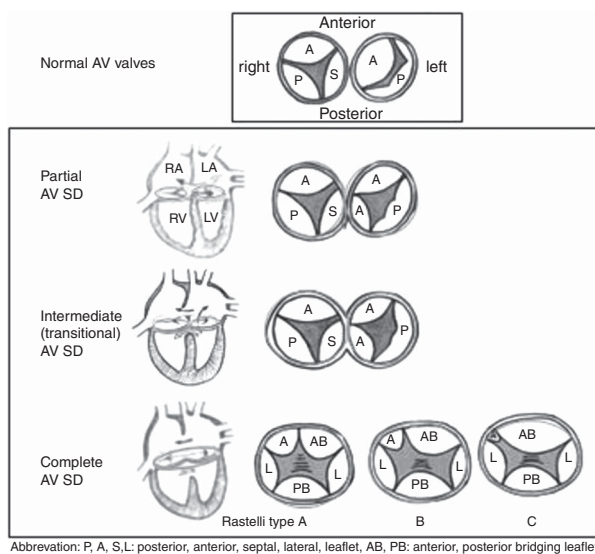


Figure 1.

Diagram describing the spectrum of AVSD, including all forms.

- Left sided superior vena cava
- Double orifice / parachute left atrioventricular valve

Associated genetic anomalies

- Present in 64% of patients; 96% of them being Trisomy 21
- About 40–50% of children with Trisomy 21 are affected with an AVSD, mainly complete form.

Differential diagnosis

- All forms of atrial septal defects
- Ventricular septal defects
- Complex lesions in association with an AVSD

Hemodynamics and clinical presentation*Partial AVSD*

- Left to right shunting at atrial level
- Pulmonary hypertension in large defects

Clinical symptoms are limited to large defects or significant left atrioventricular valve regurgitation.

Complete AVSD

- Hemodynamics of a large ventricular septal defect with equal pressures in both left and right ventricle
- shunting from left ventricle to right atrium possible
- pulmonary vascular resistance is rapidly increasing from 6 months of life onwards
- often significant regurgitation of left or right atrioventricular valve

Symptoms include tachypnea, hepatomegaly and failure to thrive. Special attention to Trisomy 21 patients: hypoventilation due to upper airways obstruction, alveolar hypoventilation and general muscular hypotonia.

Diagnostic work-up*ECG*

- left axis deviation
- prolonged PR interval
- left or biatrial enlargement
- right and left ventricular hypertrophy.

Echocardiography

Two-dimensional echocardiography is the primary imaging technique. Assessment of

- atrioventricular septum defect including size of atrial and ventricular components
- shunt conditions
- atrioventricular valves: anatomy; cleft; valve incompetence; septal insertion of papillary muscles of the left atrioventricular valve
- size of the ventricles – ventricular imbalance?
- left or right ventricular outflow tract obstruction?
- evaluation of pulmonary arterial pressure
- exclusion of coexisting lesions: left superior vena cava, tetralogy of Fallot, coarctation, heterotaxy.

Cardiac catheterization

- usually not necessary for diagnosis or management of pAVSD and of cAVSD before the age of 6 (–12) months of life
- indicated in older children when elevated pulmonary vascular resistance is suspected
- assessment of the pulmonary vascular resistance and testing of pulmonary vasoreagibility with pulmonary vasodilators.

Indications for treatment

The aim of treatment is timely surgical repair of the lesions.

Indications:

Asymptomatic pAVSD may be corrected electively after the first year of life, otherwise earlier if it is complicated by clinical symptoms or failure to thrive.

The cAVSD is usually repaired between 3 and 6 months of life; earlier repair has to be considered in infants with severe heart failure.

Management

Pharmacologic treatment

In patients with heart failure medical treatment is indicated until surgical repair.

Surgical treatment

- Partial AVSD: In pAVSD surgical repair consists of closure of the atrial defect with a patch and reconstruction of the left atrioventricular valve (including closure of the cleft).
- Complete AVSD: In cAVSD the atrial and ventricular defects are closed with patches, and the common atrioventricular valve is separated into a right and left sided atrioventricular valve.

Three different techniques are currently used for repair of the cAVSD (fig. 2):

- “single patch” technique: one patch is used for both defects; the common atrioventricular valve is divided into a left and right component which are sutured to the patch on both sides.
- “double patch” technique: separate patches are used for each defect (usually dacron for closure of the ventricular septal defect and pericardium for the atrial septal defect).
- “modified single patch” technique: may be used in smaller inlet ventricular septal defects, including the *intermediate* AVSD: the ventricular septal defect is closed by suturing the atrioventricular valve tissue down to the septal crest; the atrial septal defect is then closed by a patch.

The procedures are accomplished by repair of associated lesions.

- Consideration of additional complex lesions:
 - cAVSD and tetralogy of Fallot: In case of severe cyanosis early surgical or interventional treatment (e.g. a systemic-pulmonary shunt) may be needed in early infancy; repair is then postponed into early childhood.
 - Imbalanced ventricles: A single ventricle concept has to be considered if appropriate sizes of the ventricles are not given and an appropriate separation of the AV valves is not possible. In case of moderate RV hypoplasia a 1.5 ventricle “repair” may be considered.

Surgical complications:

Overall hospital mortality for repair of a cAVSD is about 3.0%; a bodyweight <3.5 kg and an age <2.5 months are

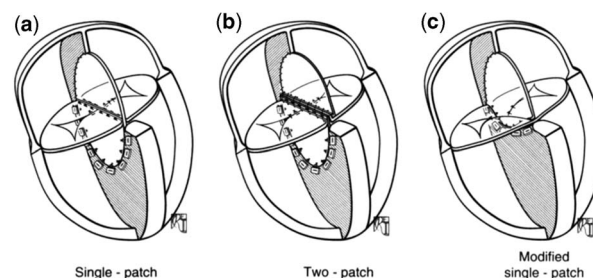


Figure 2.

Surgical repair of common AVSD

Mavroudis C, Backer CL. *Pediatric cardiac surgery*. 4. Edition 2013. Blackwell Publishing Ltd.

both associated with higher rates of mortality. The risk of complete AV block requiring permanent pacemaker implantation is below 3%.

Prognosis

Postoperative long term survival of patients with a *partial* AVSD is excellent (90% in 20 years). In *complete* AVSD survival is described to be 95% after 5 years, and 80 and 78% after 10 and 20 years, resp.

The most frequent residual lesion is left atrioventricular valve incompetence with a reoperation rate about 2 to 10% within 10 years.

Additional postoperative complications are: development or increasing left ventricular outflow tract obstruction (subaortic ridge or IVS hypertrophy, anomalous septal insertion of the anterior left AV valve).

Postoperative arrhythmias (atrial tachyarrhythmias) are rare.

Recommendations for follow up

Regular assessment is warranted for:

- residual atrial or ventricular septal defects
- left sided (rarely right sided) atrioventricular valve incompetence
- left ventricular outflow tract obstructions (see above)
- persistence of pulmonary hypertension: implementation of pulmonary vasodilators.

Conflict of interests for this guideline:

Angelika Lindinger: None

Nikolaus Haas: None

Joerg Sachweh: None.

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Persistent arterial duct (PDA)

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Objective Isolated PDA in infancy, childhood and adolescence; PDA in preterm infants excluded.

Abstract The arterial duct without pulmonary hypertension can be treated easily by interventional means; in case of elevated pulmonary arterial pressure/resistance a thoroughly diagnostic work-up is necessary.

Keywords: *Persistent arterial duct (PDA), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Postnatal persistent vascular communication between the bifurcation of the pulmonary arteries and the descending aorta just distal to the left subclavian artery (of notice: the PDA of the premature infant is not addressed in this guideline).

Epidemiology

Prevalence of the isolated arterial duct: 4/10.000 live births >4 weeks of age (corresponding to 5% of all congenital heart diseases).

Differential diagnosis

- differential diagnoses: aortopulmonary collaterals, aortopulmonary window, coronary artery fistulae, common truncus arteriosus with truncus valve regurgitation, combined aortic valve lesion, ruptured sinus Valsalvae aneurysm

- Caveats: A large PDA may mask coexistent coarctation of the aorta. Diagnosis of PDA may be difficult in patients with congenital heart disease and pulmonary hypertension.

Hemodynamics

Left to right shunt, which is related to

- diameter and length of the duct
- pressure gradient between aorta and pulmonary artery and
- pulmonary vascular resistance.

A moderate to large PDA causes left ventricular volume overload with congestive heart failure and an increase in pulmonary arterial pressure and resistance.

Clinical presentation

Clinical symptoms in the moderate to large PDA: systolic precordial thrill, widened systemic arterial pressure, bounding pulses; signs of heart failure in the majority of patients within the first year of life.

Four categories are introduced with respect to hemodynamics and clinical practice:

- Silent duct: very small, hemodynamically insignificant PDA; no murmur
- Small PDA: systolic murmur; no symptoms
- Moderately large duct: continuous murmur, heart failure often present
- Large PDA with severe heart failure symptoms and pulmonary hypertension: gradual loss of diastolic murmur with increasing pulmonary vascular resistance.

Diagnostic work up

Echocardiography

Description of size and anatomy of the duct, left atrial and left ventricular dimensions, systolic and diastolic gradients through the PDA, assessment of the pulmonary arterial pressure.

ECG

Left ventricular hypertrophy in volume overload, right ventricular hypertrophy in pulmonary hypertension.

Cardiac catheterization

Not required for diagnostic purposes; indicated in elevated pulmonary artery pressure (pulmonary arterial pressure or resistance > 2/3 of systemic level) including balloon test occlusion and application of pulmonary vasodilators, if necessary.

Indications for treatment

- *Very small, silent duct*: no treatment (no follow-up).
Rationale: the silent duct is diagnosed by chance and is not associated with any hemodynamic abnormalities; the risk for endarteritis is extremely low.
- *Small PDA*: may close spontaneously; therefore interventional closure should be performed beyond infancy.
- *Moderately large PDA*: in *asymptomatic* patients closure is recommended beyond infancy; in case of *heart failure* it should be closed at the time of diagnosis.
- *Large PDA with pulmonary hypertension*: patients older than 6 months should undergo diagnostic investigation before closure (diagnostic management see above)

Management

Interventional closure

Device closure of the PDA is the treatment of choice beyond newborn age. A variety of systems is available including coils and nitinol mesh devices. Closure rate is high (97–100%) after one year of follow-up. Procedural risks are low including device embolization, hemolysis and vascular complications.

In newborns and very small infants interventional closure is technically challenging, but feasible with modern devices. Complications include obstruction of the left pulmonary artery and the descending aorta as well as the vascular access site.

Surgical closure

Surgical closure is limited to large ducts not suitable for interventional closure. In symptomatic newborns and small infants it is still the method of choice. Rare complications are laryngeal and phrenic nerve injury, chylothorax, pneumothorax, bleeding and inadvertent ligation of adjacent vessels (left pulmonary artery, descending aorta).

Prognosis

Prognosis after closure is excellent. In arterial ducts with elevated pulmonary arterial pressure it depends on the degree of persistent pressure level.

Recommendations for follow-up

- Follow-up examinations are recommended in patients with residual shunting and complications after surgical or interventional closure.
- After complete device closure prophylaxis for endocarditis is recommended for 6 months.

Conflict of interests for this guideline

Angelika Lindinger: None

Rainer Kozlik-Feldmann: None

Robert Cesnjevar: None

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Partial anomalous pulmonary venous connection (PAPVC)

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Abstract Partial anomalous pulmonary venous connection results in recirculation of pulmonary venous blood through the lungs (left-to-right shunt) causing enlargement of the right atrium and ventricle, and dilation of the pulmonary artery. The initial diagnosis usually is made by echocardiography, and might be confirmed angiographically. Surgery is the definitive treatment of partial anomalous pulmonary venous connection which is generally associated with excellent outcome and low morbidity. Due to its association with severe pulmonary and cardiac anomalies, children with the infantile form of the scimitar syndrome continue to present a management challenge.

Keywords: *Partial anomalous pulmonary venous connection (PAPVC), Guideline, Congenital Heart Disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

In partial anomalous pulmonary venous connection, one or more, but not all pulmonary veins drain directly either into the right atrium or into a systemic vein. In 90% of cases the abnormal drainage is right-sided, in 20% an entire lung (either right or left) is involved.

Based upon their underlying anatomy, two subtypes may be differentiated (see figure 1):

- **Partial anomalous pulmonary venous drainage**
The pulmonary vein(s) connect(s) normally and directly to the anatomic left atrium; due to intracardiac defects (most often sinus venosus defects), however, oxygenated blood from the anomalous vein can either enter exclusively the right atrium or both the left and right atrium.
 - **Sinus venosus defects:** the by far most common anomaly is a deficiency of the wall that normally separates the right

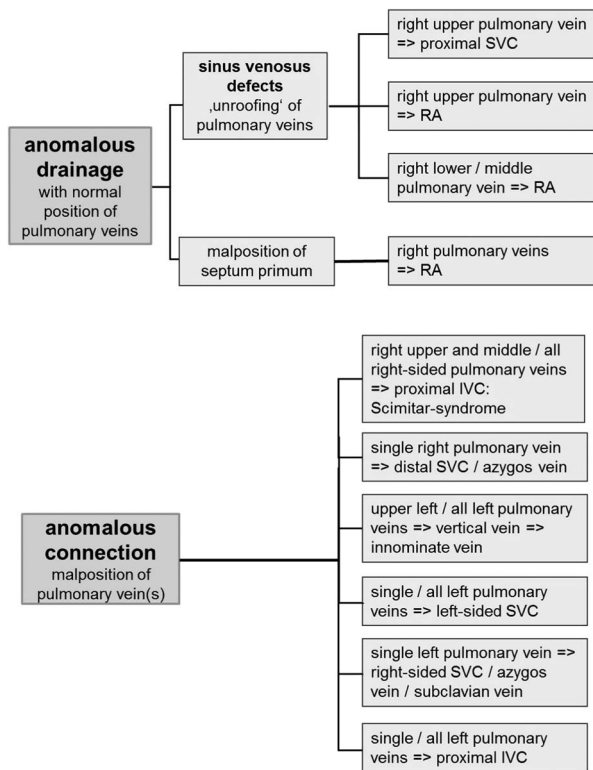


Figure 1.

Anatomic variants of partial anomalous pulmonary venous connection

SVC = superior cave; RA = right atrium; IVC = inferior vena cava.

upper pulmonary vein from the superior vena cava ("unroofing"), resulting in the right pulmonary vein draining into the superior vena cava, or in the right and left atrium

- **Malposition of the septum primum:** leftward shift of the posterior and/or superior attachment of the septum primum resulting in right pulmonary veins draining into the right atrium rather than into the left atrium due to a defect of the septum secundum

• True anomalous pulmonary venous connections

The anomalous pulmonary vein(s) directly connect(s) with a systemic vein (rarely the right atrium)

Scimitar syndrome

- Complex malformation syndrome, in which part or even the entire right lung is drained by right pulmonary veins that connect anomalously to the inferior vena cava proximal to the liver veins
- Common concomitant malformations in symptomatic infants are
 - hypoplasia of the affected lung and its associated airways
 - unusual bronchial or vascular distribution patterns
 - hypoplastic right pulmonary artery
 - pulmonary sequestration
 - aortopulmonary collateral vessels
 - pulmonary vein stenosis
- Additional cardiac defects may be present.

Epidemiology

The overall prevalence of partial anomalous pulmonary venous connection is not known. It is often reported as an incidental

finding in asymptomatic patients. Scimitar syndrome is reported in 2/100.000 live births.

Relationships to consider

Associated defects

- Coincidence with other cardiac abnormalities, most often an atrial septal defect
- Malposition of the septum primum is associated with heterotaxia and polysplenia

Differential diagnosis

- Haemodynamically relevant atrial septal defects
- 'Infantile form' of scimitar syndrome: pulmonary malformations resulting in pulmonary arterial hypertension

Haemodynamics

There is return of oxygenated blood to the right side of the heart with subsequent recirculation through the pulmonary vasculature (left-to-right shunting). Significant shunting is associated with two or more anomalous connecting veins or an associated atrial septal defect, resulting in enlargement of the right atrium and ventricle, and dilation of the pulmonary artery. Patients with single pulmonary vein involvement do not have significant haemodynamic and cardiac structural changes.

Clinical presentation

The severity of clinical signs and symptoms is related to the degree of left-to-right shunt, the presence of pulmonary hypertension, and the presence of other associated cardiac and pulmonary defects; patients with a small degree of shunt are usually asymptomatic and are often identified incidentally by a cardiac murmur. Although patients with moderate to large shunts may be asymptomatic in childhood, left-to-right shunt increases with age, and symptoms appear before 40 years of age.

Presentation of **scimitar syndrome** varies widely depending upon the age at diagnosis:

- Symptomatic infants ('infantile form') tend to present with more severe symptoms related to heart failure and pulmonary hypertension: tachypnea, poor feeding, failure to thrive, and cyanosis.
- About one-half of patients who are diagnosed after the first year of life ('adult form') remain asymptomatic; others may present with fatigue, dyspnea, and recurrent pneumonia, or might be identified by an incidental finding on chest X-ray.

Diagnostic work up

The initial diagnosis of partial anomalous pulmonary venous connection is usually made by echocardiography. It might be confirmed by magnetic resonance imaging, computed tomography, or cardiac catheterization. Other tests, including chest radiography and electrocardiogram, may provide supportive evidence for right heart enlargement or left-to-right shunting, but are not diagnostic.

Transthoracic echocardiography

The diagnosis of partial anomalous pulmonary venous connection should be considered when the echocardiography demonstrates

- unexpected right atrial or right ventricular enlargement, especially if there are no other explanations for these finding(s) (e.g., secundum atrial septal defect)
- a sinus venosus-atrial septal defect with left to right shunt (nearly 90% of cases are combined with partial anomalous pulmonary venous connection)
- fewer than four pulmonary veins connecting to the left atrium (although one must consider the possibility of a single pulmonary vein draining an entire lung)
- a dilated superior vena cava, innominate vein, or inferior vena cava (and no other clear explanations are present); in these cases the anomalous pulmonary vein(s) may connect directly to these structures

Transoesophageal echocardiography

is more sensitive than transthoracic echocardiography in detecting partial anomalous pulmonary venous connection, especially for its most common form (right pulmonary vein(s) draining into the superior vena cava).

Magnetic resonance imaging

- provides enhanced visualization of the pulmonary vasculature including the anomalous pulmonary vein(s)
- can measure the ratio of pulmonary to systemic blood flow ($Q_p:Q_s$ = shunt volume) noninvasively
- can calculate blood flow to the right and left pulmonary artery separately

Computed tomography

- might be performed to visualize the pulmonary vasculature including the anomalous pulmonary vein(s) in cases with atypical anatomy
- is the preferred imaging modality in patients with scimitar syndrome to display associated pulmonary malformations: hypoplasia of the right lung, bronchial malformations, sequestrations, aorto-pulmonary collaterals

Cardiac catheterization

- provides additional haemodynamic information, such as pulmonary vascular resistance, shunt volume calculated by oximetry, cardiac output, and ventricular pressures
- might be indicated to (re)evaluate the causes and degree of pulmonary arterial hypertension
- might be indicated to display postoperative pulmonary venous obstruction, defined as:
 - reduction of vessel diameter < 50% of maximum diameter
 - mean pressure gradient of ≥ 4 mmHg
 - slow transition time after dye injection into the ipsilateral pulmonary artery

Chest radiography

- should be performed before surgical intervention
- in scimitar syndrome, the anomalous vein is identifiable in about 50% of cases ('adult form') as it courses towards the diaphragm and the inferior vena cava, giving the characteristic 'scimitar' appearance; furthermore, hypoplasia of the right lung combined with dextroposition of the heart may be present.

Electrocardiogram

- may demonstrate evidence of right heart enlargement
- should be performed at least once before surgery.

Indications for treatment

Surgical treatment is considered in the following circumstances:

- haemodynamically significant left-to-right shunt (pulmonary to systemic blood flow ratio greater than 1.5:1) manifesting as right ventricular volume overload
- recurrent pulmonary infections, particularly with scimitar syndrome and pulmonary sequestration

Asymptomatic patients with partial anomalous pulmonary venous connection with small left-to-right shunt do not require any intervention, as they have a normal life expectancy; this includes patients with the 'adult form' of the scimitar syndrome.

Management

Surgical repair connecting the blood of the anomalous vein(s) to the left atrium is the definitive treatment in partial anomalous pulmonary venous connection.

Medical treatment

- Rarely indicated in patients with partial anomalous pulmonary venous connection
- 'Infantile form' of the scimitar syndrome: specific medication for pulmonary arterial hypertension and congestive heart failure may stabilize these patients.

Catheter interventions

- Occlusion of significant aorto-pulmonary collaterals (including the subdiaphragmatic arterial supply of sequestrations) may stabilize infants with the scimitar syndrome prior to surgery
- (Stent-)angioplasty may be indicated to treat postoperative pulmonary venous obstruction.

Surgery

Corrective surgery is electively performed in preschool age. The surgical strategy depends on the underlying anatomy:

- Single or two pericardial patches are used to redirect the pulmonary venous blood to the left atrium, combined with potential augmentation of the systemic veins and closure of the atrial septal defects; alternatively, a division of the superior vena cava distal to the anomalous venous connection with subsequent reimplantation to the right atrial appendage may be performed.
- Left-sided anomalous pulmonary venous connection may be corrected by dissecting the vertical vein cranial to the anomalous vein with its subsequent reimplantation to the left atrial appendage.
- In scimitar syndrome, an intracardiac tunnel is created between the anomalous vein and the left atrium through an atrial septal defect in most cases.

Outcome

Patients with partial anomalous pulmonary venous connection generally have an excellent outcome with low perioperative morbidity and mortality (0.4%); the risk of postoperative

pulmonary vein obstruction is considered to be below 2% after 15 years.

The exception are patients with the ‘infantile form’ of the scimitar syndrome who still have an increased risk (early mortality rate recently below 6%) despite improved perioperative management and postponing surgical repair beyond infancy; the risk of postoperative obstruction of the redirected scimitar vein is nowadays considered to be 15% after 3 years.

Other potential postoperative complications after repair of right-sided partial anomalous pulmonary venous connection are:

- stenosis of the superior vena cava (10–17%)
- sinus node dysfunction; the rate of subsequent pacemaker implantation varies considerably.

Recommendations for follow-up

Postoperative findings after surgical repair requiring further diagnostics and potential treatment:

- reobstruction of the redirected pulmonary vein(s)
- obstruction of the superior vena cava
- persisting pulmonal arterial hypertension
- unequal distribution of pulmonary blood flow (obstruction of redirected veins?)
- sinus node dysfunction with bradyarrhythmia

Besides clinical, echocardiographic, and electrocardiographic evaluation during outpatient visits, more extensive diagnostic is recommended in symptomatic patients:

- Holter monitoring in case of suspected bradyarrhythmias
- cardiopulmonary exercise test
- magnetic resonance imaging for pulmonary venous obstruction

Conflict of interests for this guideline

Harald Bertram: None

Oliver Dewald: None

Angelika Lindinger: None

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Link to all references listed in the original German long version “Partielle Lungenvenefehlmündung”: www.kinderkardiologie.org/leitlinien/

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Total anomalous pulmonary venous connection (TAPVC)

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Abstract Total anomalous pulmonary venous connection is a cyanotic congenital defect in which all pulmonary veins fail to reach the left atrium. This results in drainage of the pulmonary venous return into the systemic venous circulation, leading to right ventricular volume load and pulmonary recirculation. More than half of the patients suffer from obstruction of the pulmonary venous confluence or draining veins resulting in severe symptoms, which may lead to respiratory failure and shock within the first days of life. Treatment of choice is surgical redirection of pulmonary venous blood to the left atrium. Nowadays, perioperative mortality is below 10%, but specific demographic and anatomic characteristics remain important determinants of postoperative survival despite improved perioperative care.

Keywords: *Total anomalous pulmonary venous connection (TAPVC), guideline, Congenital Heart Disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Total anomalous pulmonary venous connection can be divided into four anatomic variants defined by the location of their connection relative to the heart: supracardiac, cardiac, and infracardiac anomalous venous connection. In the mixed type, pulmonary venous connections enter at two or more different levels.

In more than 90% of patients the pulmonary veins course to a confluent chamber that is located just posterior to the left atrium. This pulmonary confluence is connected to the systemic circulation by an anomalous venous channel (“draining vein”). In more than 50% of patients, these draining veins show different degrees of obstruction, which is of profound clinical importance. An atrial septal defect is necessary for survival in postnatal life.

Classification (relative frequency)

- **Supracardiac** (44–49%): the blood originating from the pulmonary venous confluence passes through a vertical vein

that mostly connects to the left innominate vein. Obstruction of the draining vein occurs in 45% of patients, in particular in case of connection to the superior vena cava or azygous vein

- **Cardiac** (16–21%): the pulmonary veins join a common vessel that connects to the posterior aspect of the coronary sinus (90%) or directly to the right atrium; pulmonary venous obstructions occur in less than 20% of patients
- **Infracardiac** (26–28%): the pulmonary confluence drains into a common vertical vein that courses inferiorly through the diaphragm, and commonly inserts into the portal vein; other connections include the hepatic vein, the ductus venosus or inferior vena cava. The infracardiac variant has the highest risk of pulmonary venous obstruction (85%)
- **Mixed** (5–9%): refers to any combination of connections that enter at two or more different levels, the most common arrangement is that of three pulmonary veins joining to form a single confluence and a fourth vein draining via a separate venous connection. This type is often associated with complex congenital heart defects and heterotaxia.

Epidemiology

The prevalence of total anomalous pulmonary venous connection ranges from 0.5 to 1.1 per 10,000 live births, corresponding to 0.4 to 0.9% of patients with congenital heart disease.

Relationships to consider

Associated lesions

Total anomalous pulmonary venous connection frequently occurs as an isolated cardiac lesion, but may be associated with other complex congenital heart defects (e.g. functional single ventricle) or viscerotaxial situs abnormalities (heterotaxia with asplenia). Associated genetic lesions or syndromes are present in 7% of patients, namely VACTERL-association, Goldenhar syndrome, or microdeletion 22q11 syndrome.

Differential diagnosis

- In patients with **unobstructed pulmonary veins**, other forms of congenital heart disease with cyanosis and pulmonary recirculation must be ruled out: transposition of the great arteries with ventricular septal defect, common arterial trunk
- In patients with **severe pulmonary venous obstruction**, other causes of critical illness in newborns with symptoms of respiratory failure and shock must be excluded: sepsis, pneumonia, congenital heart disease with duct-dependent systemic circulation, and metabolic disorders.

Haemodynamics

The entire oxygenated pulmonary venous return mixes with blood from the systemic venous system. The mixed, partially oxygenated blood is then shunted right-to-left into the systemic circulation (generally at the atrial level; infrequently through a patent duct), resulting in cyanosis. As a result of right atrial and right ventricular volume load, both the right atrium and ventricle are dilated.

- **Unobstructed forms** generally have a net degree of left-to-right shunting due to the relatively low resistance in the lungs leading to pulmonary overcirculation with mildly elevated pulmonary artery pressure;
- In **obstructed forms**, pulmonary venous pressure is elevated, resulting in progressive interstitial and alveolar edema;

pulmonary resistance and pulmonary artery pressure are elevated (may be suprasystemic), resulting in right ventricular dilation, hypertrophy, and right heart failure.

Clinical presentation

Clinical manifestations vary widely depending on the presence and degree of pulmonary venous obstruction:

- In **unobstructed** forms, findings are similar to other disorders with large volume left-to-right shunts. Patients may present with tachypnea, poor feeding, hepatomegaly, and failure to thrive. Although hypoxia is present, the range varies, and some patients may not appear to be cyanotic.
- Patients with **severe obstruction** generally present as critically ill newborns with cyanosis. In general, the more severe the obstruction, the earlier and more severe the clinical presentation. Elevated pulmonary artery pressure and edema result in respiratory distress and tachypnea, and diminished systemic output resulting in hypotension. There is a substantial risk of rapid decompensation with respiratory failure and shock in the first days of life.

Diagnostic work up

Diagnosis of total anomalous pulmonary venous connection is generally made by transthoracic echocardiography.

Echocardiography

Suggestive findings are:

- inability to demonstrate the normal pulmonary venous connections to the left atrium.
- dilatation of right atrium, right ventricle, and pulmonary artery
- right-to-left interatrial shunting (be aware of a restrictive interatrial communication)
- in patients with supracardiac lesions: demonstration of a common ascending collecting vein and a dilated superior vena cava
- in patients with infracardiac lesions: demonstration of a common descending collecting vein in combination with atypical flow pattern in either the hepatic or portal vein

Cardiac catheterization and angiography

- generally no longer needed to establish the diagnosis
- reserved for patients with unclear anatomy in echocardiography
- may be indicated in patients with associated congenital heart disease

Magnetic resonance imaging and computed tomography

- may be indicated if the echocardiogram is inconclusive in identifying all vascular connections
- can provide additional information about extracardiac structures (pulmonary vasculature / airways)

Chest radiography

- should be obtained before surgical intervention
- generally nonspecific; suggestive findings are:
 - 'snowman sign' in young children with supracardiac total anomalous pulmonary venous connection as a result of an enlarged heart and dilation of vertical vein, innominate vein, and superior vena cava

- in case of severe pulmonary venous obstruction, the combination of vascular congestion and pulmonary edema leads to a reticular pattern spreading out from the hilar regions (ground glass appearance)

Electrocardiography

- right atrial and right ventricular hypertrophy, nonspecific
- should be obtained before surgical intervention.

Management

In general corrective surgery for total anomalous pulmonary venous connection should be performed once the diagnosis is made and the patient is stabilized. Timing of surgery depends on the degree of pulmonary venous obstruction and the condition of the patient. In some patients with complex variants of total anomalous pulmonary venous connection associated with single ventricle physiology, complete separation of pulmonary and systemic venous blood may not be possible.

Medical treatment

- Generally not necessary in patients **without pulmonary venous obstruction**; supportive medical treatment of congestive heart failure may improve symptoms
- In patients with **severely obstructed total anomalous pulmonary venous connection**:

The clinical condition of these critically ill neonates should be optimized prior to surgery using all options of modern paediatric intensive care, including ventilation with inhaled nitric oxide, inotropic support, prostaglandin; if this management fails to stabilize the patient, extracorporeal membrane oxygenation might be used to correct severe hypoxaemia, acidosis, and haemodynamic instability allowing urgent surgical correction to be performed even in the most severely ill patients.

Catheter interventions

Palliative procedures rarely used to stabilize patients prior to surgical correction, if intensive care therapy was unable to correct hypoxaemia, acidosis, and shock:

- Atrial septostomy in patients with restrictive interatrial blood flow
- Stent angioplasty of severely obstructed pulmonary and draining veins in supracardiac forms

Surgery

Surgical techniques are depending on the underlying lesion:

- Supra- and infracardiac forms: a normal pulmonary venous connection is created by anastomosing the pulmonary venous confluence with the left atrium. The vertical vein is ligated and divided, the atrial septal defect closed by a patch.
- Cardiac forms with connection to the coronary sinus: an unroofing of the coronary sinus is performed by excising the wall between the coronary sinus and the left atrium; the atrial septal defect is closed by a patch, thereby draining the coronary sinus to the left atrium.
- Mixed lesions require an individualized approach based on the underlying anatomy.

Outcome

Improved perioperative intensive care management including preoperative catheter interventions or extracorporeal membrane

oxygenation, and refined surgical techniques have led to an improved survival rate in recent years. Overall, early mortality in isolated total anomalous pulmonary venous connection is less than 10 percent. Long-term survival rate to childhood and adolescence is about 85 percent.

Reported risk factors associated with perioperative mortality include:

- preoperatively existing pulmonary venous obstruction and younger age at operation, both reflecting the need for urgent surgery
- additional cardiac anomalies, in particular single ventricle physiology (which increases mortality rates more than 3-fold!)
- small size of pulmonary veins and confluence
- postoperative pulmonary venous obstruction and persisting pulmonary hypertension

Prognosis

Stenosis of pulmonary vein(s) and surgically created anastomoses are the main reasons for reintervention. The rate of reoperation ranges between 10 and 15 percent in patients with isolated total anomalous pulmonary venous connection. If restenosis does not occur within the first year after surgery, reoperation is rarely required.

Reported risk factors associated with postoperative pulmonary venous obstruction:

- preoperative pulmonary venous obstruction with small pulmonary vein and confluence size
- preoperative respiratory failure
- mixed total anomalous pulmonary venous connection

Recommendations for follow-up

In absence of residual pulmonary vein stenosis or pulmonary hypertension, exercise tolerance is generally normal after surgical repair regardless of the anatomic subtype. Physical activities and sports participation should not be restricted.

In symptomatic patients, however, cardiopulmonary exercise capacity should be measured every 2–3 years. Because of the known association between intracardiac surgical procedures and atrial arrhythmias, periodic Holter monitoring should be considered even in asymptomatic adolescents every 2–3 years.

Conflict of interests for this guideline

Harald Bertram: None

Oliver Dewald: None

Angelika Lindinger: None

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Pulmonary valve stenosis (PS)

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Abstract Obstruction of right ventricular outflow tract at valvar level (pulmonary valve stenosis) results in pressure overload of the right ventricle. Treatment is indicated in infants with critical stenosis, in symptomatic patients and in oligo-/asymptomatic patients if moderate or severe stenosis exists. Balloon valvuloplasty is the treatment of choice, but surgery is indicated in non-responders and especially with dysplastic valves. Although the prognosis is generally good, long-term follow-up is recommended.

Keywords: *valvar pulmonary stenosis, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- Obstruction of right ventricular outflow tract at pulmonary valve level due to morphological valve anomalies. In pulmonary valve stenosis the valve can be bicuspid or tricuspid with partially fused commissures, unicommissural or acommisural with reduced pulmonary valve opening area or dysplastic with myxomatous thickening and hypomobility of the valve tissue.

- Prevalence: 6/10.000 live births (corresponding to 6–10% of all congenital heart defects).

Relationships to consider

Associated lesions

hypoplastic pulmonary arteries, supralvalvar and peripheral stenosis (esp. in dysplastic valves)

Associated syndromes

Noonan, LEOPARD, Williams-Beuren syndrome

Differential diagnosis

Muscular subvalvar stenosis/right ventricular outflow tract obstruction, muscular subinfundibular stenosis (double chambered right ventricle), supralvalvar obstruction, flow-related stenosis in left-to-right shunt lesions.

Haemodynamics

- Pressure overload of right ventricle with consecutive concentric right ventricular hypertrophy, normal or reduced distal pulmonary artery pressure, turbulent poststenotic blood flow in main pulmonary artery with poststenotic dilatation (often absent in dysplastic valve stenosis)
- In neonatal critical (duct-dependent) pulmonary valve stenosis and patients with severe PS and atrial communication: cyanosis due to right-to-left shunt at atrial level.

Clinical presentation

- Systolic, ejection type heart murmur, maximum at left upper sternal border, transmitted into lungs and back, ejection click in early systole in mild to moderate stenosis (not with dysplastic valves), left parasternal thrill with moderate or severe stenosis
- Infants with critical pulmonary stenosis: cyanosis at birth, hypoxemia with ductal closure
- Older patients with atrial communication: cyanosis
- Rarely with severe stenosis: exertional dyspnoea, chest pain, dizziness, syncope.

Diagnostic work-up

Clinical examination

Auscultation, pulse oximetry excludes or confirms right-to-left shunting

ECG

right axis deviation, right ventricular hypertrophy, P-pulmonale, incomplete (or rarely complete) right bundle branch block

ECHO

key modality for diagnosis, must include diameter and morphology of pulmonary valve, annulus size, pulmonary valve competence or degree of regurgitation, additional sub- or supralvalvar stenosis, poststenotic pulmonary artery dilatation, diameter and function of tricuspid valve, estimation of the right ventricle, pressure of right ventricle (if tricuspid regurgitation), size, morphology (bi- or tripartite) and function of right ventricle, Doppler gradient between right ventricle and pulmonary artery, shunting at atrial level.

MRI

Generally not necessary. May be indicated in poor echo windows

Cardiac catheterization

Indicated, if intervention/treatment is planned.

Indications for treatment

- Always in critical neonatal pulmonary valve stenosis regardless of gradient across valve and in symptomatic patients of any age
- In asymptomatic infants and older children with preserved right ventricular function with a peak systolic Doppler gradient of >40–50 mm Hg.

Management

Treatment aim is the relief of the stenosis

Medical therapy

In neonates with critical pulmonary valve stenosis prostaglandine infusion to maintain or re-open the duct should be immediately commenced. Heart failure medication should not postpone causative treatment.

Catheter interventions

In patients with isolated pulmonary valve stenosis without valvular dysplasia balloon valvuloplasty is the treatment of choice. Patients with dysplastic valvar stenosis respond less well to valvuloplasty. Risks: Vascular injury at the access site (bleeding, occlusion), or pulmonary insufficiency can occur especially with balloon over-dilatation. Mortality is low, also in neonates (<0.5%),

Surgery

Indicated if balloon valvuloplasty is ineffective and should aim for relief of the stenosis while maintaining valve function. Using extracorporeal bypass open commissurotomy is performed. During surgery thickened cusps can be mobilized by shaving or partial excision. In addition, the valve annulus can be enlarged and a supra-ventricular stenosis can be relieved. A myectomy with or without right ventricular outflow tract patch plasty to relieve (reactive) subvalvar stenosis can be done.

Prognosis

The prognosis depends primarily on the degree of residual stenosis and/or on insufficiency of the pulmonary valve as well as on right ventricular and tricuspid function. In general the long-term effect is excellent. Re-stenosis can occur in 5–10% of patients, especially in the first year after intervention.

Recommendations for follow-up

- regular follow-up with intervals depending on the severity of residual findings: ECHO, ECG/Holter
- Sport: If (residual) peak Doppler gradient <40 mm Hg, no exercise restrictions required
- Endocarditis prophylaxis not routinely recommended.

Conflict of interests for this guideline

Christian Jux: Proctor and advisory board St. Jude Medical Inc.
Jochen Weil: None
Alexander Horke: None

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Pulmonary regurgitation

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Abstract Pulmonary regurgitation results from incomplete diastolic closure of the pulmonary valve leading to backflow of blood from the pulmonary artery into the right ventricle. Pulmonary regurgitation is rarely found as an isolated lesion but a common residuum after treatment of pulmonary valve stenosis or surgical repair of tetralogy of Fallot. Significant pulmonary regurgitation can lead to right ventricular enlargement and dysfunction. Right heart failure as well as atrial and ventricular arrhythmias may develop.

Cardiac magnetic resonance imaging is the method of choice to quantify pulmonary regurgitation and to assess right ventricular volume and function.

Pulmonary valve replacement can be performed by surgery or by percutaneous valve implantation. Indications are significant pulmonary regurgitation in combination with severe right ventricular volume overload, symptoms of heart failure, exercise limitations or new onset of ventricular or atrial arrhythmias.

Keywords: *pulmonary regurgitation, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Pulmonary regurgitation is defined as incomplete closure of the pulmonary valve in diastole leading to backflow of blood from the pulmonary artery into the right ventricle.

Pulmonary regurgitation can be present in

- Isolated congenital pulmonary regurgitation (very rare – absent pulmonary valve syndrome)
- Residual lesion following balloon dilatation of pulmonary valve stenosis
- Residual lesion following surgical repair of tetralogy of Fallot or similar lesions (by a transannular patch) (60–90%)
- Residual lesion after surgical or percutaneous implantation of valve-less or valved conduits.

Haemodynamics

Pulmonary regurgitation causes volume overload of the right ventricle with an increased stroke volume. Chronic volume overload results in progressive right ventricular enlargement and dysfunction with consecutive development of tricuspid regurgitation. Ventricular filling from the atrium can be impaired. Both, tricuspid regurgitation and impaired ventricular filling result in enlargement of the right atrium. Finally right ventricular dilatation may compromise left ventricular function.

Clinical presentation

- A diastolic murmur is heard at the left sternal edge. When pulmonary regurgitation is severe, a to-and-fro murmur can be heard.
- With progressive right ventricular dysfunction signs of right heart failure such as hepatomegaly, jugular venous distension and peripheral oedema may be present.
- Development of atrial or ventricular arrhythmias indicate progressive right atrial or ventricular dilatation and/or dysfunction.

Diagnostic work up

Echocardiography

- Documentation of pulmonary regurgitation.
- Grading of severity of pulmonary regurgitation.
- Estimation of right atrial and right ventricular enlargement.
- Estimation of right and quantification of the left ventricular function.
- Assessment of pulmonary artery pressure by analysis of Doppler flow
- Tricuspid regurgitation jet velocity for assessment of right ventricular pressure
- Diagnosis of additional or residual lesions
 - Pulmonary stenosis/ right ventricular outflow tract obstruction
 - Branch pulmonary artery stenosis
 - Atrial or ventricular septal defect
 - Aortic regurgitation, diameter of the ascending aorta

Magnetic Resonance Imaging

Method of choice to

- Quantify pulmonary regurgitation
- Quantify right ventricular volume: in patients with repaired tetralogy of Fallot enlargement of the right ventricle is considered severe, if the end diastolic volume approaches 150 to 160 ml/m² or the end systolic volume approaches 80–90 ml/m².
- Quantify right and left ventricular function.
- Diagnosis of additional lesions in patients with repaired congenital heart disease, in particular pulmonary branch stenosis or aortopulmonary collaterals.
- Quantify right to left ratio of pulmonary blood flow.

ECG

Detection of arrhythmias and assessment of QRS duration (in patients with repaired tetralogy of Fallot a QRS duration of >180 msec or an annual increase of 3.5 msec has been associated with a risk for sustained ventricular arrhythmias and sudden cardiac death).

Holter-ECG

Documentation of atrial or ventricular arrhythmias.

Chest-X ray

- Indicates heart size.
- May show an enlarged central pulmonary artery.
- Unilateral rarefaction of peripheral pulmonary arteries may indicate branch pulmonary artery stenosis.

Cardiopulmonary exercise testing

Indicated at regular intervals to objectify exercise tolerance and detect its decline.

Diagnostic cardiac catheterization

- Estimation of size and function of the right ventricle in patients who cannot be studied with MRI.
- Assessment of pulmonary artery anatomy and pressures if pulmonary hypertension is suspected.
- Assessment of left ventricular diastolic pressure if left ventricular dysfunction is suspected.
- Coronary angiography prior to pulmonary valve replacement to exclude significant coronary artery anomaly.

Biomarkers of heart failure

Measurement of BNP/Nt-pro-BNP is helpful to estimate the degree of ventricular dysfunction and to detect disease progression.

Indications for treatment

Replacement of the pulmonary valve is reasonable in the presence of significant pulmonary regurgitation (RGF > 30% on MRI assessment) and in the presence of one of the following findings:

- Symptoms (right heart failure, limited exercise capacity).
- Moderate to severe enlargement of the right ventricle or increase in size on serial assessment. The pulmonary valve should be replaced when ventricular volumes are below an end diastolic volume of 150–160 ml/m² and an end systolic volume of 80–90 ml/m², as a normalization in ventricular size after valve replacement is likely.
- Moderate to severe dysfunction of the right ventricle or deterioration on serial assessment.
- New onset or moderate or severe tricuspid regurgitation in the presence of right ventricular enlargement.
- New onset of symptomatic or sustained atrial or ventricular arrhythmia.
- QRS duration >180 ms or increase in QRS duration of ≥3.5 ms/year.

In patients with repaired tetralogy or similar lesions and moderate to severe pulmonary regurgitation, pulmonary valve replacement is reasonable if additional cardiac lesions need surgery.

Management

Pharmacologic treatment

Medical treatment of heart failure and arrhythmia shall not delay pulmonary valve replacement.

Catheter intervention

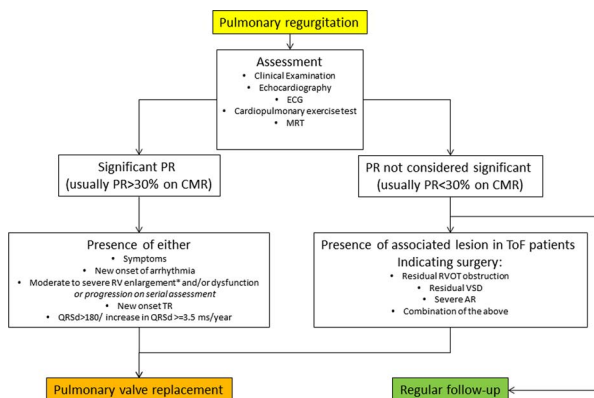
Percutaneous pulmonary valve replacement is an alternative to surgical pulmonary valve replacement if the underlying anatomy is suitable.

Surgery

- Surgical pulmonary valve replacement is the best evaluated treatment for symptomatic pulmonary regurgitation.
- There is no general consensus as to which type of pulmonary valve prosthesis should be used for pulmonary valve repair. Usually, biological valves are used.

Outcome

- Similar results have been reported using different pulmonary valve prostheses. Operative mortality is low (ECHSA congenital data base 1–3%) and depends on age, comorbidities, and number of previous operations. Freedom of re-replacement depends on age group and underlying heart disease. Freedom from reoperation in patients >1 year of age at implantation is as follows:– 60–70 % after 10 years after orthotopic implantation– 30–40 % after 10 years after extraanatomic conduit replacement



Simplified management algorithm for the treatment of pulmonary regurgitation

*In patients with repaired tetralogy of Fallot enlargement of the right ventricle is considered severe if the end diastolic volume approaches 150 to 160 ml/m² or the end systolic volume approaches 80–90 ml/m². If the pulmonary valve is replaced when ventricular volumes are below these thresholds, normalization in ventricular size after valve replacement is likely. It is therefore recommended to intervene on pulmonary regurgitation before the ventricle has enlarged above these thresholds.

Abbreviations: AR, aortic regurgitation; MRT, magnetic resonance tomography; PR, pulmonary regurgitation; RV, right ventricle; ToF, tetralogy of Fallot VSD, ventricular septal defect.

- After pulmonary valve replacement in infancy, earlier need for re-intervention is to be expected.

When performed timely, pulmonary valve replacement leads to reduction in right ventricular size, improvement in functional class and exercise tolerance.

The risk for infective endocarditis is increased after pulmonary valve replacement.

Recommendations for follow-up

Regular follow-up examinations are necessary usually in all patients who underwent pulmonary valve replacement. Focus is on any signs of degeneration of the implanted prosthesis. The usual follow-up interval is yearly.

Conflicts of interest:

Andreas Eicken: Proctor for Medtronic(Melody-valve)

Alexander Horke: None

Anselm Uebing: None

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Peripheral Pulmonary Artery Stenosis

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Abstract Peripheral pulmonary artery stenosis (PPS) is defined as congenital or postoperative narrowing or hypoplasia distal to the main pulmonary artery. Congenital PPS is often associated with genetic disorders like Alagille or Noonan syndrome.

PPS results in elevated right ventricular systolic pressure. Patient's condition may range from "free of symptoms" to right heart failure. Exact diagnosis is established mainly by angiography.

Invasive treatment is the only option, either by surgery, catheter intervention or combined (hybrid procedure). Whereas the surgeon is not able to address PPS distal to the hilus, every part of the pulmonary artery system can be approached by interventional means. The final therapeutic goal is to decrease right ventricular pressure, at its best to normal level.

Keywords: *peripheral pulmonary artery stenosis, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

PPS is a congenital hypoplasia or postoperative narrowing of one or more peripheral pulmonary arteries distal to the main pulmonary artery. It is defined by

- Reduction of the vessel diameter by $\geq 40\%$
- Reduction of $\geq 30\%$ in case of additional right ventricular volume overload (i.e. atrial septal defect or tricuspid regurgitation)

Common underlying post-operative congenital heart defects include

- Tetralogy of Fallot
- Pulmonary atresia and ventricular septal defect
- Common arterial trunc
- D-Transposition of the great arteries.

Relationships to consider

Associated lesions and syndromes

- Alagille syndrome
- Noonan syndrome
- Williams-Beuren syndrome
- Elastin defects
- German measles embryopathy

Differential diagnoses

- Primary pulmonary hypertension
- Valvar or subvalvar pulmonary stenosis

Haemodynamics

- Elevated right ventricular systolic pressure
- Subsequent pulmonary valve and/or tricuspid valve regurgitation

Clinical presentation

- Systolic murmur with propagation into the respective lung
- Patient's condition may range from "free of symptoms" to right heart failure (mild – moderate – severe)
- Rhythm disturbances

Diagnostic work-up

Echocardiography

Echocardiographic evaluation of PPS is limited to proximal lesions close to the pulmonary bifurcation.

Additional lesions to be assessed by echocardiography include

- Right ventricular outflow tract obstruction (subvalvar, valvar and supra-valvar pulmonary stenosis)
- Anomalies of the pulmonary bifurcation
- Additional septal defect
- Ascending aorta (supra-valvar aortic stenosis)

Pulse oximetry

May be decreased in patients with right to left shunting (e.g. ventricular septal defect).

ECG

Typical findings are right ventricular hypertrophy, incomplete or complete right bundle branch block.

Chest X-ray

May be helpful in cases of different perfusion of the lungs.

Cardiac Magnetic Resonance Imaging and Computed Tomography

Helpful for primary diagnosis and follow-up.

Cardiac catheterization

Angiography is still the gold-standard for detailed imaging. It is able to delineate exact anatomy of the stenosis by selective and supraseductive injections. Balloon sizing of the lesions may provide additional accuracy in terms of severity and extension of the stenotic segment. Treatment can be performed during the same session.

Indications for treatment

- Reduction of the vessel diameter $\geq 40\%$ of the regular vessel diameter.
- Reduction of the vessel diameter $\geq 30\%$ in case of additional right ventricular volume overload, i.e. atrial septal defect or tricuspid regurgitation.
- In patients after Glenn or Fontan operation a stenosis of less than 30% should be addressed.

Management

Aim is the reduction of increased right ventricular systolic pressure.

Pharmacologic treatment

There is no medical treatment for relief of PPS.

Catheter interventions

Therapeutic options are balloon dilation and stent implantation. In general balloon dilation is less effective compared to stent

implantation. A close cooperation with the pediatric heart surgeon is recommended.

- **Balloon dilation**
In small children balloon dilation should be preferred to stent implantation. In patients with PPS due to postoperative scar formation balloon dilation may be very effective. In case of severe stenosis the use of a cutting balloon catheter may be considered.
- **Stent implantation**
Stent implantation for treatment of peripheral pulmonary artery stenosis is an effective and long-lasting therapeutic option. Antiplatelet therapy is often administered over a period of three to six months following the procedure.

Surgery

Indication for surgery is relief of extrahilar pulmonary artery stenosis/hypoplasia by patch plasty. This operation is mainly performed in combination with correction of other underlying lesions or RV to PA conduit implantation.

In patients with hypoplastic pulmonary arteries implantation of an aortic-pulmonary shunt may promote vessel growth.

Hybrid procedure

A hybrid approach may be useful in selected patients.

Prognosis

Restenosis may occur after surgery as well after intervention even after repeated procedures. Reasons for re-stenosis include scar formation, intimal proliferation, inadequate radial forces of stents and/or growth of the patients.

Recommendations for follow up

Life-long regular follow-up visits are necessary. Noninvasive and/or invasive imaging is required in order to document the result of treatment and for planning further therapeutic steps.

Prevention

None

Conflicts of interests for this guideline:

Martin B.E. Schneider: None
Angelika Lindinger: None
Alexander Horke: None

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Valvar Aortic Stenosis

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Abstract A stenosis of the aortic valve is caused by an incomplete opening of the valve cusps and/ or by a hypoplastic aortic valve annulus. The obstructed flow across the aortic valve results in pressure load of the left ventricle leading to hypertrophy of the myocardium and in very severe stenosis to myocardial ischemia and failing of the left ventricle function. Patients with severe stenosis are at risk for syncope and sudden death, but many of these patients remain asymptomatic. Indication for treatment in asymptomatic patients with normal ventricular function is a mean Doppler gradient across the valve at rest of more than 40 mmHg. The first line treatment is the relief of the stenosis by interventional balloon valvuloplasty or by surgical commissurotomy/reconstruction. If these interventions are not successful a surgical replacement of the valve is indicated. Valve replacement can be performed by the Ross operation, or by implantation of a mechanical or biological valve. Life-long follow-up examinations are necessary in all patients with interventionally or surgically treated valvar aortic stenosis.

Keywords: *valvar aortic stenosis, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

A stenosis of the aortic valve is caused by an incomplete opening of the cusps and/or by a hypoplastic aortic valve annulus. The morphologic spectrum ranges from partially fused commissures to rudimentary or missing cusps resulting in a functional bicuspid or

monocusp valve. Thickened and severely dysplastic valves show a reduced opening. In very severe forms of aortic stenosis there is a pinhole opening in a thick valve structure.

A *critically valvar aortic stenosis* is defined as severe aortic stenosis in neonates resulting in a failing left ventricle due to the pressure load requiring a right to left shunt across the patent ductus arteriosus to maintain systemic circulation (duct- dependent systemic circulation).

Severe aortic stenosis in the fetus may result in a hypoplasia of the left ventricle and/or in endocardial fibroelastosis which represents a form of a restrictive cardiomyopathy.

Epidemiology

2,4/10.000 live births corresponding to 2,2% of all congenital heart defects.

Relationships to consider

Associated lesions

- Coarctation of aorta
- Subvalvar aortic stenosis
- Hypoplasia of the left ventricle

Differential diagnosis

- Isolated subvalvar aortic stenosis
- supralvalvar aortic stenosis
- Shone syndrome with mitral valve stenosis and subvalvar aortic stenosis in combination with coarctation of aorta
- hypoplastic left heart syndrome

Haemodynamics

- the obstructed flow across the stenotic aortic valve results in a pressure load of the left ventricle leading to myocardial hypertrophy and in very severe stenosis to myocardial ischemia and failing of the left ventricle.
- the severity of aortic stenosis is characterized by the pressure gradient across the aortic valve and the area of the aortic valve opening. The classification of the severity is shown in Table 1.

Clinical presentation

- auscultation
 - systolic ejection murmur loudest at the second right intercostal space radiating into carotid artery
 - often early systolic click

Table 1. Classification of severity of valvar aortic stenosis

Severity	Cardiac catheterisation Peak-to-peak gradient	cw Doppler Vmax ACC/AHA ESC	maximal instantaneous gradient Bernoulli	Mean instantaneous Gradient Bernoulli ACC/AHA ESC	Echo Aortic valve opening area ACC/AHA ESC
mild	<30 mmHg	<3 m/s	<36 mmHg	<25 mmHg	>1.5 cm ² (>1 cm ² /m)
moderate	30-50 mmHg	3-4 m/s	36-64 mmHg	25-40 mmHg	1-1.5 cm ² (0.6-1 cm ² /m)
severe	>50 mmHg	>4 m/s	>64 mmHg	>40 mmHg	<1 cm ² (<0.6 cm ² /m)

- thrill palpable in the jugulum in severe stenosis due to turbulences
- symptoms in severe stenosis:
 - dyspnea at mild exertion, angina pectoris, exercise induced syncope, sudden death
- Caveat:** patients even with severe stenosis often remain asymptomatic!
- Neonates with critical aortic stenosis:
 - often no ejection murmur audible due to severely reduced cardiac output
 - dyspnea, tachycardia and hepatomegaly as signs of heart failure with reduced ejection fraction
 - pulmonary oedema and cardiogenic shock when arterial duct closes in a duct dependent systemic circulation.

Diagnostic work-up

Echocardiography

Echocardiography is the key diagnostic imaging technique to delineate the following features:

- Morphology of the aortic valve (e.g. tricuspid, bicuspid or monocusp, mobility, dysplasia)
- Diameter of aortic valve annulus and left ventricular outflow tract
- Diameter and wall thickness of the left ventricle (e.g. hypertrophy of the posterior wall)
- function of the left ventricle
- Size and morphology of the pulmonary valve
- Doppler
 - Peak and mean gradient across the valve
 - Aortic valve regurgitation.
- In neonates with critical aortic stenosis for the decision making of biventricular repair versus univentricular palliation especially
 - length of the left ventricle(forming the apex?)
 - diameter of the mitral and aortic valve
 - endomyocardial fibroelastosis
 - patent arterial ductus.

Exclusion of other malformation such as subvalvar or supralvalvar aortic stenosis or coarctation of aorta.

ECG

Beyond early infancy, ECG may show left ventricular hypertrophy and myocardial ischemia.

Holter monitoring indicated if suspicion of ventricular arrhythmia or myocardial ischemia during exertion

Exercise test with blood pressure measurement

Useful in asymptomatic patients with moderate aortic valve stenosis for risk stratification and indication for treatment. The following findings should be evaluated:

- symptoms during exercise
- blood pressure response (insufficient increase or even decrease)
- ventricular arrhythmia
- repolarization abnormalities (myocardial ischemia).

Caveat: exercise test is contraindicated in symptomatic patients!

Cardiac catheterization

Usually indicated only when interventional procedure is planned.

Magnet Resonance Imaging/ Computer Tomography

May give additional information on aortic morphology and coronary anatomy.

Indications for treatment

In children beyond the neonatal age a relief of the stenosis is indicated in symptomatic patients or in asymptomatic patients with a severe stenosis which is characterized by

- non-invasive Doppler instantaneous gradient >64 mmHg (Flow velocity > 4 m/sec)
- non-invasive Doppler mean gradient >40 mmHg or
- invasively measured peak-to-peak gradient >50 mmHg.

These measurements are only applicable for the treatment decision if the patient has a normal left ventricular function at resting conditions. The non-invasive determination of the mean gradient by Doppler is most frequently used and most useful parameter in childhood.

In children with a moderate stenosis and gradient measurements below the above mentioned limits the treatment should be considered if further features are present such as

- ST- Segment changes at rest
- Pathological response during exercise testing
- Reduced left ventricular function
- Dilatation of left ventricle

Management

In the majority of patients, aim of management is the relief of the stenosis by catheter intervention or surgery. As a general principle catheter interventions or aortic valve reconstruction are no final solutions but of delaying character. Target of treatment in neonates, infants and children is to postpone valve replacement until adult size anatomical dimensions are reached. Valve replacement possibilities, perioperative risk and long term outcomes improve in teenagers and adults.

Critical valvar aortic stenosis

- Pharmacologic treatment
Administration of intravenous Prostaglandin to maintain patency of the arterial duct.

In neonates with acute heart failure treatment with inotropic support, reduction of preload with loop diuretics and mechanical ventilation may be necessary.

- Relief of the stenosis
Treatment options in neonates with a sufficient diameter of the valve annulus are

- catheter intervention with balloon dilation or
- surgical commissurotomy

Both methods achieve similar good results. In the hemodynamically instable neonate the interventional balloon dilation is the preferred treatment option since this approach is less invasive and better tolerated than open heart surgery.

In neonates with a sufficient left ventricular size but a very small aortic valve annulus and/or severely dysplastic cusps an interventional valvuloplasty or a surgical valvotomy is often unsuccessful. The Ross operation offers an alternative treatment option (see below).

- Univentricular palliation

In neonates with severely hypoplasia of the left heart structures

precluding a biventricular approach, only a univentricular palliation is possible.

Catheter interventions

- Valvuloplasty

Balloon valvuloplasty is an established alternative in children of all age groups to surgical commissurotomy. The results of valvuloplasty are equally good compared to the surgical ones. This intervention is a low risk procedure in children beyond neonatal age and can postpone aortic valve replacement for many years, often up to adolescence or adulthood.

Aortic regurgitation is a side effect of valvuloplasty. To avoid a severe aortic valve incompetence, the diameter of the balloon should not exceed the diameter of the aortic annulus (ratio diameter balloon/ aortic annulus 0.8–0.9). Furthermore rapid right ventricular pacing to reduce the cardiac output during balloon inflation helps to minimize the damage to the valve cusps.

Valvuloplasty should not be attempted in the presence of preexisting aortic valve regurgitation grade 2 and more.

- Percutaneous aortic valve replacement

No experiences in children available for the time being

Surgery

The following surgical procedures are possible

- Commissurotomy with and without reconstruction of cusps
- Ross Operation
- valve replacement with
 - mechanical valve
 - biological valve

- Commissurotomy with/without reconstruction of cusps

As for valvuloplasty a good valvar morphology and absence of a significant aortic valve regurgitation are prerequisites for a successful commissurotomy. Additional reconstruction of dysplastic cusps may be indicated to diminish the degree of residual stenosis and regurgitation.

If the cusps intraoperatively are considered not to be suitable for a successful repair, a valve replacement is performed in the same operation.

- Ross operation In the case of aortic stenosis the aortic root is replaced by the patient's pulmonary root (autograft) including a coronary artery transfer. A valved biological conduit (e.g. homograft/bovine valved jugular vein) is placed in pulmonary position.

The Ross operation has the following advantages:

- can be performed at any age
- anticoagulation is not necessary.

The Ross operation has the following disadvantages:

- Change of a one vessel in a two vessel disease
- the valved conduit in pulmonary position will require a replacement due to patients growth or degeneration of the valve.

This replacement can be performed by percutaneous pulmonary valve implantation in suitable patients.

- failure of the autograft (root dilatation with consecutive aortic valve regurgitation) in aortic position is common (mostly in not augmented root replacements) due to the unphysiological systemic pressure to the originally pulmonary valve.

- Mechanical valve replacement

This type of valve replacement is an established and safe

procedure with the advantage of a potentially unlimited lifespan, but has the following limitations/disadvantages:

- aortic annulus size of >15 mm
- lifelong anticoagulation
- in pregnant women the risk of anticoagulation to the fetus and mother may require a modification of the anticoagulation scheme.
- In children the size of the valve will be outgrown requiring a repeated valve replacement. Preferably a mechanical valve, therefore, should be implanted only in patients in whom an adult size valve can be used.
- Aortic homograft replacement
This type of valve replacement is performed as root replacement with coronary artery transfer. It plays a minor role in current practice. Its main indication is destructive endocarditis of the aortic valve.
Major problems:
 - supply of suitable grafts (especially in small sizes)
 - High risk reoperations after root replacement due to calcification of the whole root
- Biological valve replacement
A biological valve replacement can be performed by stented xenografts (aortic valve replacement) or as root replacement by xenoconduits with coronary transfer. The commercially available valves differ in tissue, stent design, fixation, fixation pressure and antimineralization agents, leading to great variability of durability. Only limited experiences are published in pediatric patients. Biological valve replacement has following limitations/disadvantages:
 - Aortic annulus size of >17 mm
 - anticoagulation necessary for 6 month.
 - Limited life span due to early degeneration

This type of valve replacement may be useful in growing children to bridge the time until an implantation of an adult size valve is possible or in young women in the childbearing age to avoid the risk of anticoagulation during pregnancy.

Prognosis/Outcome

Balloon valvuloplasty

The procedure related mortality of balloon valvuloplasty in critically ill neonates depends on the individual clinical status and ranges around 10%. In children beyond neonatal age, however, the mortality is very low (less than 5 in 1000 children)

Surgical valve repair

In patients with surgical valve repair the 30 days- perioperative mortality is 2%.

Valve replacement

In patients after the *Ross procedure* the perioperative mortality depends on age, ventricular function, anatomical aspects, comorbidity and number of previous operations. In a systematic meta-analysis the Ross procedure was associated with an early mortality of 4.2% and late mortality of 0.6%/year. The re-operation rate was 1.6%/year for autograft failure and 1.9%/year for pulmonary conduit failure.

In patients after *mechanical valve replacement* a systematic meta-analysis showed that this type of valve replacement was associated with an early mortality of 7.3% and late mortality of 1.2%/year. The perioperative 30 days- mortality was 1.5% and the re-operation rate 1.07%/year.

In patients after *aortic homograft replacement* a systematic meta-analysis showed that this procedure was associated with an early mortality of 12.8%; and late mortality of 1.5%/year. The re-operation rate was 5.4%/year.

In patients after *biological valve replacement* the perioperative mortality for stented xenografts is comparable to mechanical valve implantation.

Recommendations for follow-up

Life-long follow-up examinations are necessary in at least yearly intervals.

Echocardiography is the cornerstone for the assessment of aortic valve and ventricular function.

In patients after the Ross operation the function of the right ventricle to pulmonary artery conduit has to be examined beside the function of the autograft in aortic position.

In patients with a mechanical valve replacement attention should be paid to a possible paravalvar leak and/or to an impaired opening of the leaflets due to thrombotic adhesions. Monitoring of anticoagulation (INR-levels) is essential.

Patients with aortic stenosis and an indication for intervention or surgery should refrain from any sport. Those with asymptomatic moderate stenosis can exercise at all sports with low or moderate intensity. Patients with mild stenosis can perform any kind of sports, even on competition level.

All patients with a valve replacement have a high risk for endocarditis. Life-long prophylaxis for endocarditis, therefore, is indicated.

Conflicts of interest for this guideline:

Jochen Weil: None

Nikolaus Haas: None

Alexander Horke: none

Peter Ewert: None

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Aortic regurgitation

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Abstract Aortic regurgitation in childhood and adolescence is mainly a sequela after balloon dilatation or surgical intervention of aortic stenosis. Resulting left ventricular volume overload may lead to congestive heart failure. The primary diagnostic tool for diagnosis and assessment of severity is echocardiography. Indications for surgery include presence of symptoms and/or progressive left ventricular dilatation. The surgical aim is valve repair, but valve replacement may be required.

Keywords: *aortic regurgitation, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Aortic regurgitation is defined as a leakage of the aortic valve and is graded as mild, moderate or severe. Congenital aortic regurgitation is very rare. In the majority of patients it is a sequela of

- Interventional or operative valve procedures
- Bicuspid aortic valve
- Unicuspid aortic valve
- Subaortic or doubly committed VSD
- Subaortic stenosis
- Inflammatory and degenerative cardiac valve diseases (e.g. endocarditis)
- Dilatation of the aortic root (e.g. Marfan syndrome)

Differential diagnosis

- Aorto-pulmonary window
- Aortico-left ventricular tunnel
- Ruptured Sinus valsalvae aneurysm
- Coronary fistulae

Haemodynamics

Left ventricular volume overload with elevated enddiastolic pressure, increased wall tension and increased myocardial oxygen

consumption are found in hemodynamically significant aortic regurgitation. Impaired coronary perfusion due to low systemic diastolic blood pressure may aggravate myocardial dysfunction.

Clinical presentation

Symptoms

- Often mild or absent for a long time.
- In severe aortic incompetence: impaired exercise tolerance, fatigue, thoracic pain, rarely dizziness and syncope.
- In advanced stages signs of left heart failure: tachycardia and pulmonary congestion.

Clinical examination

With moderate or severe aortic regurgitation a widened pulse pressure, bounding pulses (“water hammer”) and a hyperdynamic precordium are present. Auscultation along the left sternal border reveals a diastolic decrescendo murmur.

Diagnostic work-up

Echocardiography

Echocardiography is mandatory to assess

- Anatomy, dimension and function of the aortic valve
- Grading of the severity of aortic regurgitation (mild, moderate, severe) by 2D- and Doppler echocardiography.
- Dimensions of aortic root and ascending aorta
- Size and function of the left ventricle.
- Diastolic reversal flow in descending aorta

Progressive left ventricular enlargement and/or decreased LV ejection fraction are signs of impeding left ventricular compensatory mechanism.

ECG

Progressive QRS widening and repolarization abnormalities may reflect left ventricular volume overload and dysfunction.

Cardiopulmonary exercise test

Cardiopulmonary exercise capacity, signs of myocardial ischemia during exercise and occurrence of rhythm disturbances are helpful for decision making in moderate and severe aortic regurgitation.

Laboratory tests

NT-proBNP levels are usually normal or only mildly elevated; there is only a weak correlation between left ventricular diameters and heart failure stage according to the NYHA classification.

Cardiac catheterization

Indicated if hemodynamically relevant cardiac co-morbidities need to be assessed; selective coronary angiography may be recommended prior to aortic root surgery.

Magnetic resonance imaging, computed tomography scans

Magnetic resonance imaging is indicated in patients with moderate and severe aortic valve insufficiency in case of suboptimal echocardiographic images for the assessment of left ventricular systolic and diastolic volumens, systolic function, grading of aortic regurgitant severity (regurgitation fraction: mild <30%, moderate 30–49%, severe >50%) and left ventricular volumetry. Furthermore,

dimensions of the aortic root, coronary arteries and aortic valve anatomy (e.g. bicuspid) can be evaluated.

Computed tomography

Computed tomography may be useful to visualize anatomy of aortic valve, coronary arteries, aorta and left ventricle, if echocardiographic imaging is not sufficient and/or contra-indications for magnetic resonance imaging are present.

Indications for treatment

- Surgical management is indicated if clinical symptoms are present.
- In asymptomatic patients, an operation is indicated in case of progressive left ventricular dilatation and/or impaired left ventricular function.
- Aortic root dilatation requires special attention: in adults, aortic root diameter ≥ 55 mm (Marfan syndrome ≥ 45 mm, bicuspid valve ≥ 50 mm in the presence of risk factors) is an indication for surgery irrespective of the extent of aortic regurgitation. In children, upper limits of growth-adapted aortic root diameter and/or progression of aortic dilation should be used.
- Aortic regurgitation without hemodynamic significance does not require treatment unless combined with another lesion requiring intervention (e.g. subaortic VSD, doubly committed VSD, subvalvar aortic stenosis, aortic root disease).

Management

Pharmacologic treatment

In children, afterload-reducing medication (e.g. ACE inhibitors) is often used to postpone a surgical procedure. However, there is no evidence to support its efficacy except when co-existing arterial hypertension is present. The use of β -blockers should be avoided.

Catheter interventions

Transcatheter aortic valve replacement is currently not approved in children and adolescents.

Surgery

- Aortic valve repair is the treatment of choice depending on individual anatomy.
- In patients with aortic root dilation: valve sparing aortic root replacement appears preferable in selected patients, otherwise a composite graft is used to replace the valve and the aortic root.
- For aortic valve replacement, various surgical techniques are available: insertion of a homograft, a bioprosthesis, a mechanical valve, or the Ross operation (pulmonary autograft).

Outcome:

- When performed timely, aortic valve repair/replacement results in reduction of left ventricular size, improvement of left ventricular function and exercise tolerance.
- Early postoperative mortality is mainly influenced by patient-related factors like age, preoperative haemodynamics, urgency of the procedure, and the underlying disease
- Long-term results after aortic valve repair and replacement are favourable; residual aortic regurgitation may increase over time.
- The risk for infective endocarditis is increased after aortic valve replacement.

- Valve-related complications, such as degeneration of biological prosthesis or homograft may result in the need for re-operations.

Prognosis

Patients with aortic regurgitation and normal left ventricular systolic function may remain asymptomatic and clinically stable for many years; some patients, however, may develop progressive left ventricular dilatation and functional deterioration requiring a surgical procedure.

Recommendations for follow-up

Regular follow-up studies by echocardiography are recommended in order to detect any increase in the degree of aortic valve regurgitation, progressive left ventricular dilatation and functional impairment. ECG is useful to detect signs of myocardial ischemia. Exercise testing is indicated in patients with more than mild regurgitation.

After aortic valve surgery, attention should be focussed on degeneration of native or implanted valve tissue. Life-long anticoagulation is indicated following mechanical aortic valve replacement.

Conflict of interests for this guideline

Carsten Rickers: None
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 Oliver Dewald: None

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Subaortic stenosis

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Abstract Subaortic stenosis, also called subvalvar aortic stenosis, is an obstruction of the left ventricular outflow tract below the aortic valve annulus. Leading symptom is a systolic ejection murmur. The clinical course of subvalvular aortic stenosis varies from stable asymptomatic disease to rapidly progressive obstructive severe illness. Investigation is primarily performed by echocardiography. Treatment is indicated at a Doppler gradient of >50 mm Hg maximum / >30 mm Hg mean, or in progressive aortic regurgitation. Treatment of choice is surgical relief of obstruction. Long-term follow-up is required due to its high rate of postoperative recurrence.

Keywords: *subaortic stenosis, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Subaortic stenosis, also called subvalvar aortic stenosis, is a left ventricular outflow tract obstruction below the aortic valve annulus. It may be isolated or part of more complex lesions. The four basic anatomic variants (the first two accounting for 70–80 % of cases) are:

- a thin discrete membrane consisting of endocardial fold and fibrous tissue
- a fibromuscular ridge consisting of a thickened membrane with a muscular base at the crest of the interventricular septum

- a diffuse, fibromuscular, tunnel-like narrowing of the left ventricular outflow tract
- accessory or anomalous mitral valve tissue.

The general opinion is that discrete/fibromuscular subaortic stenosis is an acquired lesion resulting from an underlying abnormality in left ventricular outflow tract architecture representing a dynamic process that continues and has consequences into adulthood.

Sometimes left ventricular outflow tract obstruction develops "secondary" after corrective surgery for congenital heart disease such as atrioventricular septal defects, some types of ventricular septal defects, and more complex lesions requiring a tunnel-like surgical connection of the left ventricle to the aorta.

Haemodynamics

Subaortic stenosis causes pressure load and hypertrophy of the left ventricle despite normal arterial blood pressure. Progressive secondary aortic regurgitation may develop due to jet damage of the valve.

Epidemiology

Subaortic stenosis accounts for approximately 1% of all congenital heart defects, or 10 to 14% of congenital left ventricular outflow tract obstructions, respectively; 2/3 of patients are male. In clinical practice, most cases of membranous and fibromuscular subvalvar aortic stenosis develop and progress over time so that most consider it an acquired rather than congenital lesion. However, there also is evidence for a congenital predisposition.

Relationships to consider

Associated lesions / syndromes

- More than half of the patients have concomitant cardiac malformations: patent ductus arteriosus, ventricular septal defect, aortic coarctation, and others
- Shone complex (combination of subaortic stenosis, mitral valve abnormalities and aortic coarctation)
- More complex congenital heart disease associated with subaortic stenosis:
- double outlet right ventricle, tricuspid atresia or interrupted aortic arch with ventricular septal defect and posterior septal malalignment)
- 'Hypoplastic left ventricular outflow tract', described as combination of tunnel-like subaortic stenosis, hypoplastic aortic valve annulus and aortic valve stenosis

Differential diagnosis

- Valvar aortic stenosis
- Hypertrophic obstructive cardiomyopathy

Clinical presentation

Patients with mild or moderate isolated subaortic stenosis may remain free of symptoms for a long time. Severe obstruction may lead to exercise intolerance, dyspnoea, heart failure and pulmonary edema. Sometimes the first symptom is exercise induced syncope or chest pain. In significant obstruction there is always a systolic ejection murmur without an ejection click and most often without a thrill in the jugular notch.

Diagnostic work-up

Aim: Morphologic description (including relation to mitral valve and aortic valve), assessment of functional severity, and exclusion or description of associated lesions.

Echocardiography is the primary diagnostic method and should delineate the following features:

- Localization, anatomic variant, and severity of subaortic stenosis.
- Aortic valve (regurgitation?)
- Mitral valve (morphological or functional subaortic obstruction?)
- Left ventricular dimension, function and myocardial hypertrophy
- Mean and maximum gradients calculated by Doppler echocardiography.

ECG may show left ventricular hypertrophy and repolarization abnormalities in advanced disease.

Indications for treatment

Asymptomatic patients with isolated subaortic stenosis and normal left ventricular function should be treated at a peak / mean Doppler gradient of >50/30 mm Hg, respectively. An additional indication might be new or progressive aortic regurgitation during follow-up.

Management

Medical and interventional treatment are not recommended.

Surgery

Treatment of choice is surgical relief of subaortic outflow tract obstruction. Intraoperative techniques thereby mainly depend on the individual morphology, involving resection of fibrous membranes or fibromuscular rings with or without myectomy and mitral valvuloplasty. Tunnel-like lesions may require patch enlargement of the left ventricular outflow tract and/or aortic valve replacement (Rastan-Konno technique and modifications).

Outcome

- Overall surgical mortality is less than 1%; however, it is higher after surgery for tunnel-like lesions operated in infancy
- Perioperative risks are aortic regurgitation, mitral valve complications requiring repair or replacement, acquired ventricular septal defect, left bundle branch block, and complete AV-block
- Late mortality nowadays is low in isolated membranous or fibromuscular subaortic stenosis, but not negligible in tunnel-like and complex forms.

Prognosis

- Native subaortic stenosis is usually progressive over time. However, mild forms with a maximal Doppler gradient of less than 30 mm Hg in infancy may remain stable over long time period. Freedom from initial operation was reported to be 60% at 5 years and 50% at 15 years from initial diagnosis
- Earlier progression to subaortic resection was associated with higher initial LVOT gradients, smaller age at diagnosis, smaller mitral annulus z-score and a faster rate of LV gradient progression
- Recurrence rates after resection and myectomy of isolated subaortic stenosis are significant (20–30%, reoperation required

in 15–20 % in 10 years follow-up) and have been associated with high preoperative outflow tract pressure gradients, tunnel-like lesions, and younger age at surgery. Recurrence rates after more complex surgical procedures (e.g. Rastan-Konno technique) are lower.

Recommendations for follow-up

Asymptomatic patients with lower gradients and without symptoms beyond infancy should be followed at 12 months intervals. Long term follow-up after corrective surgery is mandatory due to the risk of restenosis and progression of aortic regurgitation.

Conflict of interests for this guideline

Ingo Dähnert: None

Sven Dittrich: None

Narayanswami Sreeram: None

Gerardus Bennink: None

Harald Bertram: None

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Supravalvular aortic stenosis

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Abstract Supravalvular aortic stenosis is an obstruction of the ascending aorta at or above the sinotubular junction. Most commonly it is due to a genetic disorder of the elastin gene with or without the clinical features of Williams-Beuren syndrome.

Diagnostic evaluation is primarily performed by echocardiography. Obstructions of branch pulmonary arteries, coronary arteries or aortic branches are frequent and must be ruled out or documented. Other angiographic imaging techniques might be necessary.

Treatment is required in symptomatic patients, advanced left ventricular hypertrophy or mean Doppler gradient of more than 50 mm Hg. It consists in surgical relief of aortic obstruction. Long-term follow-up is required due to the risk of restenosis.

Keywords: *supravalvular aortic stenosis, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Supravalvular aortic stenosis is an obstruction of the ascending aorta. It may occur as isolated lesion or as part of more complex malformations. Most common (60–75%) is a distinct hourglass shaped obstruction at the sinotubular junction. Other forms (25–40%) consist of partial or complete hypoplasia of the ascending aorta or both the entire thoracic and abdominal aorta. Proximal obstructions of all aortic branches may occur. Rare cases of membranous supravalvular aortic obstructions have been described.

In most cases there is a structurally altered aortic wall: aortic media is thickened and dysplastic with increased numbers of smooth muscle cells and collagenous fibers while elastic fibers are decreased and elastin is abnormal. This results in increased aortic stiffness and impaired Windkessel effect.

Supravalvular aortic stenosis may also involve the distal part of the aortic valve commissures and thus also compromise the movement of the semilunar valve leaflets. Coronary artery obstructions of variable degree may be present due to aortic wall abnormalities obstructing the ostia or extending to the coronary vessel walls.

More than 50% of patients with supravalvular aortic stenosis are also affected by supravalvular pulmonary stenosis.

Epidemiology

Supravalvular aortic stenosis accounts for less than 0.05% of congenital heart defects, corresponding to less than 7% of all fixed forms of congenital left ventricular outflow tract obstructions. The estimated incidence is 1:20,000 live births. It is classically associated with the autosomal-dominant multi-system disorder Williams-Beuren syndrome (estimated prevalence: 69%), a genetic disorder caused by deletion or mutation of the elastin gene on the short arm of chromosome 7. Supravalvular aortic stenosis in these patients has been characterized as one manifestation of an ‘elastin arteriopathy’.

Haemodynamics

Supravalvular aortic stenosis results in increased pressure load and concentric hypertrophy of the left ventricle. Prestenotic dilation of the aortic root may develop.

Diastolic coronary perfusion may be impaired. The risk of subendocardial ischemia is increased even in the absence of additional coronary obstructions.

Clinical presentation

- Isolated supravalvular aortic stenosis usually remains asymptomatic for a long time.
- Higher pressure gradients cause a characteristic loud mid-systolic ejection murmur. In contrast to valvular aortic stenosis an ejection click is absent. The murmur is transmitted along the carotid vessels. Usually there is a thrill in the jugular notch.
- Asymmetric upper extremity blood pressures are often noted. In patients with Williams-Beuren syndrome it is not unusual, that the highest blood pressure can be measured at the left arm.
- Left ventricular dysfunction may lead to exercise intolerance and symptoms of heart failure. Coronary involvement can cause chest pain by myocardial ischemia.
- In Williams-Beuren syndrome there are associated syndromal findings e.g. failure to thrive, elfin face, intellectual impairment, arterial hypertension, and others.

Relationships to consider

Associated lesions

- supravalvular pulmonary artery stenosis
- coronary artery stenosis due to focal or diffuse coronary narrowing, or due to obstruction by redundant, dysplastic aortic valve leaflets
- aortic coarctation
- ostial stenoses of carotid, subclavian, renal, mesenteric, iliac, and other peripheral arteries.

Caveats

Any procedure requiring general anesthesia in patients with haemodynamically relevant supravalvular aortic stenosis is at elevated risk for severe complications and sudden cardiac death due to the impaired coronary perfusion in situations of arterial hypotension; patients at highest risk are infants with the Williams-Beuren syndrome, particularly those with biventricular outflow tract obstruction.

Diagnostic work-up

Imaging should reveal the entire thoracic and abdominal aorta as well as originating arteries. Relevant central and peripheral pulmonary artery stenoses must be ruled out. If echocardiography is inconclusive, other imaging techniques are mandatory.

Echocardiography is the primary diagnostic method. The following details should be described:

- Localization, form and severity of supravalvular aortic stenosis
- Left ventricular outflow tract, aortic valve, and ascending aorta
- Aortic arch (coarctation?), supraaortic vessels
- Left ventricular function and extent of myocardial hypertrophy
- Branch pulmonary artery stenosis
- Doppler calculations of mean and maximum gradients

MRI- and CT-angiography are equally suited for investigation of aorta and pulmonary arteries.

Abdominal ultrasound may detect stenoses of the abdominal aorta or large visceral arteries.

Diagnostic *cardiac catheterization and angiocardiology* may be indicated in selected cases, e.g. suspected coronary artery obstruction, peripheral pulmonary artery stenosis or distal systemic artery obstruction. Patients with more severe obstructions, coronary stenosis or bilateral outflow tract obstruction have a significantly higher mortality associated with catheterization.

ECG usually shows left ventricular hypertrophy. Repolarization abnormalities may occur with progressive disease or coronary involvement. Right ventricular or biventricular hypertrophy may be found in additional supravalvular pulmonary stenosis.

Genetic tests and counseling are recommended in the presence of phenotypic abnormalities or in familial forms. Prenatal genetic diagnostics may be applicable in familial forms.

Indications for treatment

- Treatment of isolated supravalvular aortic stenosis is indicated at the occurrence of symptoms like chest pain, dyspnoea or syncope.
- Other indications are progressive left ventricular hypertrophy or repolarization disorders in ECG.
- Treatment of asymptomatic patients is recommended at a peak / mean Doppler gradient of above 70 / 50 mmHg or an invasive measured peak-to-peak gradient of more than (30 to) 60 mmHg.

Management

Medical treatment and catheter interventions are not recommended

Surgery

Surgical treatment consists in an enlargement of the obstructed region. Apart from the simple patch plastic, repair by extended aortoplasty was developed. Different techniques are available.

Surgical risk is elevated in diffuse aortic hypoplasia involving the aortic arch, bicuspid aortic valve, biventricular outflow tract obstruction and in multiple vessel involvement.

Outcome

Surgical results are generally favourable. There is an elevated risk of early mortality in infancy, potentially representing more severe disease in symptomatic children. Overall mortality is reported to be 4% to 6% at 10 years, up to 15% at 15 years and between 3% and 23% at 20 years.

Surgical elimination of the obstruction does not cure the underlying general aortic wall disorder.

Prognosis

- Pressure gradients of less than 20 mm Hg in infancy generally remain unchanged during the first two decades of life, pressure gradients exceeding 20 mm Hg usually increase during childhood.
- After surgical repair, residual obstruction, aortic regurgitation and coronary complications may occur. The risk of reoperation varies between 2 and 9% in the entire group being highest after surgical repair in infancy (up to 50% after 5 years).

- Supravalvular aortic stenosis may predispose the coronary arteries to premature atherosclerotic changes.

Recommendations for follow-up

Asymptomatic patients beyond infancy with low gradients should be followed at 12 months intervals. Restriction of physical activity is not required, if the pressure gradient is less than 20 mmHg. Long term follow-up after surgical repair is mandatory for residual obstructions, aortic regurgitation and systemic arterial hypertension.

Conflict of interests for this guideline

Ingo Dähnert: None

Sven Dittrich: None

Narayanswami Sreeram: None

Gerardus Bennink: None

Harald Bertram: None

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Coarctation of the aorta

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Abstract Coarctation of the aorta is an anatomically “simple” congenital cardiac defect. Various surgical and interventional measures may relieve the stenosis, the functional result however may require lifelong medical attention due to hypertension and related problems. If untreated or inadequately treated, the life expectancy is reduced.

Keywords: *coarctation aorta, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- defined narrowing of the distal aortic arch, it may also extend to the thoracic and abdominal aorta
- prevalence: 3 /10.000 live births (corresponding to 3–5% of all congenital heart defects)

Relationships to consider

Associated lesions

Common: bicuspid aortic valve (45–83%), valvular aortic stenosis, PDA, VSD, Shone Complex, stenosis or abnormal origin of the subclavian artery. Less common: hypoplastic left heart syndrome, AVSD, d-TGA, DORV (Taussig Bing), aneurysms of the ascending aorta, cerebral aneurysms

Associated syndromes

Turner, Williams Beuren, CHARGE, VACTERL, Kabuki

Differential diagnosis

PDA, interrupted aortic arch, other left sided obstructive lesions

Hemodynamics and clinical presentation

- Critical CoA of the newborn: severe narrowing and duct dependent perfusion of the lower body (weak femoral pulses, differential cyanosis), rapid development of heart failure, shock and secondary organ failure after PDA closure (no femoral pulses).
- Non Critical CoA: prestenotic hypertension (nose bleed, head aches, LV hypertrophy), poststenotic hypotension (leg pain), weak or absent femoral pulses, systolic murmur.

Diagnostic work-up

Clinical examination

Auscultation and assessment of pulses, saturation and blood pressure at both arms and legs.

24-hr blood pressure

Helpful to define hypertension and for long term management

Ergometry

Helpful to assess exercise induced hypertension in long term management

ECC

Not helpful in small children, may show left ventricular hypertrophy later

ECHO

Key parameter for diagnosis, must include LV size and function, aortic valve anatomy, anatomy of the arch and head and neck vessels, gradient (PW/CW), flow in abdominal aorta, additional defects (i.e. PDA, LSVC, etc.)

MRI

Not necessary if ECHO is adequate. Good visualization of anatomy in older patients, good tool for planning of therapy

CT

Same as for MRI, caveat: radiation

Cardiac catheterization

No longer used for diagnostics, only if intervention is planned.

Indications for treatment

General aim is to establish a normal sized aorta without gradient. Treatment is indicated

- in every patient with critical CoA irrespective of the measured or calculated gradient.
- in all patients with a systolic blood pressure gradient > 20 mmHg
- in those patients with a systolic gradient < 20 mmHg, arterial hypertension (at rest or exercise or at 24-hr BP monitoring) and a significant narrowing of the CoA site (CoA to aorta ratio at the level of the diaphragm (CoA index < 80%))

Management

Medical treatment

in critical CoA prostaglandin is used for re-opening the PDA and establishment of body perfusion. Additional treatment for heart failure or organ impairment as necessary.

Surgery

In typical CoA neonates and children resection and extended end-to-end anastomosis (lateral thoracotomy) is treatment of choice, additional patch augmentation if needed. In older children prosthesis insertion or bypass surgery may be necessary.

When additional arch hypoplasia exist arch augmentation using medial sternotomy, cardiopulmonary bypass, hypothermia and selective perfusion of the head vessels may be necessary.

Risks: in simple CoA mortality is below 3%, in complex CoA (i.e. plus VSD, hypoplastic arch, etc.) below 10%. Other risks include nerve lesions (phrenic and recurrent nerve), chylothorax and (rarely) paraplegia. Restenosis is 5–10% in older children but higher in newborns.

Catheter interventions

- In newborns and children below 6 months of age balloon dilatation or stent implantation is not standard care due to the high rate of restenosis. It may be used as palliative therapy if surgery must be postponed (NEC, intracranial hemorrhage, sepsis, etc.).
- In children > 6 months interventions show comparable results to surgery. Balloon dilatation is favorable in short segment stenosis, stent implantation in longer segment stenosis. Balloon dilatation is the treatment of choice for all re-coarctations.
- In older children stent implantation is the treatment of choice. Covered stents may be used for tight coarctations.

Risks: Mortality <1%, vascular injury at the access site (bleeding, occlusion), injury at the CoA-site with dilatation (rupture, aneurysm, dissection) or stent implantation (dislocation, stent fracture, intima proliferation).

Prognosis

Untreated patients or patients with not adequately treated with CoA show limited life expectancy. High prevalence of hypertension especially in those with inadequate treatment, i.e. a CoA index <80%. Further risks are development of aortic aneurysms, rupture, dissection, coronary or cerebral vascular disease, retinal damage, intracranial haemorrhage. In patients with bicuspid aortic valve aortic dissection and aneurysms are possible.

Recommendations for follow-up

- Lifelong follow-up examinations are compulsory:
 - blood pressure gradients, 24-hr blood-pressure monitoring, ECHO annually
 - Exercise tests every 3–5 years
 - MRI if ECHO is unclear every 5 years
- Sport: static exercise must be avoided, endurance exercise is preferred.

Conflict of interests for this guideline

Nikolaus A. Haas: None
 Peter Ewert: None
 Alfred Hager: None
 Christian Schlensak: None

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Congenitally corrected transposition of the great arteries (cc TGA)

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Abstract Congenitally corrected transposition of the great arteries (ccTGA) is characterized by the presence of discordant connections at both the atrioventricular and ventriculoarterial junctions. Pathology and pathophysiology in patients with isolated cc TGA as well as in patients with ccTGA and typical associated lesions are presented. Diagnostic work-up and therapeutic options are summarized.

Keywords: *congenital corrected transposition of great arteries (ccTGA), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

ccTGA is due to abnormal looping of the primitive cardiac tube to the left (l-loop) instead of to the right as in the normal embryonic development of the heart (d-loop). ccTGA is a rare lesion with a prevalence of 0.03 in 1000 newborns accounting for approximately 0.05% of all congenital heart defects.

In the patient with isolated ccTGA, situs solitus and normal atrial arrangement, systemic venous blood enters the morphologic right atrium which is connected by a mitral valve with a morphologic left ventricle. This left ventricle pumps the blood into a discordantly connected transposed pulmonary artery. Pulmonary venous return is received by the left atrium which is connected via a tricuspid valve with a morphological right ventricle which supports the transposed aorta. Although ventricular

inversion is physiologically corrected by the transposed great arteries, hemodynamics and natural history are significantly different from patients with a normal heart even in patients with isolated cc TGA.

Additional anomalies/lesions are present in >90% and include:

- Situs inversus (5%)
- Dextrocardia/mesocardia (25%)
- Ventricular septal defect, perimembranous, large (>50%)
- Left ventricular outflow tract obstruction, at subvalvar/muscular and valvar level (30–50%, functionally pulmonary stenosis), usually combined with a large VSD
- Dysplasia of the tricuspid valve (90%), + /- displacement of the septal or posterior leaflets, comparable with Ebstein anomaly of the tricuspid valve
- Coronary artery anatomy: important for double switch repair (see below); in general – mirror image distribution of the coronary arteries in atrial situs solitus and cc TGA (right-sided coronary artery – morphological left coronary artery and vice versa); significant prevalence of variant patterns
- Right aortic arch
- Aortic arch obstruction
- Atrial septal defect
- Specialized conduction system: double AV nodes – a normal posterior AV node with no AV bundle, and an abnormal anterior AV node with the penetrating bundle – risk of complete AV block at birth (5–10%) and/or later in life
- Accessory atrioventricular pathways.

Differential diagnosis

Other forms of complex congenital heart defects with transposition of the great arteries.

Sophisticated work-up should be performed in specialized centres experienced in paediatric cardiology and paediatric cardiac surgery.

Clinical presentation

Patients with isolated ccTGA may be asymptomatic for a long period. Symptoms depend on age and severity of associated lesions:

- Infants: bradycardia (+/- heart failure) due to AV block, supraventricular tachycardia, cyanosis (left ventricular outflow tract obstruction and VSD), congestive heart failure (large VSD; malformed tricuspid valve with significant regurgitation – isolated or combined)
- Childhood and adolescence: heart failure due to right ventricular dysfunction due to volume overload resulting from tricuspid regurgitation.
- In patients with left ventricular outflow tract obstruction: improved prognosis with regard to right ventricular dysfunction due to altered position of the interventricular septum resulting in stabilization of the tricuspid valve geometry.

Diagnostic work-up

Echocardiography

Echocardiography is the keystone for diagnosis: the following features should be delineated:

- Definition of situs and cardiac position
- Morphology and function of the ventricles and the AV valves (atrioventricular discordance); Ebstein malformation

of the tricuspid valve and quantification of AV valve regurgitation

- Position of the great arteries (usually aorta left and anterior to pulmonary artery, parallel) and their discordant connections to the ventricles
- Location, size and extension of ventricular septal defect
- Assessment of severity and level of left ventricular outflow tract obstruction
- Branching pattern of aortic arch (left vs. right) and ductus arteriosus, coarctation of the aorta
- Imaging of coronary arteries
- Atrial septal defect.

Chest X-ray

Helpful for assessment of cardiac position and cardiomegaly

Cardiac catheterization and angiography

- Not required for primary diagnosis in the majority of patients
- In complex forms of ccTGA: assessment of detailed anatomy and hemodynamics
- Preoperative measurement of pulmonary vascular pressures and resistance, delineation of coronary artery anatomy.

Cardiac MRI

- Demonstration of atrial and abdominal situs in selected patients
- Gold standard for assessment of ventricular mass, volume and function, particularly for the systemic RV and the LV after pulmonary artery banding for training before anatomical repair
- Display of rerouting intraatrial baffles after double switch procedure

Electrocardiography

12 leads ECG: Normal p wave axis, Q waves in right precordial leads, absent in left precordial leads, AV block?

Holter monitor: Mandatory during follow-up (pre- and postoperative)

Pulse oximetry

For assessment of cyanosis, normal in isolated ccTGA

Exercise testing

- For assessment of cardiopulmonary performance
- May help in timely detection of deteriorating right ventricular function
- Evidence for ventilation/perfusion mismatch
- Repeat testing in 3-year interval strongly recommended during follow-up

Management

Medical therapy of chronic congestive heart failure

See DGPK guideline for the treatment of chronic heart failure (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/LL_Chronische_Herzinsuffizienz_Nov_2015_Rickers.pdf)

Surgical/interventional therapy

Due to the wide spectrum of anatomical and hemodynamic abnormalities in patients with ccTGA, no uniform recommendations can be given. At the present time, there is no evidence that patients with isolated ccTGA will profit from prophylactic anatomical repair. In addition, in patients with ccTGA and left ventricular outflow tract obstruction presenting with balanced hemodynamics, surgical therapy may be postponed.

Surgical procedures/catheter interventions may be categorized as follows:

- Palliative procedures: aorto-pulmonary shunt, stenting of the arterial ductus, PA banding, creation/enlargement of atrial septal defect
- Anatomical correction with utilization of the morphological left ventricle as systemic ventricle and the mitral valve as systemic atrioventricular valve (double atrial and arterial switch procedure); left ventricle training by pulmonary artery banding may be achieved in patients <15 years for anatomical repair; patients with preoperative tricuspid regurgitation and right ventricular dysfunction may benefit most from anatomical repair
- Conventional surgery with preservation of the right ventricle as systemic ventricle (i.e. atrial and ventricular septal defect closure, resection of left ventricular outflow tract obstruction or implantation of left ventricle-pulmonary artery conduit, reconstruction or replacement of the tricuspid valve)
- Univentricular palliation
- Pacemaker implantation including CRTD therapy

Recommendations for follow-up

Children with cc TGA should be seen by a pediatric cardiologist on a regular basis.

- Unoperated patients with isolated ccTGA: focus on right ventricular function, tricuspid insufficiency, supraventricular and ventricular tachyarrhythmias, atrio-ventricular conduction
- Unoperated patients with ccTGA and associated lesions as well as patients after conventional surgical procedures preserving the right ventricle as systemic ventricle: see above as well as focus on associated defects and residual lesions (left ventricular outflow tract obstruction, ventricular septal defect)
- Patients with ccTGA after anatomical correction: see recommendations for follow-up of patients with d-TGA after atrial and arterial switch procedures (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/25%20LL%20D-Transposition%20der%20grossen%20Arterien.pdf)

Conflict of interests for this guideline

Thomas Paul: None
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Ebstein's anomaly of the tricuspid valve

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Abstract Ebstein's anomaly of the tricuspid valve is defined by apical displacement of the coaptation level into the right ventricle and its rotation towards the outflow tract. The morphological mechanisms are absent or incomplete separation of parts of the tricuspid valve from its myocardial origin leading to displacement of the functional tricuspid valve annulus into the right ventricle. By that part of the right ventricle augments the right atrium (atrialised portion of the right ventricle) whereas the apical part works as a true right ventricular pumping chamber (functional right ventricle). The malformed valve is predominantly regurgiting and rarely stenotic or a combination of both. The pulmonary forward flow is reduced due to incompetence of the tricuspid valve and right ventricular dysfunction.

Surgical repair is indicated at any age when symptoms appear or if exercise capacity declines. Rhythm disturbances are frequent with a high incidence of pre-excitation and Wolff-Parkinson-White-Syndrome in up to 44%.

Keywords: *Ebstein's anomaly, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Ebstein's anomaly of the tricuspid valve is defined by:

- Failure of tricuspid leaflet delamination of tricuspid leaflets from the myocardium
- Most often the septal and the posterior leaflets are involved, whereas the anterior leaflet is often mobile
- Morphological consequence is an apical displacement of the functional tricuspid valve annulus and a rotation towards the outflow tract
- Part of the right ventricle augments the size of the right atrium (atrialised right ventricle), the other part forms the effective right ventricle (functional right ventricle)
- Right ventricular dilatation is common
- Patent foramen ovale or atrial septal defect
- In severely symptomatic neonates, functional pulmonary atresia with duct-dependent pulmonary circulation may be present

Epidemiology

Prevalence: 4/1000 of all patients with congenital heart defect (0,4%) diagnosed within the first year of live.

Relationships to consider

Associated lesions

- Accessory atrio-ventricular pathways
- Rarely ventricular septal defect

Differential diagnoses

- Dysplasia of the tricuspid valve
- Tricuspid regurgitation due to prolapse and restriction of the leaflets
- Congenitally corrected transposition with Ebstein like malformation of the tricuspid valve
- Pulmonary atresia with intact ventricular septum with Ebstein like malformation of the tricuspid valve
- Cor triatriatum dextrum
- Neonatal cyanotic congenital heart defects

Haemodynamics

- Volume overload and dilatation of the right ventricle resulting in reduced function
- Diminished pulmonary forward flow, reduced preload of the left ventricle
- Reduction of cardiac output
- Dependent on the degree of tricuspid regurgitation rise of right atrial pressure

- Right-to-left shunt at rest or under stress resulting in arterial desaturation
- High pulmonary vascular resistance aggravating cyanosis in severely symptomatic neonates

Clinical presentation

Leading symptoms depend on age at first presentation and morphology

- Neonates: severe cyanosis with duct dependant pulmonary circulation
- Infants: right heart failure with tachycardia, tachypnoe and enlarged liver
- Preschool and school age children: systolic heart murmur
- Adolescents and young adults: tachyarrhythmia
- Adults may be oligo-symptomatic for long time and present with reduced exercise tolerance
- Rarely adults present with cyanosis as a marker of end stage right heart failure
- **Life threatening complications:** severe cyanosis at any age, thromboembolic events (cerebral and pulmonary), tachycardia

Diagnostic work-up

Echocardiography

Echocardiography is the key diagnostic imaging technique requiring the delineation of the following features:

- Detailed analysis of insertion, mobility, tissue quality and attachments of all three tricuspid leaflets
- Apical displacement of the functional tricuspid annulus in the four chamber view (more than 8 mm/m² in adults)
- Morphological classification of severity (see figure 1)
- Quantification of the tricuspid regurgitation by colour Doppler echocardiography (parasternal/subcostal short axis and four chamber view)
- Proof or exclusion of right to left shunt on atrial level by colour Doppler or contrast echocardiography

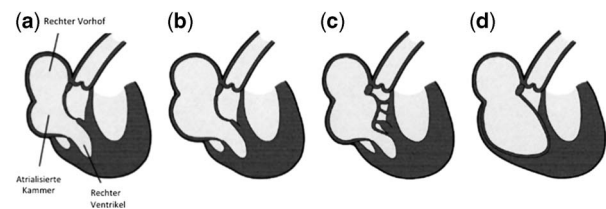


Figure 1.

Morphological grading (Carpentier classification):

Classification of the severity of Ebstein's anomaly from a surgical point of view into type A (slight form) to type D (severe form):

Type A has only moderate displacement of the anterior and septal leaflet with a versatile part. The right ventricle is small.

Type B has a clear shift of the valve closure which means that the septal and posterior leaflets are shifted and the atrialized chamber is large with thin, non- contractile wall.

In type C the anterior leaflet is also restrictive; the septal leaflet may be severely hypoplastic. The effective functional RV is rather small with limited function.

For type D the anterior leaflet is strongly adherent to the front wall, with the atrialized RV portion acting like a "tricuspid sac".

- Assessment of size and function of the right ventricle
- Assessment of size and function of the left ventricle
- Estimation of right ventricular pressure
- Assessment of flow across the right ventricular outflow tract and pulmonary valve

Pulse oximetry

- Measured at each follow-up examination at rest
- Continuous measurement essential in neonatal cases to titrate prostaglandin therapy
- In children, adolescents and adults measurements during exercise test to detect right to left shunt at atrial level

Chest-X ray

Cardiomegaly reflecting right heart enlargement and / or signs of reduced pulmonary perfusion. Indicated prior to operation.

ECG and 24-hours ECG

In the majority of patients abnormalities are present in the ECG:

- Enlarged p wave as a marker of right atrial dilatation,
- Prolonged PQ interval indicating conduction abnormalities in the proximal parts of the conduction system.
- QRS deformation resembling complete or incomplete right bundle branch block
- delta wave and shortened PR-interval indicate ventricular pre-excitation
- 24 hours ECG to document supraventricular tachycardia

Diagnostic cardiac catheterization

Usually not indicated for routine preoperative diagnostic evaluation

Magnetic Resonance Imaging

Important tool for long term follow-up

- Gold standard for assessing right and left ventricular volumes and function
- Direct measurement of pulmonary and systemic flow
- Quantification of right-to-left-shunt

Computed Tomography

Indicated in selected cases, if MRI is not feasible

Exercise testing

- Peak oxygen uptake and maximum working capacity are predictive for long term outcome
- Decline in exercise capacity may precede clinical symptoms and be an important tool for timing of surgery
- Should be done routinely every 1-2 years
- Pre- and postoperative evaluation are recommended to document the individual benefit of surgery

Indications for treatment

Symptoms such as cyanosis, heart failure, arrhythmias are clear indications for treatment. In asymptomatic patients with decline in working capacity and peak oxygen uptake treatment should be considered.

The timing of a surgical intervention in asymptomatic patients is still unclear. A multimodal diagnostic approach with detailed echocardiography, treadmill testing, Magnet Resonance Imaging seems to facilitate the individual decision.

Management

Pharmacologic treatment

- Intravenous administration of prostaglandin in cyanotic neonates
- Preload reduction of the right ventricle in patients with heart failure using diuretics
- Antiarrhythmic drugs according to underlying arrhythmias

Interventional treatment

- Stenting of the arterial duct in cyanotic neonates in order to avoid long term prostaglandin therapy
- Closure of interatrial communication only if tricuspid regurgitation is mild or less and there is predominant left-to-right shunt even during exercise
- Catheter ablation of accessory pathways in symptomatic patients with tachycardia

Surgical treatment

- Palliation
 - Aorto-pulmonary shunt to improve pulmonary blood flow and decrease cyanosis
 - Cavo-pulmonary anastomosis to unload the right ventricle. This may be done as part of a one and a half ventricle "repair" together with reconstruction of the tricuspid valve or as staged definitive palliation with total cavo-pulmonary connection as last step.
 - Corrective: reconstruction of the tricuspid valve and closure of interatrial communication – if not possible replacement of the tricuspid valve and closure of the interatrial communication.

Surgery

Reconstructive surgery should be preferred whenever technically feasible. The aim of the intervention is to re-establish a competent tricuspid valve at its original atrio-ventricular connection.

There are two main principles:

- Autologous reconstruction with plication of the right ventricle and its dilated tricuspid valve annulus (Cone-Operation). Cone-Operation: Extensive surgery with detachment and mobilisation of all valvular tissue. Creation of a 'new' tricuspid valve at the level of the true tricuspid valve annulus and plication of parts of the right ventricle.
- Single stitch technique: Less extensive surgery creating a monocusp tricuspid valve by mobilisation of the papillary muscle of the anterior leaflet and re-fixation of the tension apparatus without plication.

The surgical technique is often influenced by the quality and amount of valve tissue.

If tricuspid valve reconstruction is not possible for technical reasons, valve replacement is indicated. Bioprosthesis show a longer durability than mechanical prosthesis and should be implanted predominantly. There is initial experience implanting trans catheter valves in dysfunctional bioprostheses.

Outcome

Mortality rates between the two different operations are comparable. Early mortality rates of 1.2% are published in a big series of more than 100 patients with monocusp technique. Da Silva reported on 3% early mortality for the cone technique.

Prognosis

Without surgery, patients with moderate or severe tricuspid valve regurgitation frequently experience a decline in exercise capacity and arrhythmias in later life. There is a high risk for severe congestive heart failure which may preclude surgery.

Patients after an initially successful operation may face the following complications:

- Reoccurrence of tricuspid valve regurgitation with possible need for reoperation
- Right ventricular dilatation and dysfunction
- Supraventricular and ventricular tachyarrhythmias
- Need for pacemaker implantation

Recommendations for follow-up

Regular follow-up examinations are necessary usually once a year.

The positive effect of any type of operation is characterized by an increase of maximum working capacity and oxygen uptake of the patient, proven by cardiopulmonary exercise testing earliest 10 months after surgery. These benefits seem to last even in mid- and long-term follow up.

The following examinations should be performed:

- Echocardiography
- ECG
- Holter-monitoring – every year in patients with rhythm disorders, otherwise every two years
- Cardiac Magnet Resonance Imaging 1 year after the operation, then on clinical indication
- Cardio-pulmonary exercise testing – at least every other year

A prophylaxis of endocarditis is indicated in:

- Patients with reconstructive surgery 6 months after the operation
- Patients with tricuspid valve replacement
- Patients with a previous history of endocarditis
- Patients with cyanosis

Conflicts of interests for this guideline:

Manfred O. Vogt: None
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Thomas Paul: None

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Tetralogy of Fallot

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Abstract Tetralogy of Fallot is defined by the anterior and superior deviation of the outlet septum resulting in a subvalvar pulmonary stenosis, malalignment ventricular septal defect, overriding aorta and right ventricular hypertrophy. Surgical repair is achieved by closure of the ventricular septal defect and relief of the right ventricular outflow tract obstruction in the majority of the patients at the age between 4 and 12 months. If a primary repair is not feasible, palliative surgical or interventional procedures are indicated to increase pulmonary blood supply. The long-term follow-up after correction can be complicated by pulmonary valve regurgitation, residual right ventricular outflow tract obstruction, right ventricular dysfunction and arrhythmia.

Keywords: *Tetralogy of Fallot, Guideline, Congenital Heart Disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Tetralogy of Fallot is defined by:

- anterior and superior deviation of the outlet septum
- subpulmonary (infundibular) stenosis

- non-restrictive malalignment ventricular septal defect
- overriding aorta and
- right ventricular hypertrophy.

Common additional lesions are:

- Atrial septal defect
- stenosis and hypoplasia of the pulmonary valve as well as central and peripheral pulmonary arteries
- patent arterial duct and/or aortopulmonary collaterals in severe right ventricular outflow tract obstruction (functional pulmonary atresia)
- right aortic arch (25%)
- anomalous left anterior descending coronary artery arising from the right coronary artery and crossing the right ventricular outflow tract (5%).

Epidemiology

2.7/10.000 live births (corresponding to 2.5% of all congenital heart defects).

Relationships to consider

Associated lesions

- tetralogy of Fallot with absent pulmonary valve
- tetralogy of Fallot with complete atrioventricular septal defect
- disconnected left pulmonary artery

Syndromes

- microdeletion of the chromosome 22q11 in 9–17% of patients with uncomplicated Tetralogy of Fallot and in 60–70% of patients with Tetralogy of Fallot and right aortic arch.

Differential diagnosis

- pulmonary atresia with ventricular septal defect
- double outlet right ventricle
- valvar pulmonary stenosis with ventricular septal defect.

Haemodynamics

- right-to-left shunt at ventricular level due to right ventricular outflow tract obstruction resulting in systemic cyanosis
- cyanosis depends on the degree of right ventricular outflow tract obstruction
- some patients with only mild right ventricular outflow tract obstruction do not show cyanosis (“pink Fallot”).

Clinical presentation

- systolic murmur caused by the right ventricular outflow tract obstruction
- systemic cyanosis.
- **life-threatening complication:** hypoxic spell characterized by increased cyanosis, tachycardia, lower intensity or absence of the systolic heart murmur and sometimes unconsciousness and seizures. In infancy spells may be triggered by stress, hypovolaemia and fever.
- drumstick fingers, squatting and cerebral abscess in the long-term follow-up of patients with untreated tetralogy of Fallot

Diagnostic work-up

Echocardiography

Echocardiography is the key diagnostic imaging technique requiring the delineation of the following features:

- Right ventricular outflow tract obstruction (subvalvar, valvular and supravalvar pulmonary stenosis)
- Pulmonary artery bifurcation and branch pulmonary arteries
- Malalignment ventricular septal defect (additional muscular ventricular septal defect?)
- Overriding aorta, aortic arch (right aortic arch? aberrant right subclavian artery?)
- Coronary arteries (anomalous left anterior descending coronary artery crossing the right ventricular outflow tract?)
- Additional anomalies (e.g. atrial septal defect? complete atrioventricular septal defect? left persistent superior vena cava?)

Transcutaneous O₂-saturation:

- measured at each follow-up examination.
- Persistent saturations of <80% require therapeutic intervention

ECG

No diagnostic value, typical findings are right ventricular hypertrophy, incomplete right bundle branch block.

Chest-X ray

No diagnostic value; necessary only once before operation

Diagnostic cardiac catheterization

- Usually not indicated for routine preoperative diagnostic evaluation.
- may be indicated if there is suspicion of multifocal pulmonary artery blood supply, anomalous origin of pulmonary artery or an anomalous left anterior descending coronary artery crossing the right ventricular outflow tract.
- often necessary after previous interventional or surgical palliation

Magnet Resonance Imaging

Usually not indicated for the routine preoperative diagnostic evaluation.

Computer Tomography

Same as for Magnet Resonance Imaging, cave: radiation

Genetic examination

Fluorescence-in-situ hybridization for microdeletion 22q11.2 recommended in all patients.

Indications for treatment

Aim is the anatomical repair with relief of the right ventricular outflow tract obstruction and closure of the ventricular septal defect. Elective surgical repair is recommended in infants aged 4–12 months if clinically stable. In deeply cyanotic patients or patients with cyanotic spells, repair can be performed earlier with a slightly increased morbidity and mortality (mortality 7.8% in neonates versus 0.9% in infants). If a repair cannot be

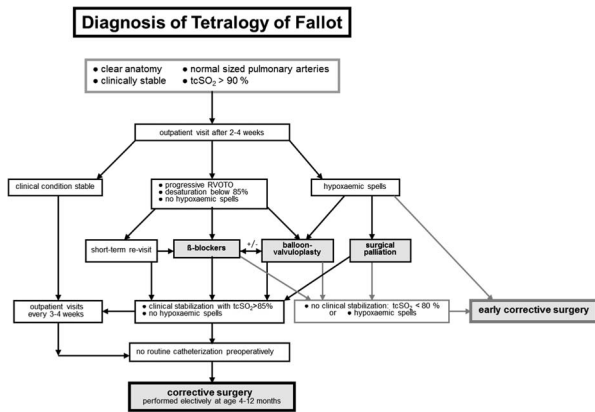


Figure 1.

Management algorithm in neonates with Tetralogy of Fallot being clinically stable during the first weeks of life. Progressive right ventricular outflow tract obstruction (RVOTO) leading to severe cyanosis or even hypoxic spells may require medical, interventional or surgical palliation or early corrective surgery (yellow boxes). Otherwise elective surgical repair is performed beyond 4 month of age.

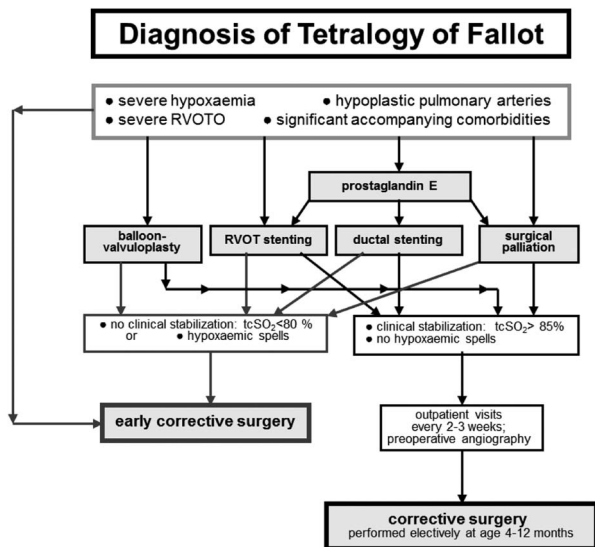


Figure 2.

Management algorithm in highly symptomatic neonates with unfavorable anatomy (red frame) who require an urgent palliation with drugs and interventional or surgical procedures (yellow boxes). If palliative treatment fails to stabilize the patients or as primary treatment with higher perioperative risk, early surgical repair should be considered (blue arrows). In case of clinical stabilization after palliation, corrective surgery is performed electively.

performed a palliation to increase pulmonary blood flow is indicated (Fig 1 and 2).

Modes of palliative procedures:

- medical treatment
- interventional catheterization
- surgical procedures.

None of these procedures has a proven superiority.

Management

Drug treatment for palliation

- Infusion of Prostaglandin in newborns with a duct-dependent pulmonary artery perfusion up to an interventional or a surgical procedure.
- Oral administration of a beta blocker, e.g. Propranolol (2–6 mg/kg/day) for the reduction of right ventricular outflow tract obstruction, if the transcutaneous oxygen-saturation remains <85%. If oxygen-saturations remain <80% despite medical treatment with the highest tolerated dosage of propranolol, an invasive procedure is indicated.

Cave:

The **hypoxic spell** is an emergency situation requiring immediate treatment to increase systemic vascular resistance and pulmonary blood by

- Chest- knee position of the patient
- Administration of O₂
- Sedation with Morphine (s.c or i.v. 0.1–0.2 mg/kg), Midazolam (i.v. 0.1 mg/kg, rectally or intranasally 0.5 mg/kg) or Ketamine (i.v. 1–2 mg/kg, i.m. 5 mg/kg)
- Administration of volume (repeated bolus of 10 ml/kg)
- If necessary: intubation, i.v. noradrenaline and beta-blocker on intensive care unit.

Interventional cardiac catheterization

- To improve pulmonary artery perfusion the following interventions can be performed as palliation in selected patients (Fig 1):
 - Balloon valvuloplasty in symptomatic neonates and young infants with severe and predominantly valvar pulmonary artery stenosis
 - Stenting of patent arterial duct in neonates with duct depend pulmonary artery perfusion
 - Stent implantation into the right ventricular outflow tract
- Occlusion of haemodynamically significant major aorto-pulmonary collaterals with coils or vascular plugs (dual blood supply mandatory).

Surgery

Surgical palliation to increase the pulmonary artery blood flow is nowadays rare since primary correction is the preferred procedure in the majority of patients.

Palliation (Fig 2)

Options:

- Aortopulmonary shunt, e.g. modified Blalock–Taussig shunt
- Patch enlargement of the right ventricular outflow tract.

Repair

Surgical repair includes:

- Closure of ventricular septal defect

Patch closure of the ventricular septal defect via transatrial approach or by ventriculotomy.

- Right ventricular outflow tract enlargement Mode of enlargement depends on the diameter and the morphology of the pulmonary artery valve.

- If the diameter of the pulmonary valve annulus $> -2SD$, a pulmonary valvotomy with preservation of the pulmonary valve annulus is usually performed.
- If the annulus $< -2SD$, usually a transannular patch is inserted.
- To minimize the extent of pulmonary valve regurgitation an enlargement of the annulus as small as possible is recommended even if a mild to moderate residual right ventricular outflow tract obstruction persists.
- Relief of subvalvar pulmonary stenosis by a transatrial myectomy or by ventriculotomy with patch enlargement.

The result of the repair should be assessed intraoperatively by transoesophageal echocardiography for a residual ventricular septal defect or right ventricular outflow tract obstruction.

Prognosis

Possible complications in the long-term follow-up:

- pulmonary valve regurgitation,
- residual stenosis of the right ventricular outflow tract
- right ventricular dilatation and dysfunction
- supraventricular- and ventricular arrhythmia.

Recommendations for follow-up

Regular follow-up examinations are necessary usually once a year. The following examinations should be performed:

- Echocardiography
- ECG (Increased risk of ventricular tachycardia if QRS-duration >180 msec or with an increase of QRS-duration of >3.5 msec/year)
- Holter-monitoring (every 3 years in asymptomatic children >10 years of age due to the risk of non-sustained tachycardias)
- Cardiac magnetic resonance imaging (to quantify right ventricular dilatation/dysfunction and/or severity of pulmonary valve regurgitation)
- Cardio-pulmonary exercise testing (in patients aged >10 years every 5 years. Many patients overestimate their exercise capacity).

A prophylaxis of endocarditis is indicated in:

- non- corrected patients,
- patients after surgical repair for 6 months
- patients with percutaneous or surgical pulmonary valve replacement (high-risk group).

Conflicts of interest for this guideline:

Jochen Weil: None

Harald Bertram: None

Jörg S. Sachweh: None

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Double Outlet Right Ventricle

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Abstract The term 'double outlet right ventricle' refers to any cardiac anomaly in which both the aorta and the pulmonary trunk originate, predominantly or entirely, from the morphologic right ventricle. In this situation, the left ventricle mainly ejects through an interventricular communication, usually referred to as a ventricular septal defect, which is almost invariably present. A broad spectrum of malformations occurs; the clinical picture varies widely, depending on the associated abnormalities. The treatment of choice is surgical repair with separation of systemic and pulmonary circulation, which can be achieved in the majority of patients. In rare cases, unfavourable anatomy still requires univentricular palliation according to the 'Fontan principle'.

Keywords: double outlet right ventricle (DORV), guideline, congenital heart disease

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Double outlet right ventricle is defined by the following morphological criteria:

- Both great arteries arise completely (100%+100%) or predominantly (one artery with 100%, the other with more than 50% of its circumference) from the right ventricle [$>150\%$ rule', fig. 1].
- Infundibular morphology, the spatial relationship of the semilunar cusps, and coronary artery anatomy are variable.
- The left ventricle is connected to the right ventricle through an interventricular communication, usually referred to as a ventricular septal defect (fig. 2). Its location in relation to the great arteries is important for the clinical condition, and of major significance concerning the surgical repair strategies.

The following classification of the VSD location is commonly used:

- subaortic (65%)
- subpulmonary (20–25 %)
- doubly committed (3%)
- non-committed (7%)
- Any arrangement of the atrial appendages, or situs, and any atrioventricular connection may be present
- Multiple combinations of associated malformations are possible, resulting in a heterogeneous morphological appearance of hearts with double outlet right ventricle.

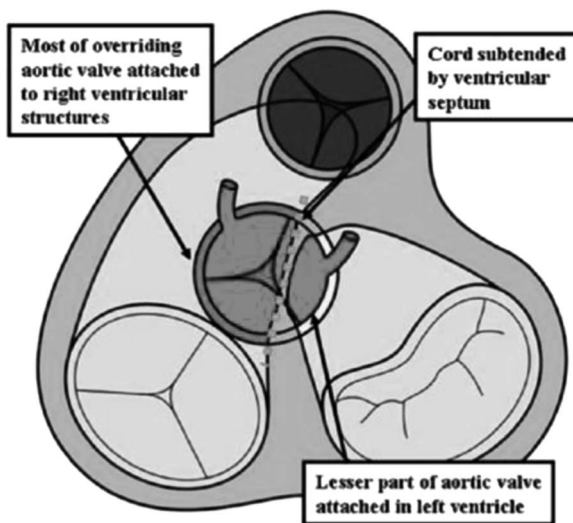


Figure 1.

An overriding valve, in this instance the aortic valve, is assigned to one or other ventricle according to the location of the cord subtended by the plane of the ventricular septum relative to the overriding valvar orifice (blue dotted line). In this instance, the greater part is supported by the right ventricle (green surround), with the lesser part (yellow surround) remaining in the left ventricle. The effective ventriculo-arterial connection, therefore, is double outlet right ventricle.

[modified from Mable et al. *Cardiol Young* 2008; 18(Suppl. 3):39–51].

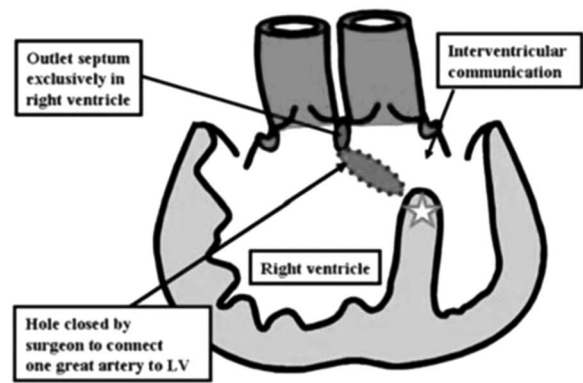


Figure 2.

Both arterial trunks arise predominantly from the right ventricle. The star shows the crest of the muscular ventricular septum.

[modified from Mable et al. *Cardiol Young* 2008; 18(Suppl. 3):39–51].

Epidemiology

- Prevalence: 1.1 per 10,000 live births, corresponding to 1.3% of all congenital heart defects
- However, the overall frequency is not well documented as the condition overlaps with a range of defects, and is often not categorized solely as double outlet right ventricle.

Differential diagnosis

- Tetralogy of Fallot
- Transposition of the great arteries
- Isolated malalignment ventricular septal defect
- Complex cyanotic heart disease

Haemodynamics

The heterogeneous morphologic appearance of hearts with double outlet right ventricle results in a broad spectrum of different haemodynamic situations. Pulmonary recirculation may lead to congestive heart failure on the one hand, and transposition-like arterial arrangements or severe pulmonary stenosis / atresia may result in severe cyanosis on the other. Associated aortic coarctation usually leads to early onset of heart failure.

Clinical presentation

Despite of its morphologic heterogeneity, the clinical picture in most patients with double outlet right ventricle can be categorized into the following common variants (fig. 3):

- **VSD type:** Subaortic interventricular communication without pulmonary/subpulmonary stenosis. The clinical picture resembles that of a large, isolated ventricular septal defect with pulmonary hyperperfusion. Cyanosis is usually mild or absent, and symptoms of congestive heart failure usually develop gradually over the first few months of life.
- **Fallot type:** Subaortic interventricular communication and pulmonary / subpulmonary stenosis. The clinical presentation is indistinguishable from classic Tetralogy of Fallot with a systolic murmur in the newborn period, and cyanosis which develops and progresses during the early months of life. Hypoxic spells may occur.

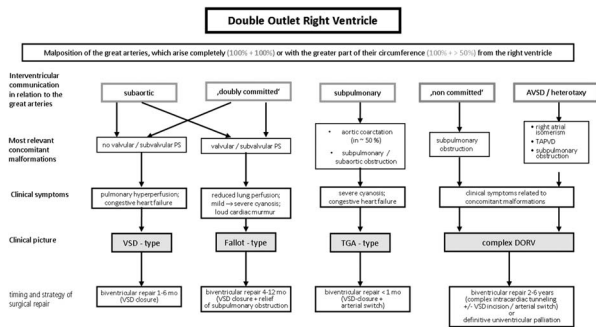


Figure 3.

Most frequent anatomic variants, clinical symptoms and surgical strategy in patients with double outlet right ventricle (see text) [VSD = ventricular septal defect; PS = pulmonary stenosis; AVSD = atrioventricular septal defect; TAPVD = total anomalous pulmonary venous return; TGA = transposition of the great arteries; RVOTO = right ventricular outflow tract obstruction].

- **TGA type:** Subpulmonary interventricular communication (Taussig-Bing malformation). The clinical picture is essentially the same as that of transposition of the great arteries with ventricular septal defect. The newborn presents often with cyanosis and early development of symptoms of congestive heart failure.
- **Complex DORV:** Heterogeneous morphology without clear assignment of the interventricular communication to one of the great arteries. The clinical picture is determined by concomitant malformations.

Diagnostic work-up

Echocardiography

Primary diagnostic tool to describe intracardiac anatomy for differentiation of morphologic subtypes with respect to potential surgical biventricular repair. It should address:

- Position, size, interrelationship, and course of the great arteries including the aortic arch
- Morphology and size of the interventricular communication / the ventricular septal defect in relation to the diameter of the aortic valve
- Location and severity of subpulmonary or subaortic obstruction
- Morphology and size of both ventricles and atrioventricular valves
- Atrial arrangement, systemic and pulmonary venous return (especially in heterotaxy syndromes).

ECG and chest X-ray

The findings are not specific and depend on the underlying anatomic variants. Chest X-ray is recommended prior to therapeutic interventions.

Pulse oximetry

Important tool to document the degree of cyanosis and thereby the potential need for therapeutic interventions.

Management

Pharmacologic treatment

- Prostaglandin E in duct dependent patients with severe subpulmonary obstruction / pulmonary atresia
- Diuretics, β -blockers, mineralocorticoid antagonists in pulmonary hyperperfusion with congestive heart failure.

Catheter interventions

- Fallot type: balloon valvuloplasty; ductal stent; RVOT stent
- TGA type: balloon atrioseptostomy.

Surgery

• Surgical palliation in infancy

- **Fallot type:** modified BT-Shunt if primary repair is not suitable or considered 'high risk'
- **non-committed VSD:** pulmonary artery band to delay complex intraventricular repair

• Surgical repair:

- **VSD type:** VSD closure at the age of one to six months; some patients may need enlargement of the interventricular communication (target size: VSD diameter > 4/5 aortic annulus; risk of AV block)
- **Fallot type:** VSD closure and relief of subpulmonary obstruction are usually performed at the age of 4 to 12 months, combined with muscular incision or patch enlargement of the right ventricular outflow tract if indicated.
- **TGA type:**
 - neonatal corrective surgery with arterial switch and VSD closure; if indicated, combined with resection of aortic coarctation / aortic arch reconstruction or resection of subaortic infundibulum, respectively;
 - in patients with subpulmonary ventricular septal defect with valvular or subvalvular pulmonary stenosis, several complex intracardiac surgical approaches have been performed:
 - Kawashima operation
 - Rastelli type repair
 - REV procedure (reparation a l'étage ventriculaire)
 - aortic translocation.
- **Complex DORV**
 - biventricular repair at the age of 2 to 6 years (after neonatal palliation): complex intraventricular baffling of the LV to one of the great arteries, combined with enlargement of the VSD and / or arterial switch
 - univentricular palliation in patients not suitable for biventricular repair.

Outcome

- Biventricular repair is achievable in most patients with low operative mortality (<5%).
- Depending on the individual morphology, a staged repair might be indicated in selected patients.
- In complex morphologic variants, the risk of perioperative mortality for complete repair is higher (~20%).
- Specific risk factors for increased perioperative morbidity and mortality are:
 - aortic arch obstruction
 - multiple VSDs
 - age less than one month at the time of repair
 - discordant atrioventricular connections
 - common atrioventricular valve
 - hypoplasia of one ventricle
 - straddling atrioventricular valves
- In complex anatomy, univentricular palliation according to the 'Fontan principle' is associated with lower operative risk.

Prognosis

The following residual findings may require reinterventions:

- Fallot type: pulmonary valve incompetence; residual subpulmonary obstruction;
- RV-PA-conduits: definite reoperation for conduit replacements as a result of either conduit stenosis due to degeneration / patient outgrowth or incompetence of the conduit valve; alternatively, percutaneous pulmonary valve implantation may be performed in adolescents
- complex intracardiac baffling: subaortic obstruction requiring reoperation.

In complex DORV patients biventricular surgical repair has a much higher rate of surgical or interventional reinterventions than univentricular palliation.

Recommendations for follow-up

- Lifelong follow-up by a pediatric cardiologist is mandatory;
- Frequency of outpatient visits is determined by residual findings after surgical repair; they are recommended at least annually;
- Cardiopulmonary exercise testing and Holter monitoring from school-age onwards every 2–3 years.

Conflict of interests for this guideline

Harald Bertram, MD: None

Jochen Weil, MD: None

Jörg S. Sachweh, MD: None

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Link to all references listed in the original German long version “Double Outlet Right Ventricle (DORV)”: www.kinderkardiologie.org/leitlinien/

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Pulmonary Atresia with Ventricular Septal Defect

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Abstract: Pulmonary atresia with ventricular septal defect is the most severe form of Tetralogy of Fallot, with absent antegrade flow into the pulmonary arteries. Pulmonary blood flow is either originating from a patent arterial duct usually in combination with well developed pulmonary arteries, or from major aorto-pulmonary collaterals with extremely hypoplastic central pulmonary arteries.

The therapeutic goal is a biventricular circulation with complete separation of systemic and pulmonary circulation and a biological valve in pulmonic position. Initially, surgical or inter-ventional palliation is necessary for oxygenation. In patients with multifocal blood supply several complex surgical procedures (unifocalization of collaterals) may be necessary to achieve a biventricular circulation. After surgical repair using valved conduits, these conduits need to be regularly replaced surgically, or by catheter interventional means. In some patients with unfavorable anatomy, only a palliation to relief cyanosis without closure of the ventricular septal defect is achievable.

Keywords: *pulmonary atresia, ventricular septal defect (VSD), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- Anterior and superior deviation of the right ventricular outlet septum
- Malalignment ventricular septal defect with overriding aorta
- Atresia of the pulmonary valve, possibly combined with atresia of the outflow tract or the pulmonary trunk
- Unifocal (patent arterial duct, single collateral) or multifocal (aorto-pulmonary collaterals) pulmonary blood supply
- Concomitant malformations:
 - Different degree of hypoplasia up to absence of the central pulmonary arteries
 - Right aortic arch (25%)
 - Coronary arterial abnormalities (left anterior descending coronary artery form right coronary artery) in up to 4%
 - Atrial septal defect

Epidemiology

Prevalence: 7 per 100,000 live births (1% of all congenital heart defects)

Relationships to consider

Associated lesions

- Microdeletion 22q11 syndrome in 23–40%, especially if right aortic arch is present
- Alagille syndrome (JAG1-gene) with severe and multiple segmental pulmonary arterial stenoses

Differential diagnosis

- Tetralogy of Fallot
- Common arterial trunk

Haemodynamics

- Central cyanosis due to right-to-left shunt at ventricular level
- Duct-dependent pulmonary blood flow or pulmonary blood flow via collaterals Systemic RV-pressure
- In case of multifocal pulmonary blood supply, there may be considerable variations of the pulmonary arteries with regional hyper- or hypoperfusion.

Clinical presentation

- Neonates/infants:
 - cyanosis due to right-to-left shunt at ventricular level; duct closure may cause acute severe cyanosis
 - if multiple aorto-pulmonary collaterals are present, patients may be in congestive heart failure as a result of pulmonary recirculation
- Older children:
 - the degree of cyanosis depends on the amount of pulmonary blood flow, which varies considerably
 - after successful biventricular repair, the clinical picture depends on the anatomy of the pulmonary vasculature (size, distribution, residual stenoses) and the resulting pressure load of the right ventricle.

Diagnostic work up

Echocardiography

Key imaging modality to establish the diagnosis with the emphasis on the delineation of the following anatomic features:

- Type of right ventricular outflow tract obstruction on sub-, valvar and supravalvar level
- Central and branch pulmonary arteries
- Pulmonary blood supply (unifocal?, multifocal?)
- Overriding aorta, right sided arch? Aberrant right subclavian artery?
- Depiction of coronary arteries.

ECG

Right axis deviation, right ventricular hypertrophy. Not diagnostic; should be performed at least once before intervention.

Chest X-ray

- Assessment of pulmonary blood flow.
- Up-turned cardiac apex,
- Right aortic arch?
- Pulmonary disease?

Should be performed at least once before intervention.

Pulse oximetry

Important tool for assessment of pulmonary perfusion and the need for therapeutic interventions, especially during follow-up.

Cardiac catheterisation

Necessary in all patients in whom the pulmonary artery anatomy and/or the pulmonary blood supply cannot be clearly seen by echocardiography.

Cardiac magnetic resonance imaging/ computer tomography

May give additional information on pulmonary artery anatomy in patients with complex changes of the pulmonary artery tree and pulmonary artery blood supply.

Management

Therapeutic goal is biventricular surgical repair with a biological valve in pulmonic position and complete separation of systemic and pulmonary circulation. A proposed treatment algorithm is given in figure 1.

Medical treatment

In a duct-dependent pulmonary blood perfusion administration of intravenous prostaglandin is the first line of palliation to secure pulmonary blood supply.

Infants with multiple aorto-pulmonary collaterals may suffer from pulmonary overperfusion requiring symptomatic treatment for congestive heart failure.

Catheter interventions

To secure an adequate pulmonary blood supply the following palliatives procedures can be performed:

- Stenting of the arterial duct in unifocal pulmonary blood supply

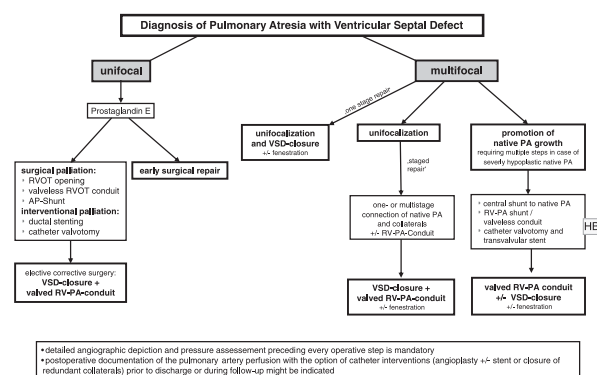


Figure 1.

Treatment algorithm for patients with pulmonary atresia with ventricular septal defect (abbreviations: PA = pulmonary artery; VSD = ventricular septal defect; RV-PA = from right ventricle to pulmonary artery; AP = aorto-pulmonary; RVOT = right ventricular outflow tract).

- Catheter valvotomy of the atretic valve with subsequent balloon dilation

In the presence of a hyperperfusion of lung segments due to a dual blood supply an occlusion of redundant aorto-pulmonary collaterals is indicated.

Surgery

- **Unifocal** perfusion of normal sized pulmonary arteries
In most cases surgical palliation is performed by an aorto-pulmonary shunt or surgical augmentation of the right ventricular outflow tract leaving the VSD open or a valveless conduit between right ventricle and central pulmonary artery. Surgically or interventionally palliated patients are re-evaluated when residual cyanosis intensifies with growth. Corrective surgery is usually performed with a valved conduit and closure ventricular septal defect at the age of 6 to 12 months. Alternatively early corrective surgery may be performed in selected patients.
- **Multifocal** pulmonary artery perfusion with diminutive central pulmonary arteries
 - The treatment algorithm is more complex (see figure 1) and there are two main concepts: In newborns, usually central pulmonary arteries are present, which later on may not be visible anymore; pulmonary arterial growth may be stimulated with a neonatal central aorto-pulmonary shunt. This is followed by MAPCA-closure, RV-PA conduits and VSD closure often in a multi-stage approach.
 - Alternatively, unifocalization procedures are scheduled during the first year of life, with the intention to restore as many lung segments as possible for further steps to achieve separate circulations by corrective surgery (usually performed as 'staged repair').

The opportunity of achieving a complete separation of pulmonary and systemic circulation by VSD-closure is difficult to assess, as the abnormal pulmonary vascular bed may develop severe forms or pulmonary arterial hypertension with increased vascular resistance and multiple peripheral pulmonary artery stenoses; indications to close the VSD vary, and often a fenestrated patch closure is performed.

Outcome

Overall survival in these patients primarily depends on the anatomy and size of the pulmonary arteries. Cumulative survival since birth of the whole patient group (untreated, palliated or corrected) is about 75% after ten years. After surgery the 5-year-survival rate from first operation is 70% in repaired patients compared to 60% in palliated patients.

Prognosis

- Patients with **unifocal** lung perfusion usually have a good prognosis and often have normal quality of life after surgical correction.
- In patients with **multifocal** collateral perfusion of the pulmonary arteries the prognosis is worse depending on the number of lung segments recruited and if these lung segments develop an increase of pulmonary vascular resistance. A patient palliated with aorto-pulmonary shunts and adequate arterial oxygen saturation may have a better quality of life than a "corrected" patient with systemic, or suprasystemic right ventricular pressure due to pathological pulmonary arteries. The treatment plan needs to be tailored individually. Many surgical and catheter

interventional steps may be necessary to help these patients to have a good quality of life. In some patients, VSD closure will never be possible

- After placement of valved conduits in the right ventricular outflow tract, further procedures will be necessary due to degeneration; these conduits need to be exchanged surgically, or by percutaneous pulmonary valve implantation
- If corrective surgery was not possible and the patients are desaturated at long term follow-up problems of hyperviscosity may arise.
- In case of very diminutive pulmonary arteries adequate treatment is still not possible

Recommendations for follow-up

- After corrective surgery annual visits with are recommended.
- Palliated patients need to be seen more frequently if their oxygen saturation is low, or they need interventional steps needed.
- Right ventricular pressure during somatic growth needs to be monitored for timing of reinterventions for relief of RVOT re-obstruction.
- Objective examinations of exercise capacity and Holter ECGs every 2-3 years.
- Psychosocial care for severely ill patients in the long-term follow-up.

Conflict of interests for this guideline

Andreas Eicken: proctor for Medtronic (Melody valve®)
Harald Bertram: None
Jörg Sachweh: None
Jochen Weil: None

Link to all references listed in the original German long version "Pulmonalatresie mit Ventrikelseptumdefekt": www.kinderkardiologie.org/leitlinien/

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Pulmonary Atresia with Intact Ventricular Septum

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Abstract Pulmonary atresia with intact ventricular septum is a disease entity with a morphologically extremely variable right heart hypoplasia. Therapeutic options depend on neonatal right ventricular size as well as subsequent growth and function.

The tricuspid valve diameter may be used as a parameter for right ventricular size and accurate echocardiographic determination of the tricuspid valve diameter is an important parameter for

categorization of the individual patient into different treatments groups.

Besides standard surgical treatment, catheter based interventions including valvotomy of the atretic valve and stenting of the arterial duct have been established, which may allow to postpone surgery in some patients to school age. However, the therapeutic goal of biventricular repair with separation of systemic and pulmonary circulation can only be reached in about half of the patients. At the other end of the anatomical spectrum, unfavourable anatomy still forces to univentricular palliation according to the 'Fontan principle'.

Keywords: *pulmonary atresia, intact ventricular septum, children, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Malformation of right-sided cardiac structures with heterogeneous morphological appearance:

- 75–80% isolated membranous atresia of the pulmonary valve; combined with infundibular atresia in 18–25%
- intact ventricular septum
- significant hypertrophy of the right ventricle
- hypoplasia of tricuspid valve, right ventricle, and pulmonary artery trunk
- usually normal sized branch pulmonary arteries
- right atrial enlargement
- considerably variable anomalies of the coronary arteries (stenoses, sinusoids, fistulas) with clear correlation to the grade of right ventricular hypoplasia

Epidemiology:

Prevalence: 30 to 45 per 10.000 live births, corresponding to 0.3% of all congenital heart defects

Relationships to consider

Associated lesions

- concomitant Ebstein's malformation (8–9%)
- myocardial sinusoids or small fistula between right ventricular cavity and coronary arteries (40–50%)

Caveats

- In rare cases (5–10%) with severe tricuspid regurgitation, right ventricle and tricuspid valve annulus are dilated, with poor systolic right ventricular function.
- right ventricle-dependent coronary circulation (7–9%)

Differential diagnoses

- tricuspid atresia
- critical pulmonary stenosis
- Ebstein's anomaly
- severe tricuspid regurgitation
- pulmonary atresia with ventricular septal defect
- "functional pulmonary atresia" (in pulmonary hypertension and / or failing right ventricle)

Haemodynamics

- duct-dependent pulmonary blood flow
- inadvertent interatrial communication with right-to-left shunt
- suprasystemic right ventricular pressure
- tricuspid regurgitation (variable)

Clinical presentation

- Neonates/infants: mild cyanosis due to right-to-left shunt at atrial level; duct closure causes acute severe cyanosis
- Older children: duct dependent cyanotic heart disease

Diagnostik work up

For complete imaging, both echocardiography and additional catheterization are usually required.

Echocardiography and cardiac catheterization

The following details must be clarified before interventional or surgical treatment is performed:

- Membranous vs. infundibular atresia
- Extent of right ventricular hypoplasia and hypertrophy
 - absence of trabecular part and apex?
 - tricuspid valve annulus diameter in apical and subcostal 4-chamber view in end-diastole (the size of the tricuspid valve correlates well with RV size); including Z-score calculation and relation to mitral valve annulus diameter
 - degree of tricuspid regurgitation; calculation of right ventricular pressure;

- Size of pulmonary valve annulus, pulmonary artery trunk and proximal branch arteries
- Size and morphology of the arterial duct (potential ductal stenting!) and atrial septal defect (restriction?)
- Extent of coronary anomalies:
 - myocardial sinusoids / small coronary fistulas
 - potential right ventricle-dependent coronary circulation, defined as

- large coronary fistulas or ectasia, possibly without aorto-coronary connection
- stenoses or discontinuations of large coronary artery branches
- coronary perfusion of relevant portions of the left ventricle being supplied by the right ventricle

ECCG

Right atrial and left ventricular hypertrophy

Pulse oximetry

Important tool for assessment of pulmonary perfusion and the need for therapeutic interventions

Management

First step of therapy is stabilization with prostaglandin. Further management depends on right ventricular size and function and coronary anatomy as well as subsequent growth of cardiac structures.

A proposed management algorithm is given in figure 1.

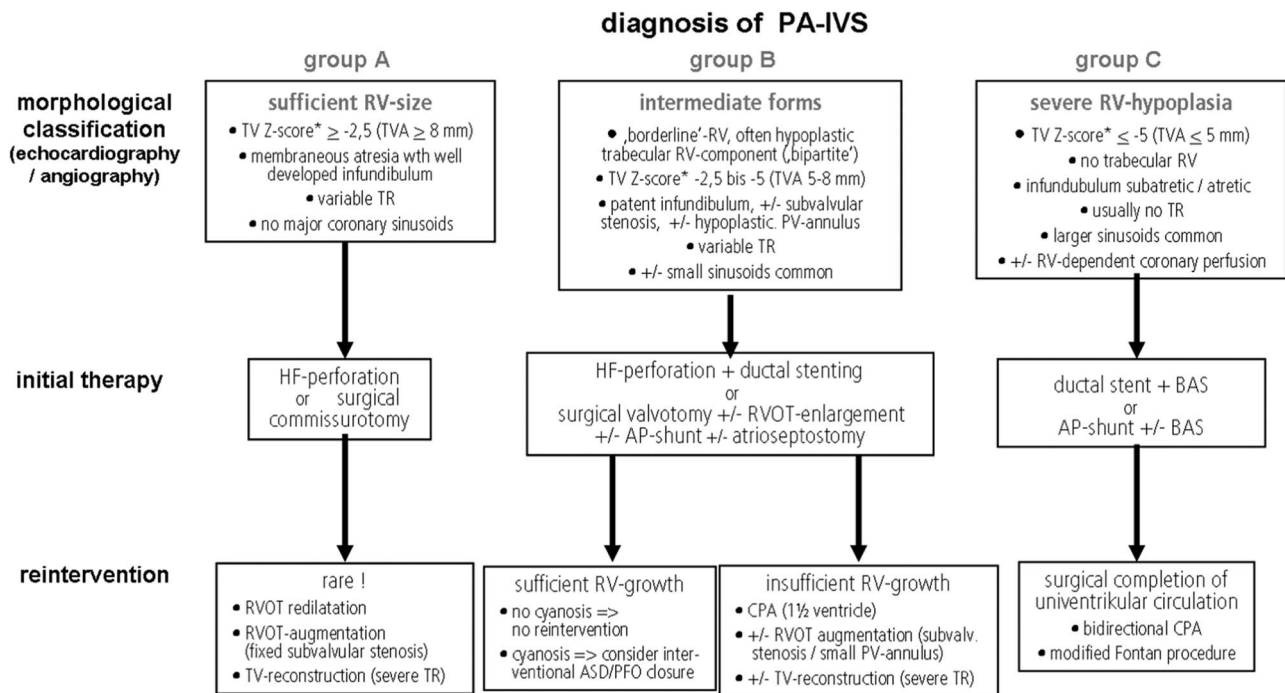


Figure 1.

Management algorithm for patients with pulmonary atresia and intact ventricular septum based on morphological criteria (RV=right ventricle; TV=tricuspid valve; TVA=tricuspid valve annulus; TR=tricuspid regurgitation; HF=high frequency; BAS=balloon atrioseptostomy; AP=systemic-pulmonary; RVOT=right ventricular outflow tract; PV=pulmonary valve; CPA=cavo-pulmonary anastomosis; ASD=atrial septal defect; PFO=persistent Foramen ovale; +/-=with or without); 8 reference values for the calculation of TV Z-scores from Pettersen et al., J Am Soc Echocard 2008; 21(8):922-34.

Neonatal decompression of the right ventricle with interventional or surgical valvotomy is a prerequisite for subsequent right ventricular growth. It allows for antegrade pulmonary blood flow, decreasing right ventricular pressure and reducing pre-existing tricuspid regurgitation. Additional pulmonary perfusion through the duct is still needed in some patients to allow for adaptation of the right ventricle, so that low-dose prostaglandin infusion might be continued.

In significant right ventricular hypoplasia with functional absence of the trabecular part, perforation of the atresia should be combined with a stent implantation in the arterial duct or surgical right ventricular outflow tract-patch and shunt insertion, respectively. Besides, balloon atrioseptostomy should be considered with respect to a longer period of right ventricular adaptation.

In patients with infundibular atresia or right ventricle-dependent coronary circulation, perforation of the atresia is not indicated. Shunt insertion combined with septectomy or balloon atrioseptostomy is the preferred treatment.

Reinterventions during follow-up focus on residual right ventricular outflow tract stenosis, surgical treatment of severe tricuspid regurgitation or closure of the atrial septal defect to achieve separation of systemic and pulmonary circulation. In the long-term, a pulmonary valve replacement will be necessary in most patients.

Insufficient growth of the right ventricle leads to surgical removal of residual outflow tract obstruction combined with a bidirectional cavo-pulmonary anastomosis ('1½-ventricle repair'); closure of the atrial septal defect might be added.

In case of severe right ventricular hypoplasia, a permanent univentricular palliation according to the 'Fontan principle' is established.

Prognosis

- Perioperative morbidity and mortality is considerable in infancy (highest in patients with low birth weight, concomitant Ebstein's anomaly, dilated thin-walled right ventricle, and right ventricle – dependent coronary circulation)
- Positive prognostic factors to achieve biventricular repair with separation of systemic and pulmonary circulation (~50% of cases) are:
 - larger initial tricuspid valve and pulmonary valve annulus diameter
 - a ratio of tricuspid valve annulus to mitral valve annulus of >0.5
 - transcutaneously measured oxygen saturation >90% at the age of one year
- Despite absolute growth of tricuspid valve and right ventricle, a real catch-up growth (increase in tricuspid valve Z-score of >2 points) is observed in only 10–15 % of cases
- Exercise capacity will be reduced in most patients even after biventricular repair due to the ongoing restrictive right ventricular physiology with diastolic dysfunction

Recommendations for follow-up

- Regular follow-up visits should assess growth and function of right ventricle and tricuspid valve (Z-scores, identical reference group!).
- objective examinations of exercise capacity and Holter ECGs every 2–3 years
- residual findings, which might require therapy:
 - reobstruction of the right ventricular outflow tract
 - atrial septal defect
 - high-grade pulmonary regurgitation
 - tricuspid regurgitation

Conflict of interests for this guideline

Harald Bertram: none
Michael Hofbeck: none
Alexander Horke: none

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APPENDIX:

Handling of Z-scores

General remarks:

- **Z = [measured value - mean of the population] / standard deviation of the normal population**
The Z-value quantifies the position of an individually measured size within a sample population; the measurement unit is the standard deviation above (+) or below (–) the population's mean value.
- The validity of the values is limited to a comparable normal population with normal distribution of the data, using the same measurement method for the definition of the mean and standard deviation. To be used in the clinical setting, the sample size should have been large enough (especially in young infants!). Mean and standard deviation may vary between the various investigators or with body size.
- For observational studies the same reference data and the same method of measurement should be used throughout.

Implications for the PA-IVS guideline:

- Most of the (older) surgical publications use reference data of inadequately small autopsy series, which should no longer be applied today.
- The best validated reference data for the calculation of tricuspid valve Z-scores are from Pettersen et al. ['Detroit data': J Am Soc Echocard 2008; 21(8):922-34], which are available for online calculation under www.parameterz.com.

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Transposition of the great arteries (TGA)

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Abstract Transposition of the great arteries is a common cyanotic congenital cardiovascular malformation characterized by ventriculo-arterial discordance resulting in parallel orientation of the systemic and pulmonary circuits. An interatrial communication is essential for survival. Additional shunting at ventricular or ductal level improves systemic saturation. Echocardiography, as the primary diagnostic tool defines the relationship between the great vessels, associated anatomical lesions and the coronary pattern. In patients with severe cyanosis, percutaneous atrial septostomy can improve blood oxygenation. The arterial switch operation (or its variants) is the method of choice for surgical repair of transposition of the great arteries. Overall, excellent outcomes are seen with an operative mortality below 6%.

Keywords: *transposition of great arteries (TGA), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition and characteristics

Transposition of the great arteries (also referred to as dextro-transposition of the great arteries; D-TGA) covered in this guideline is defined by atrio-ventricular concordance and ventricular-arterial discordance with the aorta arising from the morphologic right ventricle and the pulmonary artery from the morphologic left ventricle. Most commonly the aorta is in right anterior malposition relative to the pulmonary artery. This morphology results in a pathologic circulation in which systemic and pulmonary circuits are in parallel.

When no other heart defects (except ASD, PDA or insignificant VSD) are present it is called *simple TGA* (approx. 70%); when other defects are present it is called *complex TGA*.

Prevalence

Transposition of the great arteries accounts for approximately 2.5% of all newborns with congenital heart disease and is the second most common cyanotic malformation (after Tetralogy of Fallot). Boys are affected two to three times more often as girls. The condition is more common in children of diabetic mothers, poor nutrition or maternal intake of alcohol during pregnancy.

Relationships to consider

Associated lesions:

- Ventricular septal defect (approx. 20%)
- Outflow tract obstruction (approx. 5%)
- Coronary artery abnormalities (approx. 25%):
 - origin of the circumflex artery from the right coronary artery (approx. 14%)
 - single coronary ostium (approx. 6%)
 - intramural course of the coronary artery (<1%).
 - others (<5%)
- coarctation of the aorta (approx. <5%)
- interrupted aortic arch (<1%)
- abnormal veno-atrial connection (<1%; often with atrial isomerism)

The coronary arteries always arise from one or both of the aortic sinuses which are adjacent to the pulmonary trunk ("facing" sinuses) but their relative positions are variable. The Leiden convention is most commonly used for defining the coronary pattern independent of the position of the arterial trunks (figure 1).

Most commonly (>70% of cases) the left anterior descending artery and the circumflex artery arise from sinus 1 and the right coronary artery from sinus 2.

Differential Diagnosis:

- Other cyanotic heart diseases (e.g. Tetralogy of Fallot, see also appropriate guideline)
- Univentricular hearts with TGA (see also guideline univentricular hearts)
- Taussig-Bing-complex: characterized by a subpulmonary malaligned VSD and overriding pulmonary artery with increasing connection of the pulmonary valve (>50%) within the right ventricle. Frequently associated with pulmonary stenosis or coarctation of the aorta. (see also guideline: double outlet right ventricle).
- The entity congenitally corrected transposition (cc TGA) of the great arteries (also referred to as L-TGA) is characterized by

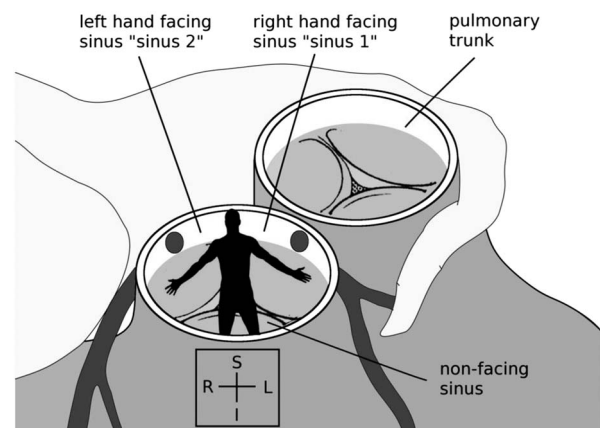


Figure 1.

Numbering of the aortic sinuses according to the Leiden convention for defining the coronary patterns according to the view from the non-coronary sinus. The sinuses supporting the coronary arteries are then located to the right (sinus 1) and left hands (sinus 2) of the observer standing in the non-coronary sinus and looking towards the pulmonary artery trunk.

atrioventricular and ventriculoarterial discordance. This non-cyanotic heart defect has to be distinguished from D-TGA and is not discussed in this article (see also guideline ccTGA).

Haemodynamics

- The systemic and pulmonary circulations are in parallel. The venous return is recirculated to the body, whilst the pulmonary venous return is recirculated to the lungs.
- Early neonatal survival depends upon persistence of the arterial duct, and an interatrial communication.
- The degree of cyanosis is mainly depending on mixing through an atrial septal defect, to a lesser degree at ductal or ventricular level.
- With low pulmonary vascular resistance and non-restrictive interatrial communication, the shunt direction is from the aorta to the pulmonary arteries.
- A restrictive interatrial communication may lead to shunt reversal and requires urgent decompression with atrial septostomy.

Clinical presentation

- The leading symptom in newborns is cyanosis without respiratory distress unresponsive to oxygen inhalation. It can be mild in patients with a large VSD.
- Infants with poor mixing may develop severe arterial hypoxemia, acidosis and circulatory failure which is aggravated by ductal closure
- Auscultatory findings are nonspecific. In newborns with intact ventricular septum no heart murmur is audible. In such cases, only a single and loud second heart beat is indicative.
- A systolic murmur is audible in cases with associated VSD or pulmonary stenosis.
- Weak femoral pulses may indicate an aortic coarctation

Diagnostic work-up

Echocardiography

Confirms the basic diagnosis and is the imaging modality of choice. The following anatomical and functional findings are important for further therapeutic decisions:

- Identification of concordant atrioventricular and discordant ventriculo-arterial connections
- The position of the aorta in relation to the pulmonary artery
- Size of the atrial septal defect and assessment of shunt flow
- Size of the arterial duct
- Origin and course of the coronary arteries
- Presence of
 - Aortic arch anomalies / coarctation
 - ventricular septal defects
 - outflow tract obstructions

Pulse oximetry

determines the degree of cyanosis (should be measured preductal at the right arm)

Labs

Blood gas analysis helps to assess systemic arterial oxygenation and circulatory failure.

ECG

Often normal in newborns; later signs of right ventricular hypertrophy with an upright T wave in V1 and right axis deviation

Chest-X-ray

may show cardiomegaly with an egg-shaped cardiac silhouette with a narrow superior mediastinum.

Blood pressure

Difference between arms and legs indicates coarctation.

Diagnostic cardiac catheterization

No longer routinely performed, only occasionally in complex anatomy or to assess pulmonary vascular resistance in patients presenting late.

Magnetic resonance imaging and computed tomography

may be helpful in selected patients, e.g. in the assessment of the mass of the left ventricle in the infant presenting late.

Management

Pharmacologic treatment

Intravenous infusion of prostaglandin E1 should be commenced immediately after delivery. It may be discontinued in the presence of adequate intracardiac mixing.

Atrial septostomy: In case of a restrictive ASD and low systemic oxygen saturation (despite prostaglandin infusion), balloon atrial septostomy (Rashkind procedure) is needed. The procedure can be performed under echocardiographic guidance on the intensive care unit.

Surgical management

- Simple TGA and TGA with VSD: Today, the arterial switch operation is the procedure of choice for anatomical correction with low complication and mortality rates (30-day mortality rate < 5%). In simple TGA the operation should be performed within the first 2 weeks of life, because the LV pressure should be near systemic levels at the time of surgery. The procedure, as currently performed involves transfer of the coronary arteries into the pulmonary trunk and translocation of the pulmonary artery in front of the ascending aorta (Lecompte manoeuvre).

In presence of a non-restrictive VSD the operation may be delayed.

In patients with a de-conditioned LV, PA banding and/or placement of a systemic-to-pulmonary arterial shunt is an option to retrain the LV before ASO.

In few cases, it is not possible to perform an arterial switch (because of late diagnosis, or rarely because of contraindicative coronary pattern). In such cases a physiological correction by an atrial switch operation should be considered using either the method as described by Senning or Mustard. Both methods involve removal of the atrial septum and creating a baffle to redirect oxygenated and deoxygenated blood flow to the aorta or PA respectively. The RV remains the systemic ventricle.

Their greatest relevance relates to the significant population of adolescents and adults who were operated in the transition period of the eighties and nineties and before.

- **D-TGA with VSD and left ventricular outflow tract obstruction (LVOTO):**

LVOTO is usually associated with a large VSD. In severe LVOTO, if ASO is not possible, the Rastelli procedure is the most frequently used option. The principle of this procedure involves creation of an intraventricular tunnel from the VSD towards the aorta. An extracardiac conduit is placed between the RV and the pulmonary arteries. The pulmonary valve is oversewn.

A similar operation, the Réparation à l'Étage Ventriculaire (REV procedure) involves extensive resection of the muscular outlet

septum, resulting in a more straight intraventricular tunnel from the LV to the aorta. After lecompte manoeuvre, a direct valveless connection of the PA to the RV is performed.

The rarely performed operation described first by Bex and Nikaidoh is characterized by a biventricular outflow tract reconstruction resulting in a true anatomical correction. The aortic root together with the coronary arteries is harvested from the RV and sutured into the left ventricular outflow tract. A straight left ventricular outflow tract is created. The pulmonary arteries are brought above the right ventricle and the right ventricular outflow tract is reconstructed using conduits.

Prognosis

Nearly all patients die within the first year of life without surgery. Surgical 30-day mortality of anatomical correction is about 5.6% (EACTS database). Risk factors are prematurity, birth weight < 2.5 kg, VSD or aortic pathologies.

Recommendations for follow-up:

Patients with TGA after surgery should be followed by a specialist lifelong. Routine follow-up intervals depend on age, type of surgery, and residual findings. Late problems and relevant issues are listed below:

After arterial switch operation:

- coronary insufficiency (occlusion or stenosis)
- supravulvar pulmonary stenosis
- aortic insufficiency
- dilatation of the aortic root
- residual VSD
- Obstruction of the ventricular outflow tracts

After atrial switch operation (Mustard/ Senning):

- Dysfunction of the systemic RV
- Regurgitation of the tricuspid valve
- Obstruction of the left ventricular outflow tract
- Obstruction of systemic or pulmonary venous return
- Interatrial Shunts (baffle-leaks)
- Rhythm disturbances: Atrial brady- and tachyarrhythmias, sinus node dysfunction, ventricular tachyarrhythmias.

Prevention

Primary prevention of TGA is not possible. Prenatal scanning and counselling should be offered. Delivery in a specialized congenital cardiac center with the option for emergency atrial septostomy or surgery is recommended.

Conflict of interests for this guideline

Carsten Rickers: None

Thomas Paul: None

Alexander Horke: None

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Common arterial trunk (CAT)

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Abstract A common arterial trunk is a rare cardiac malformation consisting of a single, common large arterial vessel (trunk) originating from both ventricles and usually a ventricular septal defect. The trunk feeds the body as well as the pulmonary arteries, the arch may be stenotic or interrupted. Surgical repair includes the closure of the ventricular septal defect, using the truncal valve as aortic valve and connecting the pulmonary arteries to the right ventricle usually with a conduit and reconstruction of the arch if necessary. Long term follow-up has to focus on the function of the aortic valve, restenosis of the arch if affected and stenosis or incompetence of the pulmonary conduit.

Keywords: *common arterial trunk (TAC), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

A common arterial trunk (CAT) is a malformation consisting of a single, common large arterial vessel (trunk) originating from both ventricles and usually a ventricular septal defect. The trunk feeds the body as well as the pulmonary arteries, the arch may have additional stenosis or even interruption. The valve may have various numbers of cusps, they may be dysplastic and stenotic or incompetent. The classification is based on the existence of a ventricular septal defect. (A = with VSD, B = without VSD, extremely rare) and the anatomy of the pulmonary arteries (see fig. 1 and table 1).

Prevalence

About 1/10.000 live births (corresponding to 0.5 to 0,8% of all congenital heart defects)

Relationships to consider

This malformation has a quite variable anatomy (see table 2) and in addition a high rate of associated lesions:

Associated lesions

- Right aortic arch (25–60%), atrial septal defects (70%)
- Stenosis of the pulmonary arteries, variations of the coronary arteries (15–30%)
- Left sided superior vena cava
- Anomalous drainage of the pulmonary veins

Associated syndromes

22q11 deletion (in 70%)

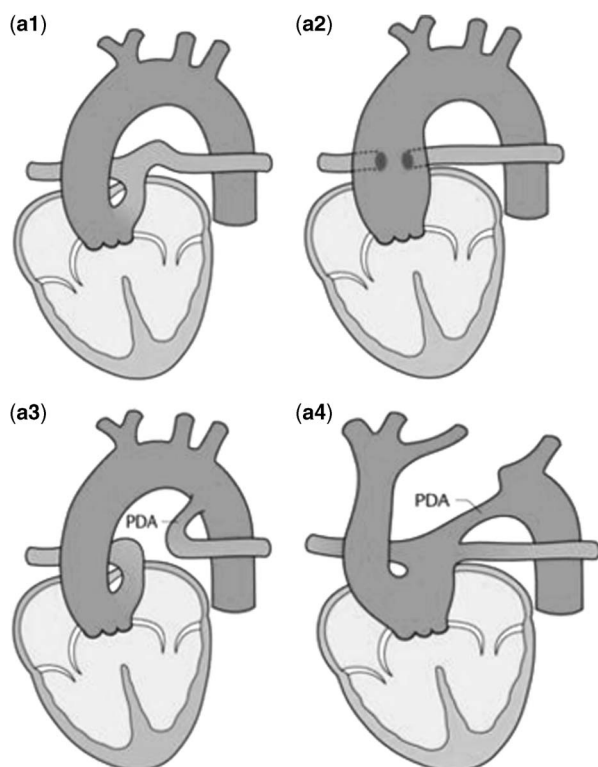


Figure 1.

Classification of the common arterial trunk according to van Praagh

Pictures from: Haas/Kleideiter, Pediatric Cardiology, Thieme Verlag, Stuttgart, 2015.

Table 1. Classification of the common arterial trunk according to van Praagh

- A1:** Aorta and main pulmonary artery originate from a single large common trunk.
 - A2:** Both pulmonary arteries originate together or closely related from the posterior wall of the common trunk.
 - A3:** One pulmonary artery comes from the common trunk. The other is perfused via a patent duct or via collaterals from the aorta (also called “hemitruncus”).
 - A4:** There are additional malformations of the aortic arch (coarctation, interrupted arch, etc.) The lower body is perfused via the patent duct.
- Note:** The most common form has a short main pulmonary artery and is therefore classified as trunk type “A 1 1/2”. Typically type A1 and A2 have no patent duct.

Table 2. typical lesions in patients with common trunk

As part of the common trunk	Associated lesions
- Malalignment VSD	- univentricular heart
- Incompetence of the truncal valve	- left superior vena cava
- Stenosis of the truncal valve	- atrial septal defect
- Variability of the cusps (2-6, most 3-4)	- anomalous pulmonary venous drainage
- Right aortic arch	- coronary abnormalities
- Stenosis of the pulmonary arteries	
- Coarctation, arch anomalies (A4)	

Differential diagnosis

Pulmonary atresia with VSD

Caveats

In arch obstruction perfusion of the lower body via the duct, in type A3 perfusion of one lung via a patent duct, both require prostaglandin infusion

Haemodynamics

Based on the large ventricular septal defect and the common trunk there is unrestricted blood flow to the lungs leading to pulmonary overflow, congestion and heart failure within the first weeks of life. The ventricles are volume loaded, this may be aggravated by an incompetence of the truncal valve. A patent duct (via prostaglandin infusion) is necessary to maintain pulmonary (A3) or systemic perfusion (A4).

Clinical presentation

The leading symptom in newborns is heart failure, cyanosis is usually mild but will be detected by neonatal pulse oxymetry screening. Closure of the duct may be without symptoms (A3) or result in shock (A4). Severe cyanosis may be a leading symptom in patients with bilateral stenosis of the pulmonary arteries.

Table 3. Echocardiography in common arterial trunk

Cardiac structure	Echocardiographic assessment
Truncal valve	- number and morphology of cusps - stenosis, regurgitation
Ventricular septal defect	- size, position, multiple defects ?
Ventricles	- size, function, volume load ?
Aortic arch	- left or right sided ? - branching vessel anatomy - diameter -> interruption?
Pulmonary arteries	- main PA yes/no - stenoses
Systemic veins	- left sided vena cava
Pulmonary veins	- drainage
Coronary arteries	- origin and course

Diagnostic work-up

Echocardiography

Diagnostic modality of choice (see table 3): The main and typical finding is a large ventricular septal defect and a single overriding vessel.

Blood pressure

Usually high pulse pressure, sometimes difference between arms/legs in coarctation/arch anomaly.

Pulse oximetry

Usually mild cyanosis (Sa O₂ > 90%) that is detected by routine neonatal pulse oxymetry screening

ECG

Biventricular hypertrophy; not diagnostic

Chest-X-ray

Volume loaded heart with pulmonary congestion, rarely different perfusion of one lung; not diagnostic.

Cardiac catheterization

Usually not required for diagnostic purposes in newborns, may be necessary in late presentation (>6 months) to evaluate pulmonary vascular reactivity. Used for interventional treatment postoperatively.

CT/MRI

Ultra fast low dose radiation CT may help in complex pre-operative anatomy. MRI assessment of right ventricular size and function postoperatively, in addition, arch anomalies and conduit anatomy and function.

Pathology

Genetic analysis for 22q11

Indications for treatment

Surgical treatment with repair is nearly always indicated, in late presentation with fixed pulmonary hypertension surgical repair is contraindicated.

Management

Symptomatic therapy

Only for stabilization of the newborn medical antifailure treatment (diuretics, ACE Inhibition, beta blockers), prostaglandin if A3 or A4. In late presentation with pulmonary hypertension see PAH treatment.

Note: Oxygen therapy is generally contraindicated despite cyanosis.

Surgery

Surgery should aim for a neonatal repair.

- Palliative surgery
Rarely used, may be required in selected cases with severe cyanosis (->shunt) or heart failure in very small or sick newborns and contraindications for repair (->bilateral banding of pulmonary arteries in preterm babies).
- Surgical repair:
Consists of closure of the VSD and connecting the pulmonary arteries to the right ventricle usually with a valved conduit (see figure 2). Arch abnormalities are repaired in the same session as well as abnormalities of the truncal valve.

Mortality: Without surgery nearly all patients die within the first year of life. Surgical mortality is about 10 to 17% (EACTS database) and higher in type A4, A4; 10 year survival about 80%.

Recommendations for follow-up

Follow-up is necessary lifelong, the relevant issues are listed in table 4. Assessment routinely by ECHO, MRI and exercise test.

Conduit

May require conduit change due to the absolute size (usually within 5 years after neonatal repair). Check for RVOT obstruction or incompetence and right ventricular volume load (see guideline pulmonary stenosis and pulmonary incompetence).

Pulmonary arteries

Check for supraventricular stenoses, right ventricular pressure load and differential lung perfusion caused by scarring after reconstruction.

Aortic valve

Incompetence caused by anatomic variations and malformation may require reconstructive surgery or replacement.

Aortic arch

Enlargement of the ascending aorta may develop necessitating replacement (>5 cm). Recurrent arch obstruction may occur after coarctation/interrupted arch.

Arrhythmias

Atrial arrhythmias may develop caused by atrial scarring (i.e. SVES, SVT, IART), ventricular arrhythmias caused by VSD patch closure (rare) or infundibular scarring. Holter Monitoring is required every 1-3 years.

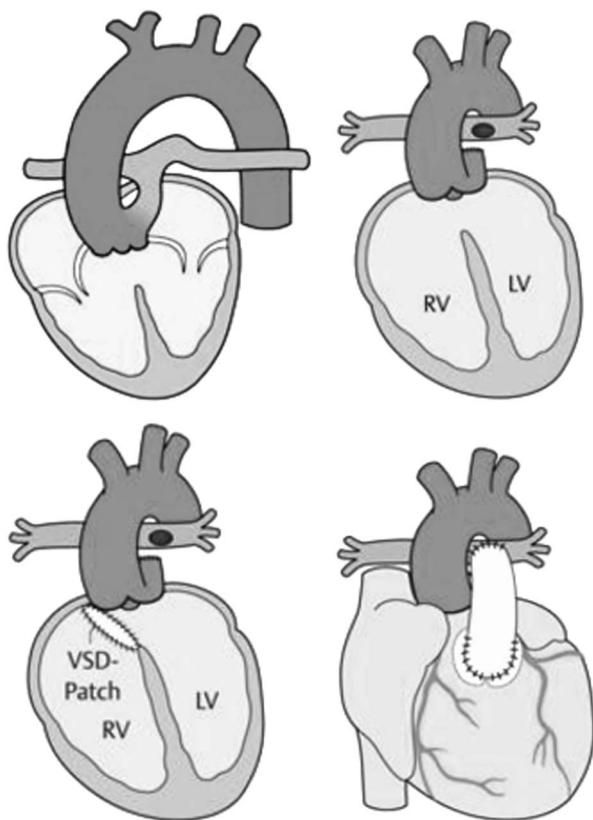


Figure 2.

Surgical repair in a common arterial trunk type A1. The main pulmonary artery is dissected, the VSD is closed with a patch and the pulmonary arteries are connected to the right ventricle with a valved conduit

Pictures from: Haas/Kleideiter, Pediatric Cardiology, Thieme Verlag, Stuttgart, 2015.

Table 4. Follow-up recommendations after surgical repair in common trunk

Truncal valve	- repair/replacement - NOT timing of conduit change
Right ventricle	- size and function
Conduit	- stenosis, regurgitation, change?
Pulmonary arteries	- stenoses, different perfusion
Aorta	- dilatation of the aorta
Arrhythmias	- arch obstruction
	- stenosis monitoring every 1-3 years
Exercise capacity	- cardio-pulmonary exercise test every 3-5 years
Psychomotoric development	- microdeletion 22q11
Pulmonary hypertension	- selective vasodilators

Pulmonary Hypertension

May persist in late presentation and after late repair. Treatment includes pulmonary vasodilators (i.e. Sildenafil, Bosentan, etc.) and possible early referral to a transplantation clinic (see guideline).

Cardiopulmonary Exercise Testing

For assessment of the impact of pulmonary regurgitation/stenosis and/or aortic regurgitation and ventricular function cardiopulmonary exercise testing is required every 3-5 years.

Prevention

Prenatal scanning and counselling should be used.

Conflict of interests for this guideline

Nikolaus A. Haas: None

Jörg Sachweh: None

Ingo Daehnert: None

Link to all references listed in the original German long version "Truncus arteriosus communis": www.kinderkardiologie.org/leitlinien/

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Hypoplastic Left Heart Syndrome (HLHS)

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Abstract Hypoplastic left heart syndrome (HLHS) is a rare but severe cardiac malformation that will result in neonatal death if untreated. Treatment options consist after stabilization of the newborn of compassionate care, classical Norwood operation, hybrid therapy or cardiac transplantation. Long-term management in univentricular palliation is associated with a high morbidity and mortality. Neurodevelopmental outcome, exercise capacity and life expectancy are impaired.

Keywords: *hypoplastic left heart syndrome (HLHS), guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Classical hypoplastic left heart syndrome consists of a combination of malformations with hypoplasia of the left ventricle, the mitral and aortic valve and the ascending aorta. The following subgroups may be distinguished (see figure 1):

- Mitral atresia/aortic atresia (MA/AoA)
- Mitral stenosis/aortic atresia (MS/AoA)
- Mitral stenosis/aortic stenosis (MS/AoS)
- Mitral atresia/aortic stenosis (plus VSD) (MA/AoS)

In addition the term hypoplastic left heart complex is used for those cases where the aorta is hypoplastic and the left ventricle is

small or based on the anatomy or physiology not able to support the systemic circulation. In these patients a univentricular palliation is required. Typical examples are:

- Double outlet right ventricle (DORV) with small left ventricle
- cc TGA with large VSD and small left ventricle
- Imbalanced AVSD with small left ventricle
- long segment subaortic stenosis with aortic stenosis, VSD and arch hypoplasia

Epidemiology

HLHS contributes to 1–2% of all congenital heart defects. 2/3 of the patients are male. HLHS is one of the most important reasons for heart failure in term newborns and the main reason for neonatal death if untreated.

Relationships to consider

Associated lesions

VSD, anomalous pulmonary venous drainage, coronary abnormalities, sinusoids, tricuspid valve abnormalities, abnormalities of the systemic veins

Associated syndromes

Genetic syndromes are present in about 10% of patients with HLHS including Turner syndrome, Noonan, trisomy 13, 18, 21, Holt-Oram syndrome, CHARGE association, and Jacobsen syndrome.

Differential diagnosis

Cyanotic congenital heart defects, critical aortic stenosis, critical coarctation, cardiomyopathy, sepsis, persistent fetal circulation

Haemodynamics

Based on the hypoplasia of the left ventricle and the inability to support cardiac output, systemic perfusion depends on the patent duct (PDA) and the right ventricle. The PDA and an adequate mixing at the atrial level are essential for postnatal survival. A restrictive or closed foramen ovale will result in pulmonary oedema and hypoxia. Intrauterine congestion will lead to morphologic changes of the pulmonary veins and lymphangiectasis.

Two changes in physiology will lead to postnatal decompensation:

- closure of the PDA will reduce systemic perfusion and retrograde perfusion of the aortic arch, i.e. the head and coronary arteries.
- decrease of pulmonary vascular resistance reduces right-to-left shunt across the PDA, increases pulmonary overflow, impairs systemic perfusion and activates sympathetic tone to enable adequate blood pressure. The volume load of the right ventricle often increases tricuspid regurgitation. This results in severe metabolic acidosis and finally in cardiogenic shock.

Clinical presentation

- Early warning signs in the newborn are tachypnoea without typical auscultation or murmur. Subsequent cardiogenic shock with weak or absent pulses, low blood pressure, tachydyspnoea, low saturation and progressive metabolic acidosis will lead to secondary renal and liver failure and necrotising enterocolitis.

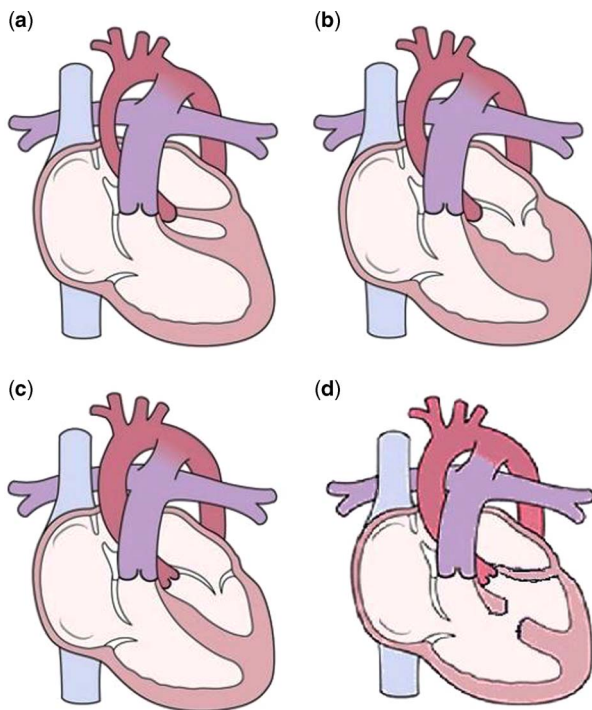


Figure 1. Various types of hypoplastic left heart syndrome. a. MA/AoA; b. MS/AoA; c. MS/AoS, d. MA/AoS and VSD.

Diagnostic work up – see table 1

Indications for treatment

If untreated a hypoplastic left heart syndrome will lead to neonatal death in nearly all cases.

Management

Medical treatment for stabilisation of the newborn

- **Prostaglandin:** to secure or enhance systemic perfusion a continuous infusion of prostaglandin (initial dose 10–20 nanogram/kgbw/min) is started. Echocardiographic assessment of the PDA is warranted, dose adaptation may be needed. (caveat: apnoea, pulmonary hyperperfusion)
- **Afterload reduction:** to increase systemic perfusion afterload reduction using Na-nitroprusside or phentolamine may be used (check blood pressure). After decompensation, volume boluses or milrinone may be required, rarely catecholamine support is necessary.
- **Ventilation:** spontaneous breathing is preferred. Supplemental oxygen therapy is contraindicated. If ventilation is required, a saturation of 75–85%, a paO_2 of about 40 mmHg and a paCO_2 of 40 (–45) mmHg should be obtained.
- **Hemoglobin:** a Hb of > 14 g/dl will optimize oxygen delivery
- in neonates with shock, adequate resuscitation according to APLS guidelines including aggressive normalization of acidosis using large quantities of sodium bicarbonate is required.

Management options in patients with HLHS after stabilization of the newborn

Table 2 summarizes treatment options in HLHS.

Surgical treatment

Basic principle of surgical therapy consists of unrestricted pulmonary venous return to the right heart (atrioseptectomy,

Table 1. primary diagnostic evaluation for HLHS

Blood pressure:	low at all 4 extremities
Saturations:	usually > 80% dependent on pulmonary perfusion
ECG	nonspecific
Chest x-ray	nonspecific
Echocardiography:	method of choice to confirm diagnosis, anatomy, function
	LV - size, function
	Aortic arch: - anatomy, lusoric artery
	ASD - restrictive ?
	PDA - stenosis, shunting
	RV - function, tricuspid regurgitation
Pulmonary veins	- drainage, stenosis?
Systemic veins	- anatomy
Cardiac catheterization	- only if pulmonary vein anatomy is unclear
	- for balloon atrial septostomy or hybrid treatment
MRI/CT	usually not necessary
Laboratory tests	acidosis, secondary organ damage, infection, coagulation tests

Table 2. Management options in patients with HLHS after stabilization of the newborn:

- Surgical treatment: Norwood operation (in a broader sense)
- Hybrid approach
- Cardiac transplantation
- Compassionate care

if necessary correction of anomalous pulmonary vein drainage) as well as an unrestricted outflow to the body from the right heart with adequate perfusion of the vessels of the arch and coronaries. Pulmonary perfusion must be balanced to enable adequate oxygenation and avoid excessive pulmonary blood flow. The 30-day mortality is about 20%, in experienced centers about 10%. First surgical step should be performed within the first week of life.

• Classical Norwood operation with modified BT shunt

After creation of a neo-aorta (patch augmentation of the pulmonary trunk and the ascending aorta) pulmonary perfusion is secured via a modified 3–4 mm Blalock–Taussig shunt mainly to the right pulmonary artery (see Figure 2a).

• Norwood operation using a Sano shunt

After creation of a neo-aorta pulmonary perfusion is secured via a 5–6 mm Gore-Tex tube between the right ventricular outflow tract and the main pulmonary arteries (see Figure 2b).

• 2nd step – superior cavopulmonary anastomosis:

Usually at 4–6 months of age the shunt is removed and the superior vena cava is connected to the (right) pulmonary artery (see guideline univentricular heart). Tricuspid regurgitation, if hemodynamically significant, should be addressed at this time (see Figure 3).

• 3rd step – total cavopulmonary anastomosis

Usually at 2 to 3 years of age complete separation of the venous and arterial circulation is achieved using an extracardiac or intracardiac Fontan procedure. A fenestration has been shown to be advantageous concerning early postoperative morbidity (see guideline univentricular heart; see Figure 4).

Hybrid approach

Especially for patients with risk factors or contraindications for a Norwood procedure (i.e. prematurity, intracranial haemorrhage, etc.) a hybrid approach (i.e. “Giessen procedure”) can be performed. It consists of bilateral banding of the pulmonary arteries, a stent implantation into the PDA and creation of an unrestricted communication at the atrial level (stent, Rashkind procedure) (see figure 5). This approach avoids cardiopulmonary bypass and major cardiac surgery in the (sick) newborn. Downsides are a higher risk for pulmonary artery stenosis or recurrent coarctation necessitating multiple additional interventional or surgical

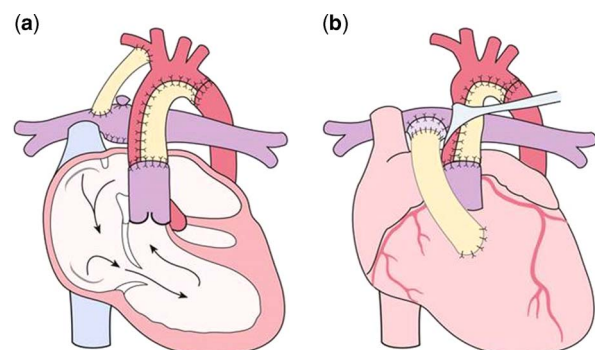


Figure 2.

a. classical Norwood operation with modified BT shunt; b. Norwood-operation using RV-PA (Sano-) Shunt.

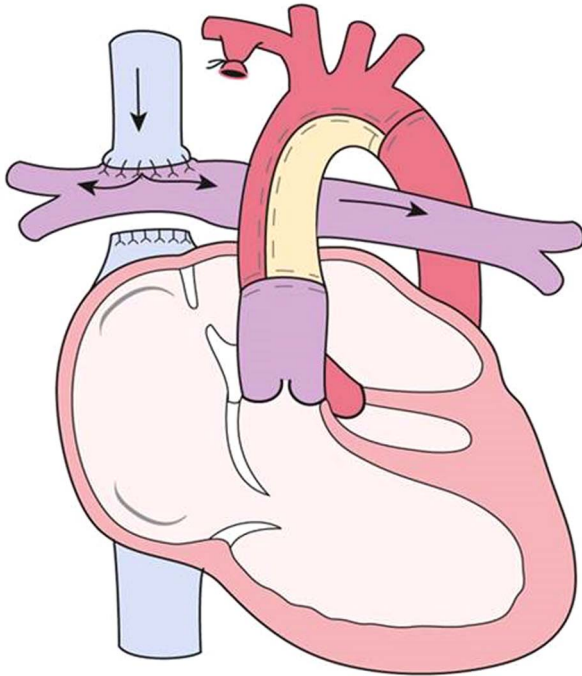


Figure 3.
HLHS after Norwood and superior cavopulmonary anastomosis.

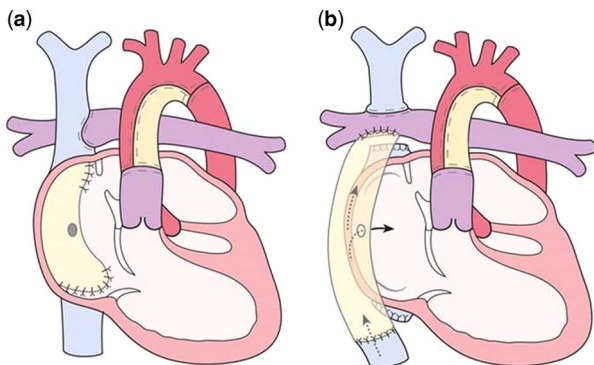


Figure 4.
HLHS and total cavopulmonary anastomosis a. intracardiac Fontan with fenestration; b. extracardiac conduit with fenestration.

procedures. The published results show however no benefit regarding mortality or neurodevelopmental outcome.

The second step of this approach is the comprehensive stage II surgery – i.e. combining a stage I Norwood with a stage II superior cavopulmonary anastomosis.

This hybrid approach offers advantages such as bridging to transplant, allowing growth of the underdeveloped left ventricle in borderline left ventricles, late Norwood operation.

Cardiac transplantation:

Due to the significant organ shortage in most countries, this treatment option is not realistic in newborns and restricted to patients with failing right ventricles lateron and in patients with failing Fontan circulation.

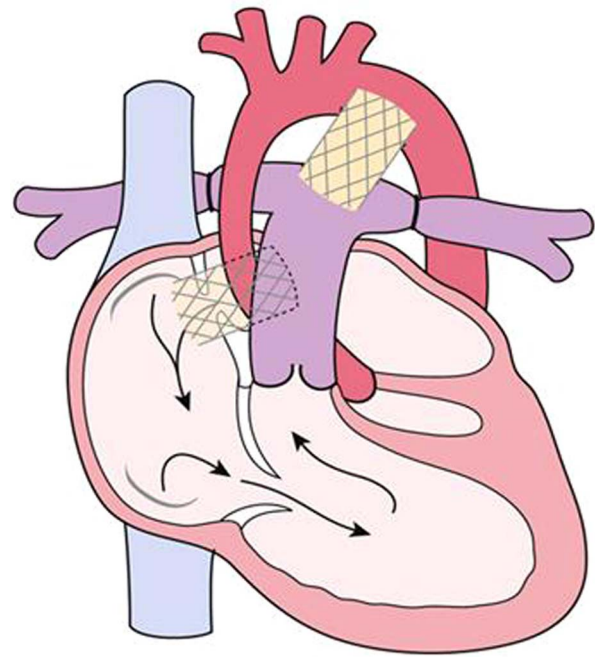


Figure 5.
Hybrid approach for palliation of HLHS. A stent is placed to maintain the patent duct, the left and right pulmonary artery is banded and a stent is placed in the PFO to maintain unrestricted flow.

Compassionate care:

Despite a high procedural success of surgical and/or hybrid treatment there is a considerable late morbidity and mortality and impaired neurodevelopmental outcome. Therefore, compassionate care has to be offered to all the families and still is an equivalent treatment option. If chosen, the patient will not receive any surgical or interventional treatment, prostaglandin infusion is stopped and adequate sedation and pain relief are administered. The family has to be supported adequately during this palliative care process.

Recommendations for follow-up

Despite the improving results of surgical and interventional procedures the overall outcome of patients with HLHS is not completely encouraging. A 5-year survival of 50–80% has been published, and some centers have reached a 10-year survival of 50–70%. Survival however does not adequately describe the very high morbidity in these patients.

Follow-up after stage 1

A close and careful follow-up is recommended focussing mainly on possible recoarctation (up to 25%), heart failure and tricuspid regurgitation. The optimal oxygen saturation is between 75 and 85%. After hybrid-stage I additional careful investigation regarding the PA banding, the PDA stent, the interatrial communication and retrograde arch perfusion is warranted. ASS is the usual anticoagulation. Some patients require additional feeding programs.

There is a risk of about 15% interstage mortality of unknown aetiology and dedicated home monitoring programs may reduce this risk.

Late follow-up

Somatic and physical development in children with HLHS is impaired. In general there are limitations in exercise capacity, life expectancy as well as in the professional development.

Based on published data children with HLHS show many additional defects of various degrees in their psychomotor development. Modern management modalities (i.e. selective cerebral perfusion, prenatal diagnosis, etc.) may theoretically give hope for improved neurological outcome later on, but the results are so far not encouraging. Children with Fontan pathway require follow-up according to the guidelines univentricular heart, those after transplantation management in a transplant centre.

Prevention

Prevention is not possible. Genetic counselling as well as prenatal diagnosis using fetal echocardiography should be offered.

Conflicts of interests for this guideline:

Nikolaus, A. Haas: None

Christian Jux: None

Joachim Photiadis: None

Hans -H. Kramer: None

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Univentricular heart and functional univentricular heart

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Abstract Univentricular hearts and functional univentricular hearts include congenital heart defects, in which biventricular repair cannot be achieved. The therapeutic aim is to establish a total cavopulmonary connection in usually three stages in early childhood. Long-term outcome may be complicated by “Fontan failure” which is characterized by reduced exercise capacity, arrhythmias, formation of collateral vessels, thromboembolic events, liver dysfunction, protein losing enteropathy, and/or plastic bronchitis.

Keywords: *univentricular heart, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition (figure 1)

- Univentricular hearts are defined as a group of congenital defects that do not have two ventricles with an appropriate inlet and pumping portion (double inlet left ventricle, tricuspid atresia, hypoplastic left heart syndrome).
- Functional univentricular hearts are defects, in which a biventricular repair seems impossible (ventricular imbalance, straddling, hypoplastic, or in infancy severely regurgitant atrioventricular valve, very large ventricular septal defects ...).
- Prevalence of univentricular hearts and functional univentricular hearts: 3.0/10.000 live births (corresponding to 2.8% of all congenital heart defects for each).

Associated syndromes

Heterotaxy, which is a risk factor for morbidity and mortality in univentricular hearts.

Haemodynamics and clinical presentation

- *Initial parallel circulation:* There is mixing of systemic and pulmonary venous return on venous, atrial and/or ventricular

level. The single ventricle supplies both the pulmonary and systemic circulation. The clinical condition is mainly defined by the ratio of pulmonary to systemic blood flow. If systemic blood flow dominates, cyanosis is severe. If pulmonary blood flow is excessive, heart failure symptoms predominate with no or only mild cyanosis. In addition, atrioventricular valve regurgitation may exaggerate heart failure. Pulmonary vein stenosis or a hypoplastic atrioventricular valve with restrictive interatrial connection may cause venous congestion.

- **Partial cavopulmonary connection:** The venous return from the superior caval vein is the only pulmonary blood supply. Cyanosis is mild to moderate.
- **Fontan circulation:** The venous return of the superior and inferior caval veins is rerouted to the pulmonary arteries. The pulmonary blood flow depends mainly on ventilation, central venous pressure and diastolic ventricular function.

The coronary sinus ($\pm$ fenestration) is the only residual shunt, cyanosis is virtually eliminated. The single ventricle serves the systemic circulation. For a good function of a Fontan circulation an unobstructed flow into the lungs, a low pulmonary vascular resistance, an unrestrictive pulmonary venous flow, a good myocardial and valvar function, and an unrestrictive outflow into the aorta are essential.

Diagnostic work up

Echocardiography establishes the primary diagnosis including all anatomical details. During follow-up ventricular and valvular function must be regularly assessed and ventricular outflow tract stenosis should be ruled out. A ventilation dependent venous flow profile confirms unrestricted blood flow in the pulmonary arteries. A residual shunt can be detected.

Pulse oximetry quantifies cyanosis which is initially an estimate for pulmonary to systemic blood flow ratio. During long-term follow-up cyanosis discloses a right-to-left shunt.

ECG has no role in establishing the diagnosis. However, it may detect arrhythmias, like preexcitation/Wolff-Parkinson-White syndrome in tricuspid atresia, or atrial flutter/atrial reentrant tachycardia during follow-up.

Holter monitoring is crucial to define arrhythmias during follow-up. **Cardiac catheterization** is usually not indicated for primary diagnosis. Indications include unusual vessel connections in complex defects or when an intervention (arterial duct stent) is intended. When partial and total cavopulmonary connections are planned, a full haemodynamic cardiac catheterisation with complete angiography is strongly recommended to evaluate pulmonary artery diameters and pressures, pulmonary vascular resistance, as well as ventricular function. In patients with Fontan failure cardiac catheterization is indicated. Again, evaluation of pulmonary pressures, vessel diameters, ventricular function, and collateral vessels may help to establish an appropriate therapy according to the underlying anatomical/haemodynamic condition.

Cardiopulmonary exercise testing (CPET) measures objective exercise capacity and may detect deteriorating hemodynamics at an early stage. In addition, CPET may uncover arrhythmias, right-to-left shunting, and ventilation/perfusion mismatch. CPET is a strong predictor for morbidity. Beyond the age of 10 years, CPET should be performed at least every three years.

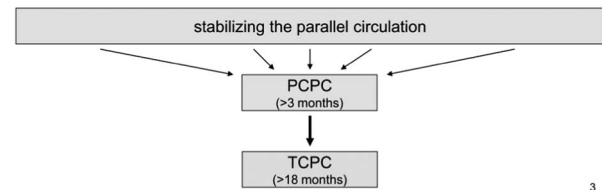
Cardiac magnetic resonance (CMR) imaging is the gold standard to assess ventricular volumes and function, as well as blood flow. It can properly demonstrate the cavopulmonary pathway and connections.

Blood tests are recommended at least annually during follow-up. Liver enzymes, BNP or NT-proBNP, as well as protein or albumin help to detect Fontan failure early. Anaemia should be omitted.

univentricular heart
only 1 ventricle with complete inlet and pumping part

DILV, TA, HLHS

„functionally“ univentricular heart
biventricular repair is impossible
hypoplastic ventricle, straddling av-valves, severe neonatal av-valve regurgitation, coronary fistulae in CAVSD, PA-IVS, AS, MS, MA, DORV, VSD, EBS, ...



3

Figure 1.

Sequential management for univentricular heart

DILV double inlet left ventricle; TA tricuspid atresia; HLHS hypoplastic left heart syndrome; CAVSD complete atrioventricular septal defect, PA-IVS pulmonary atresia with intact ventricular septum; AS aortic stenosis; MS mitral stenosis; MA mitral atresia; DORV double outlet right ventricle, VSD ventricular septal defect; EBS Ebstein anomaly; PCPC partial upper cavopulmonary connection; TCPC total cavopulmonary connection.

Indications and timing for treatment (figure 1)

- **<3 months (parallel circulation):** If the patient has a balanced pulmonary and systemic blood flow, an unrestricted venous return and left ventricular outflow, as well as normal valve function, no interventions are necessary at that time.
- usually beyond the age of 3 months: upper partial cavopulmonary connection
- usually beyond the age of 18 months: total cavopulmonary connection

Management

The sequential management plan for univentricular heart is outlined in figure 1.

Therapeutic options to improve parallel circulation:

- In order to improve pulmonary blood flow, prostaglandin E may be used to keep the arterial duct open for initial stabilisation before definite assurance of pulmonary perfusion, which may be established by arterial duct stenting or by a surgical aorto-pulmonary shunt.
- For reduction of pulmonary blood flow, the pulmonary valve is closed (or anastomosed to the ascending aorta) and the pulmonary perfusion is re-established by an aorto-pulmonary shunt. Pulmonary arterial banding is not recommended due to high risk of subsequent ventricular hypertrophy.
- In case of left ventricular outflow tract stenosis a Damus-Kaye-Stansel procedure with dissection of the pulmonary trunk below the bifurcation, anastomosis of the proximal pulmonary trunk to the ascending aorta, and an aorto-pulmonary shunt needs to be performed.
- Aortic coarctation requires timely repair. In case of a hypoplastic ascending aorta/aortic arch a Norwood procedure needs to be performed.
- For improvement of venous return a balloon atrioseptostomy or surgical atrioseptectomy may be necessary.
- In order to improve atrioventricular valve regurgitation, one sided atrioventricular valve closure or atrioventricular valve repair may be necessary.

- In patients with an aorto-pulmonary shunt, antiplatelet therapy is recommended.

In upper partial cavopulmonary connection the superior caval vein is end-to-side anastomosed to the right pulmonary artery, and the inflow into the right atrium is closed. Alternatively a Hemi-Fontan operation is performed with an anastomosis of the superior caval vein to the right pulmonary artery and a right atrial patch closure of the inflow from the superior caval vein. If there are two superior caval veins, bilateral cavopulmonary anastomoses need to be created.

Total cavopulmonary connection should be performed at an age of 18–24 (–36) months. It is considered as definitive palliation in patients with a univentricular heart. An intraatrial lateral-tunnel is created with a patch, leading the venous return of the inferior caval vein to the former orifice of the superior caval vein, which itself is connected to the right pulmonary artery. Alternatively an extra-cardiac conduit is placed between the inferior caval vein and the right pulmonary artery. Fenestration of the intraatrial tunnel or a veno-venous shunt may be used to reduce pleural effusion and postoperative hospitalization.

During follow-up the fenestration may be closed by catheter intervention.

Prognosis

Primary survival before the total cavopulmonary connection depends on the underlying defect. In patients with total cavopulmonary connection 20 year-survival is estimated to be 87%.

Recommendations for follow-up

After total cavopulmonary connection life-long regular follow-up examinations are recommended for detection early signs of Fontan failure.

Haemodynamic reasons and treatment options for Fontan failure include:

- Stenosis involving the cavo-pulmonary conduit and anastomoses, which should be treated by balloon dilatation, stent implantation or surgical patch enlargement.
- Even mildly increased pulmonary vascular resistance has a major impact on the passive pulmonary perfusion in the Fontan circulation. In Fontan failure an off-label use of pulmonary vasodilators may be an option.
- Pulmonary vein stenosis, valve dysfunction, ventricular outflow tract obstruction and aortic coarctation should be addressed immediately.
- Systolic or diastolic ventricular dysfunction must be treated according to the guidelines for heart failure. The bailout option is heart transplantation.
- Diaphragmal hernia and scoliosis often impair ventilatory enhancement of venous return and should be timely addressed.

Symptoms of Fontan failure and treatment:

- Reduced exercise capacity is often not based on haemodynamic sequelae alone, but may also result from muscular atrophy, malperfusion, and dysfunction. An active life style is recommended.
- Supraventricular arrhythmias are often based on atrial suture lines and a dilated right atrium. Treatment should be directed according to current guidelines.

- Residual shunts and collateral vessels may cause cyanosis and should be closed preferably by catheter intervention. Recurrence rate, however, is considerable.
- Prevalence of thromboembolism is high (up to 23%). Up to now, no consensus has been reached whether cerebrovascular events or (silent) pulmonary embolism should be prophylactically addressed by antiplatelet therapy or oral anticoagulation.
- Liver dysfunction can be detected by appropriate laboratory tests including serum liver enzyme activities (e.g. gamma-glutamyl transferase). Specific treatment is unknown.
- Protein losing enteropathy is characterized by reduced serum albumin, oedema, pleural effusions and ascites. It has a high 5-year mortality reaching up to 49–59%. Symptomatic therapy with steroids or heparin may be beneficial at short-term. During long-term follow-up side effects predominate in most patients affected. Oral budesonide is a new and promising option in some patients.
- Plastic bronchitis is characterised by bronchial casts. It is life-threatening and recurrent. Acute treatment with repeated bronchoscopy and intrabronchial rhDNAse rinsing or inhalation with tissue plasminogen activator and urokinase has been reported.

General preventive measurements:

- RSV prophylaxis at least in the first year of life
- Extended vaccination programme including annual flu shots and pneumococcal vaccine
- Neurodevelopmental surveillance
- Career counselling considering reduced exercise capacity
- Counselling for contraception for females
- Pregnancy surveillance by a cardiologist, as pregnancy is usually associated with an increased risk for mother and child.

Conflict of interests for this guideline

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Robert Cesnjevar: None

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Link to all references listed in the original German long version “Univentriculäres Herz”: www.kinderkardiologie.org/leitlinien/

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Supraventricular and ventricular tachyarrhythmias in children

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Abstract Common types of tachyarrhythmias in infants and children with and without congenital heart defects are presented together with contemporary recommendations on diagnosis and therapy.

Keywords: *supraventricular, ventricular tachyarrhythmias, children, guideline*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Tachyarrhythmias are defined as paroxysmal or chronic elevation of the atrial and/or ventricular rate. They are caused by abnormal impulse generation (automaticity or triggered depolarization), abnormal impulse propagation (reentry, block, aberrant conduction) and/or abnormalities in autonomic influence. In general, they can be differentiated into supraventricular and ventricular arrhythmias. In clinical practice, it is useful to separate reentrant tachycardias from those due to triggered depolarizations or abnormal automaticity.

Supraventricular tachycardias are the most frequent symptomatic tachyarrhythmias in children, adolescents and patients with a congenital heart defect. Supraventricular tachycardias are defined as tachyarrhythmias involving critical tissue proximal to the bifurcation of the His bundle, whereas ventricular tachyarrhythmias arise from the specialized ventricular conduction system or ventricular myocardium independent from atria and His bundle. Single impulses are named “extrasystoles”, two consecutive beats are called a couplet. Supraventricular or ventricular tachycardia is present in ≥ 3 consecutive impulses.

Clinical presentation

Clinical presentation depends on age of the patient, individual cardiac anatomy, and the type of tachyarrhythmia. In otherwise healthy newborns and infants paroxysmal supraventricular tachycardia with heart rates $>250/\text{min}$. may result in rapidly developing heart failure due to shortened ventricular filling. In children and adolescents with paroxysmal supraventricular tachycardia, main symptoms are palpitations and dizziness. Syncope is a rare event and may be due to atrial flutter or atrial fibrillation with rapid AV conduction via an accessory atrioventricular pathway in WPW syndrome.

Pediatric patients with chronic supraventricular tachycardias do not develop acute symptoms as heart rate is often not critically elevated. Due to long-standing elevated heart rates, however, these children may present with tachycardia-induced cardiomyopathy and severely impaired left ventricular function.

Ventricular tachycardia in children is associated with palpitations, dizziness, and syncope and may finally result in sudden cardiac death.

Diagnostic work-up

Documentation of the tachycardia on 12-lead ECG allowing diagnosis and proper treatment; assessment of frequency and duration of tachycardia episodes as well as tachycardia symptoms; imaging techniques to establish cardiac anatomy and function.

Electrocardiography

12-lead ECG with rhythm strip, Holter monitor, treadmill test, transesophageal ECG, epicardial ECG via temporary pacing wires after cardiac surgery, registration of late potentials, cardiac event monitor (external or implantable), internal ECG of pacemaker or ICD generators

- Spontaneous tachycardia on 12-lead ECG, Holter monitor in chronic or frequently recurring tachycardias, treadmill test (exercise-induced tachycardia, abrupt loss of ventricular preexcitation, QTc in Long QT syndrome, efficacy of β -blockade), event monitor in syncope of undetermined origin
- Transesophageal or epicardial recording for identification of P waves
- Search for markers of pathological impulse propagation or repolarization patterns on ECG: preexcitation syndrome, long QT syndrome, short QT syndrome, Brugada syndrome, arrhythmogenic right ventricular dysplasia/cardiomyopathy, congenital heart defects (tetralogy of Fallot with QRS duration $>180\text{ ms}$).

Pharmacological challenges

- a) Adenosine for differential diagnosis in primary atrial tachycardia and wide QRS complex tachycardia
- b) Ajmaline/flecainide challenge in Brugada syndrome
- c) Epinephrine in long QT syndrome type 1 and catecholaminergic polymorphic ventricular tachycardia

Electrophysiological testing

Programmed atrial and ventricular stimulation and endocardial mapping during electrophysiological study:

- In reentrant supraventricular and ventricular tachycardia for establishing diagnosis allowing catheter ablation
- Risk stratification in children >8 years with permanent preexcitation pattern
- Useful for risk stratification in selected patients with congenital heart defects (tetralogy of Fallot)
- Undetermined benefit in Brugada syndrome and dilated cardiomyopathy
- Not recommended in long QT syndrome, short QT syndrome, catecholaminergic polymorphic ventricular tachycardia, and hypertrophic cardiomyopathy

Cardiac imaging

Echocardiography for anatomy and cardiac function, chest X-ray, cardiac magnetic resonance imaging (myocarditis – scarring and late enhancement, arrhythmogenic right ventricular dysplasia/cardiomyopathy), cardiac catheterization with selective coronary angiography and endomyocardial biopsy (arrhythmogenic right ventricular dysplasia/cardiomyopathy, dilated cardiomyopathy, myocarditis)

Molecular genetic testing

- Recommended in long QT syndrome, short QT syndrome, catecholaminergic polymorphic ventricular tachycardia, Brugada syndrome, and arrhythmogenic right ventricular dysplasia/cardiomyopathy for identification of mutation carriers including asymptomatic family members; for risk stratification (long QT syndrome)
- Not helpful congenital heart defects

Laboratory tests

Biomarkers of myocardial injury, thyroid hormones

Differential diagnosis

Accelerated rhythms (atrial, junctional and ventricular), defined as heart rate up to 20% faster than underlying rhythm

Sophisticated work-up should be performed in specialized centres experienced in paediatric cardiology, electrophysiology and paediatric cardiac surgery.

Management

Documentation of tachycardia and rhythm post intervention by 12-lead ECG whenever possible

- Aims: termination of tachycardia, prevention of tachycardia recurrence
- Urgent therapy indicated in any child with symptomatic or life-threatening tachycardia
- Therapeutic options for tachycardia termination (see Figure 1): vagal maneuvers in reentrant supraventricular tachycardia (ice bag on face), pharmacological intervention (see Table 1a), transesophageal or endocardial stimulation, external cardioversion/defibrillation
- Therapeutic options for prevention of tachycardia recurrence: antiarrhythmic drugs (see Table 1b), permanent pacemaker in

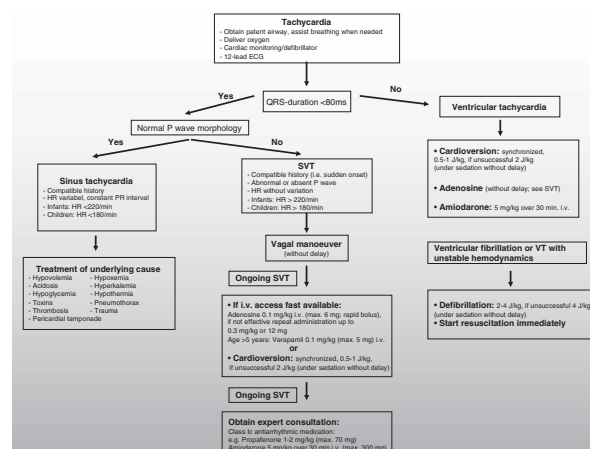


Figure 1.

Algorithm for diagnostic and therapeutic management of tachyarrhythmias in childhood.

Table 1a. Therapy of an SVT episode with regular RR intervals and stable hemodynamics

Agent	Dosage
Adenosine	0.1-0.3 mg/kg, max. 12 mg i.v. as rapid injection
Verapamil >5 years of age*	0.1 mg/kg i.v., max. 5 mg
Propafenone	1-2 mg/kg i.v.
Flecainide	1 mg/kg i.v.
Amiodarone	Start with bolus 5 mg/kg i.v. (max. 300 mg) for 30 min.; continuous infusion 10 - max. 20 mg/kg/d, bolus may be repeated
Esmolol	Initially 0.1-0.5 mg/kg/min. i.v., continuous infusion 50-200 µg/kg/min. (caution: hypotension)

(*Caution: do not administer in newborns and infants)

Table 1b. Oral pharmacological therapy for SVT (alphabetic order)

Agent	Dosage
Amiodarone	3-5 mg/kgxd (max. 200 mg/d)
β-blocker (propranolol)	2-6 mg/kgxd (max. 240 mg)
Flecainide	3-7 mg/kgxd (max. 200 mg)
Propafenone	10 mg/kgxd (max. 450 mg/d)
dl-Sotalol	2-6 mg/kgxd (max 320 mg/d)
Verapamil*	4-10 mg/kgxd
Digoxin*	0.2 mg/m ² xd

(* do not use in preexcitation syndrome)

bradycardia-induced tachycardias (long QT syndrome), surgical procedures (left cardiac sympathetic denervation)

- Catheter ablation of the tachycardia substrate
- Surgical ablation during open heart surgery (Fontan conversion)

Choice of antiarrhythmic agent for tachycardia prevention is mainly dependent on experience of the responsible pediatric cardiologist.

Indications for treatment in specific arrhythmias

Supraventricular tachycardias

- Supraventricular tachycardias due to accessory atrioventricular pathways (including preexcitation syndrome)
 - Accessory atrioventricular pathways most common substrate for supraventricular tachycardia in childhood
 - In patients with antegrade conduction during sinus rhythm (delta wave) and supraventricular tachycardia – Wolff-Parkinson-White syndrome
 - Various types of supraventricular tachycardia: orthodromic (most often), antidromic, orthodromic with bundle-branch block pattern, tachycardia via Mahaim fiber, atrial fibrillation/atrial flutter with rapid conduction via accessory pathway (risk of sudden cardiac death; no digoxin and verapamil, as both may promote conduction via pathway)
 - Newborns and infants: high chance of tachycardia recurrence but high probability of tachycardia cessation >12 months – antiarrhythmic treatment
 - In children >5 years with recurrent supraventricular tachycardia: low chance of spontaneous tachycardia cessation – catheter ablation of accessory pathway treatment of choice (success >95%, complication rate <1%: see Table 2)
 - In asymptomatic children >8 years and persistent delta wave (exercise testing): electrophysiologic testing for assessment of anterograde refractory period of accessory atrioventricular pathway; if <240 msec or multiple pathways and low risk of AV block – ablation recommended
 - In pediatric patients with supraventricular tachycardia and congenital heart defect: catheter ablation prior to surgery if vascular or cardiac access will be impaired postoperatively
 - For further details concerning antiarrhythmic treatment and catheter ablation in specific situations see Table 2
- AV nodal reentrant tachycardia

Second most common form of supraventricular tachycardia in childhood, based on dual AV nodal physiology – fast conducting pathway with long refractory period and slow conducting pathway with short refractory period. Symptoms include palpitations and dizziness, no increased risk of sudden cardiac death.

 - For acute termination of AVNRT and pharmacological treatment see Tables 1a and b; there are no restrictions concerning use of digoxin
 - For further details concerning antiarrhythmic treatment and catheter ablation in specific situations see Table 2.
 - Treatment is based on symptoms and individual psychological stress
 - Aim of electrophysiological study and radiofrequency current application: selective ablation or modulation of the slow conducting pathway; success rate in children and adolescents >96%, risk of AV block <2%, recurrence rate between 3 and 5%
 - Cryoablation/modulation of slow conducting pathway: almost no risk of AV block, but higher recurrences rates compared to radiofrequency current
- Permanent form of junctional reciprocating tachycardia (PJRT) and focal atrial tachycardia (FAT)
 - PJRT
 - Anatomical substrate: accessory atrioventricular pathway with solely retrograde conduction and

Table 2. Indications for treatment of SVT due to a accessory pathway or AVNRT

Indication	Pharmacological treatment Class	Catheter ablation
I	Recurrent sustained symptomatic SVT	- WPW after aborted sudden cardiac death - Syncope with short (≤ 240 ms) antegrade effective refractory period of the accessory pathway or short preexcited RR interval (≤ 240 ms) during atrial fibrillation - Recurrent SVT, refractory to drug therapy, decreased LV function or impaired hemodynamics
IIa	Recurrent non-sustained SVT, CHD	- SVT, body weight >15 kg, as an alternative to an effective pharmacological therapy - SVT, intended heart surgery, limited postoperative approach to the heart (i.e. prior Glenn or Fontan) - Delta wave, no SVT/no symptoms, delta wave persistence during exercise and on Holter-ECG, age >8 years, antegrade effective refractory period of the accessory pathway ≤ 240 ms or multiple accessory pathways, low risk of AV block - WPW, LV dysfunction
IIb	None	Non-sustained SVT, no CHD
III	Loss of delta wave during exercise Non-sustained SVT, no CHD	- Loss of delta wave during exercise, normal LV-function - SVT, effective pharmacological therapy, body weight ≤ 15 kg

CHD = congenital heart defect

decremental conduction properties, mainly located within the right posteroseptal space

- ECG: long RP tachycardia, deep negative P waves in ECG leads II, III, and aVF, isoelectric or positive in lead I
- Adenosine response: termination (by both antegrade or retrograde block) with instant reinitiation

- FAT

- Atrial focus with enhanced automaticity and centrifugal spread of activation
- ECG: P wave morphology “non-sinus” according to focus location; primary atrial tachycardia with “warm-up” and “cool-down” at spontaneous start and termination
- Both tachycardias usually present with chronically elevated heart rates on Holter monitors (particularly during night time), which may result in tachycardia-induced cardiomyopathy and heart failure; sometimes paroxysmal or frequently recurrent tachycardias
- Adenosine response: induction of AV block with persistence of FAT (termination rare)

Table 3. Indications for treatment of permanent junctional reciprocating tachycardia (PJRT) and focal atrial tachycardia (FAT)

Indication	Pharmacological therapy class	Catheter ablation
I	Chronic-permanent and paroxysmal PJRT/FAT, increased mean heart rate, LV-SF <30%	- Chronic-permanent PJRT/FAT, LV-SF <30%, body weight >15 kg - PJRT/FAT, refractory to pharmacological therapy or side effects, LV-SF <30%
IIa	None	- Chronic-permanent, body weight >15 kg, LV-SF >30%, as alternative to effective drug therapy - Intended heart surgery, limited postoperative access to the heart
IIB	Chronic-permanent PJRT/FAT, increased mean heart rate, LV-SF >30%	
III	Infrequent episodes of PJRT/FAT, LV-SF >30%, no symptoms	Body weight < 15 kg, as an alternative to drug therapy, infrequent episodes, normal heart rate, normal LV-SF

LV-SF = left ventricular shortening fraction

For further details concerning antiarrhythmic treatment and catheter ablation in specific situations see Table 3.

As congenital junctional ectopic tachycardia and multifocal atrial tachycardia are rare forms of supraventricular tachycardia in childhood, no structured recommendation can be offered.

- Postoperative junctional ectopic tachycardia
Occasional tachycardia immediately/early after surgical procedures for various types of congenital heart defects (TOF repair, VSD closure, AVSD repair) in infants and small children
 - Temporary tachycardia due to mechanical irritation of AV node and/or His bundle, lasting from several hours up to 3 days
 - High mortality if uncontrolled due to high ventricular rates (>230/min) and atrioventricular dissociation during early postoperative period
 - Aim of therapy: hemodynamic stabilization by slowing of ventricular rate – pharmacological therapy (i. e. amiodarone infusion), moderate body surface hypothermia (33–35 °C), and restoration of AV synchrony by cardiac pacing via epicardial wires until spontaneous cessation of tachycardia
 - There is virtually no indication for His bundle ablation
- Intraatrial reentrant tachycardias (IART) including atrial flutter
 - Atrial tachycardias: caused by a reentry mechanism around surgical and/or anatomic scars within the atria independent from the AV node and ventricular myocardium; in children after surgical repair of congenital heart disease associated with increased late morbidity and mortality
 - Atrial flutter: special form of IART with typical ECG pattern of 'sawtooth' P waves in inferior leads; right atrial macro-reentrant tachycardia with critical isthmus between

Table 4. Indications for treatment of intraatrial reentrant tachycardia (IART) and atrial flutter (AFL) in congenital heart defects (pre- and postoperative)

Indication	Pharmacological therapy class	Catheter ablation
I	Recurrent, sustained IART/AFL	- IART/AFL with haemodynamic compromise - Side effect during pharmacological therapy - Recurrent IART/AFL, refractory to pharmacological therapy
IIa	None	- Rare recurrences during pharmacological therapy
IIB	None	
III	Non-sustained tachycardia, rare occurrence	- Non-sustained tachycardia, rare occurrence

tricuspid valve and inferior vena cava; may be idiopathic or occur after surgical repair of congenital heart disease

- Isolated neonatal atrial flutter: low risk of recurrence following electrical cardioversion (transesophageal or external), no medical treatment necessary
- Adenosine does not terminate IART or atrial flutter, but may unmask P waves by induction of short lasting high grade AV block and aides in establishing the diagnosis
- Treatment:
 - Ongoing IART with hemodynamic compromise – external cardioversion with 0.5–1 J/kg (Figure 1); in stable hemodynamics – overdrive pacing (transesophageal, endocardial, external); pharmacological intervention not recommended because of low success rate and potential side effects
 - In IART with stable hemodynamics lasting >48 hrs: transesophageal echocardiography to exclude intracardiac thrombi prior to cardioversion; in case of thrombus formation: anticoagulation for 4–6 weeks, slowing of AV nodal conduction with β -blockers and digoxin
 - After successful cardioversion of IART: antiarrhythmic treatment required, but often not efficient (Table 1b)
 - Catheter ablation of IART requires identification of the critical isthmus of the IART in the individual post-operative anatomy; modern 3D mapping systems have significantly improved success; recurrence rates not insignificant
 - In patients with IART recurrence after catheter ablation continuation of antiarrhythmic treatment reduces arrhythmia burden significantly (for detailed indications see Table 4).
 - In selected patients after failed catheter ablation and particularly in patients with IART who need re-operation for hemodynamic reasons – intraoperative surgical ablation +/- pacemaker implantation highly successful
 - Catheter ablation can treat atrial flutter with a success rate of >90% and is recommended as preferred treatment
- Atrial fibrillation
Atrial fibrillation is a left atrial tachyarrhythmia with atrial rates of 400–600/min resulting in an irregular ventricular rhythm.
 - High risk of intracardiac thrombi with consecutive thromboembolic events
 - Most common symptomatic arrhythmia in adults

- Extremely rare in children and adolescents
- Diagnostic and therapeutic procedures similar to IART/atrial flutter (see above)

Ventricular tachycardias

Definition:

- ≥ 3 consecutive depolarisations from any region below the His bundle with a rate $>20\%$ higher than the immediately preceding intrinsic rate
- QRS morphology different from baseline rhythm; BBB or wide QRS complex
- Atrio-ventricular dissociation or 1:1 retrograde conduction

Types:

- Monomorphic – single QRS morphology
- Polymorphic – two or more QRS morphologies
- Non-sustained – duration ≤ 30 seconds, not leading to cardiovascular collapse
- Sustained – duration >30 seconds or leading to cardiovascular collapse

Mechanisms:

- Reentry
- Abnormal automaticity
- Triggered activity

Ventricular fibrillation is caused by completely chaotic ventricular activation. The ECG shows fine or coarse fibrillation waves.

Acute therapy

Dependent from the clinical situation:

- Synchronized cardioversion or defibrillation (Figure 1; Tables 5 and 6)
- Overdrive pacing in case of available pacing wires or electrodes
- Primary use of intravenous antiarrhythmic drugs should be avoided because of potential serious side effects
- Adenosine may be used for differential diagnosis in wide QRS complex tachycardia (SVT, VT with adenosine-sensitive mechanism)

Long-term treatment

See Table 6 and DGP guideline for ICD implantation: http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/14%20LL%20Tachykardie%20Herzrhythmusstoerungen%20Indikationen%20zur%20ICD%20TherapieAS.pdf

Specific types of ventricular tachycardias:

Idiopathic ventricular tachycardia

Characteristics:

- Monomorphic ventricular tachycardia in the absence of structural heart disease, cardiomyopathy or an inherited primary arrhythmia syndrome
- Low risk of sudden cardiac death
- Risk of tachycardia-induced cardiomyopathy in case of an incessant or persistent tachycardia

For further details see Table 7.

Table 5. Recommendations on energy use for external cardioversion/defibrillation

	Energy [J/kg]	
	First shock	If not effective
Cardioversion (synchronized)	0.5-1	2
Defibrillation	2-4	4

Table 6. Indication for treatment of ventricular tachycardias (VT)

Indication	Pharmacological therapy class	Catheter ablation
I	- Recurrent, idiopathic VT - s/p ICD-implantation, recurrent discharges - β -blocker for LQTS and CPVT - Mexiletine for LQT 3	- Sustained VT, haemodynamic compromise, substrate accessible for ablation - s/p ICD-implantation, recurrent discharges
IIa	- Quinidine for SQTS and BRS - Flecainide for CPVT	- Trigger ablation for idiopathic VF
IIB	None	
III	- Idiopathic PVC/non-sustained idiopathic VT w/o hemodynamic compromise - VT, CHD without ICD	- Idiopathic PVS without hemodynamic compromise - Non-sustained, asymptomatic idiopathic VT - VT in channelopathies and CPVT

BRS = Brugada syndrome; CHD = congenital heart defect; CPVT = catecholaminergic polymorphic ventricular tachycardia; ICD = implantable cardioverter defibrillator; LQTS = Long-QT syndrome; PVS = premature ventricular contraction; SQTS = Short-QT syndrome; VF = ventricular fibrillation; VT = ventricular tachycardia

Inherited primary arrhythmia syndromes

Caused by gene mutations altering the ion flows inside the cardiac cell or between the cell and the extracellular space leading to life-threatening ventricular arrhythmias (polymorphic ventricular tachycardia and ventricular fibrillation). Major symptoms are palpitations, syncope, cardiac arrest and sudden cardiac death. ICD therapy is used for primary and secondary prevention of sudden cardiac death.

Congenital long QT syndrome (LQTS)

Etiology: Mostly autosomal dominant mutations of genes coding the proteins of potassium or sodium channels of the cardiac cell leading to prolongation of action potential duration and life-threatening ventricular arrhythmias caused by triggered activity.

Diagnosis:

- Phenotypic: modified Schwartz criteria, major criterion: length of corrected QT (QTc) interval (for normal limits see Table 8)

Table 7. Typical types of idiopathic ventricular tachycardia in childhood

Type of tachycardia	ECG during SR/VT	Acute therapy	Permanent drug therapy	Clinical Significance
Idiopathic RVOT-VT (triggered activity)	SR: normal VT: inferior axis, LBBB, monomorphic	Adenosine	β -blocker (RF ablation)	VT non-sustained, rare symptoms
Idiopathic LV-VT (reentry)	SR: normal VT: superior axis, RBBB, monomorphic	Verapamil	β -blocker Calcium channel blocker (RF-ablation)	Often symptomatic
Differential diagnosis: Idioventricular rhythm	SR: normal slow "VT", <120% of normal rhythm	None	None	Benign

(LBBB = left bundle branch block, LV = left ventricular, RBBB = right bundle branch block, RF = radiofrequency current, RVOT = right ventricular outflow tract, SR = sinus rhythm, VT = ventricular tachycardia)

Table 8. QTc limits according to age and gender (Bazett formula)

	1–15 years	Boys >15 years	Girls >15 years
Normal	<440	<430	<450
Borderline	440–460	430–450	450–470
Prolonged	>460	>450	>470

- Genetic: search for specific gene mutations including family members at risk

Therapy:

- Acute treatment of life-threatening arrhythmias in LQTS: beta-blocking agents, magnesium, lidocaine i.v., ventricular pacing to increase baseline heart rate
- Long-term prophylaxis: beta-blocking agents, mexiletine (LQT type 3), left cardiac sympathetic denervation
- Life style modifications and avoidance of QT prolonging and adrenergic drugs (list available at www.crediblemeds.org)
- ICD therapy

Short QT Syndrome (SQTS)

Arrhythmia syndrome characterized by shortened action potential duration due to inherited ion channel disease with a QTc <340 ms; much less frequent than LQTS; may affect children from newborn age and may be associated with atrial fibrillation.

Therapy:

- Quinidine
- ICD therapy

Brugada Syndrom (BRS)

Caused mainly by sodium channel mutations. Characterized by ST elevations in right precordial leads which may be induced or pronounced by sodium blocking agents (ajmaline or flecainide challenge). Baseline ECG often without any obvious abnormalities

Therapy:

- Acute: isoprenaline i.v.
- Long-term prophylaxis: quinidine
- ICD therapy
- Life style modifications (aggressive fever treatment) and avoidance of Brugada drugs (list available at www.brugada-drugs.org)

Catecholaminergic polymorphic ventricular tachycardia (CPVT) Caused by mutations of genes affecting calcium cycling between the sarcoplasmic reticulum and cytosol of the cardiac cell. Leads to calcium overload of the cardiac cell initiating life-threatening ventricular arrhythmias.

Typical ECG signs and arrhythmias are exercise triggered:

- Polymorphic premature ventricular beats
- Focal or chaotic atrial tachycardia
- Bidirectional ventricular tachycardia
- Ventricular fibrillation

Therapy:

- Long-term prophylaxis: beta-blocking agents, flecainide, left cardiac sympathetic denervation
- ICD therapy
- Life style modifications and avoidance of adrenergic drugs

Cardiomyopathies

Arrhythmogenic right ventricular cardiomyopathy (ARVC)

Autosomal dominant disease causing changes in desmosomal proteins, leading to progressive right ventricular fibrosis with fatty infiltration and life-threatening ventricular arrhythmias. Left ventricular involvement may also be present.

Therapy:

- Live style modifications – avoidance of strenuous exercise
- Primary preventive ICD therapy in case of risk factors
- Catheter ablation for recurrent ventricular tachycardia

Hypertrophic obstructive (HOCM) or non-obstructive (HNCM) cardiomyopathy

Primary preventive ICD therapy is a class IIa indication in the presence of one or more risk factors (see DGPK guideline for ICD implantation: http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/14%20LL%20Tachykardie%20Herzrhythmusstoerungen%20Indikationen%20zur%20ICD%20TherapieAS.pdf).

Dilated cardiomyopathy (DCM)

For the indication of primary preventive ICD therapy see DGPK guideline for ICD implantation (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/14%20LL%20Tachykardie%20Herzrhythmusstoerungen%20Indikationen%20zur%20ICD%20TherapieAS.pdf). For cardiac resynchronization therapy see DGPK guideline for the treatment of chronic heart failure (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/LL_Chronische_Herzinsuffizienz_Nov_2015_Rickers.pdf).

Congenital heart disease

The main mechanisms for ventricular tachycardia are:

- Reentry (substrate is formed by surgical scars and/or anatomical conduction barriers)
- Abnormal automaticity

Therapy:

- Correction of hemodynamic residua (not protective unless tachycardia substrate is targeted)
- Surgical or catheter ablation
- ICD therapy (see DGPK guideline for ICD implantation (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/14%20LL%20Tachykardie%20Herzrhythmusstoerungen%20Indikationen%20zur%20ICD%20TherapieAS.pdf))
- Prophylactic use of antiarrhythmic drugs is not supported by available data

Recommendations for follow-up

Children with tachycardias should be seen by a pediatric cardiologist on a regular basis. Patients with incessant tachycardias should be followed in 3 to 6 months intervals, whereas young patients with pacemakers or ICD at least every 6 months.

Comments to level of evidence

Class I

Conditions for which there is evidence and/or general agreement that a given procedure or treatment plan is beneficial, useful, and effective.

Class II

Conditions for which there is conflicting evidence and/or divergence of opinion about the usefulness/efficacy of a procedure or treatment.

- Class IIa

Weight of evidence/opinion is in favor of usefulness/efficacy

- Class IIb

Usefulness/efficacy is less well established by evidence/opinion

Class III

Conditions for which there is general agreement that a procedure or treatment is not useful/effective.

Conflicts of interests for this guideline

Thomas Paul: none
 Roman Gebauer: none
 Thomas Kriebel: none
 Heike Schneider: none
 Jan Janousek: none

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Link to all references listed in the original German long version “Tachykardie Herzrhythmusstörungen”: www.kinderkardiologie.org/leitlinien/

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Bradycardias and cardiac pacing in children

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Abstract Common types of bradycardias in infants and children with and without congenital heart defects are summarized together with contemporary recommendations on diagnosis and treatment. Indications for treatment of bradycardias in children are mainly based on symptoms and functional status of the child rather than on critical lower heart rate limits alone. In addition age specific limits have to be considered.

Keywords: *bradycardias, cardiac pacing, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Bradycardias are defined as a temporary or permanent heart rate less than normal for age. Clinically relevant forms of bradycardias include sinus bradycardia, sinus node dysfunction including brady-tachy syndrome and high grade AV block (>2:1 AV conduction). Most important issue in bradycardias is distinction of functional slowing of heart rate from those forms caused by a disease of the specialized conduction system.

Clinical presentation

Typical symptoms related to bradycardia include reduced exercise tolerance, presyncope and syncope, congestive heart failure, as well as sudden cardiac death. Children with bradycardia may be completely asymptomatic.

Diagnostic work-up

Documentation of bradycardia on 12-lead ECG allowing diagnosis and proper treatment; assessment of frequency and duration of bradycardia episodes as well as bradycardia symptoms; imaging techniques to establish cardiac anatomy and function.

Electrocardiography

12-lead ECG with rhythm strip, Holter monitor, treadmill test, cardiac event recorder (external or implantable)

Imaging

Echocardiography, chest X-ray

In selected patients

Electrophysiological study (in patients with residual bifascicular block after transient postoperative complete AV block; see below)

Others

Neurological assessment, specific testing for metabolic disorders

Laboratory studies

SS-A and SS-B antibodies in newborns with congenital AV block (and their mothers!); thyroid hormones, borrelia antibodies

Differential diagnosis

Myocarditis in new onset bradycardia (see guideline DGPK: http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/21%20LL%20Myokarditis_20120824.pdf), blocked atrial bigemini, apnoea, seizures, influence of drugs.

Sophisticated work-up should be performed in specialized centres experienced in paediatric cardiology, electrophysiology and paediatric cardiac surgery.

Management

Children with symptomatic and/or life-threatening bradycardias need cardiac pacing. Medical therapy with isoproterenol, orciprenaline, epinephrine and ipatropiumbromide is only temporarily effective.

Specific situations

Congenital AV block

- Indications for permanent pacing are well defined (see below).
- Adams-Stokes attacks and heart failure may develop in children at any age
- Sudden cardiac death may be first symptomatic episode

Postoperative AV block

- Risk of AV block 1–3% after surgical procedures for congenital heart defects

- Spontaneous resolution of AV block may occur, usually within first 10 postoperative days with favourable prognosis
- Poor prognosis in non-paced patients with persistent AV block
- Residual bifascicular block after transient postoperative complete AV block – increased risk of recurrence of high grade AV block and sudden cardiac death

Sinus node dysfunction

- Mainly in patients after major cardiac surgery involving the atria
- Indication for permanent pacing in symptomatic patients after other causes have been ruled out
- In Mustard/Senning and Fontan patients with sinus node dysfunction cardiac pacing to restore chronotropic competence may prevent occurrence of atrial tachycardias/atrial flutter
- Cardiac pacing is required in patients with brady-tachy syndrome when treated with antiarrhythmic drugs; electrophysiological study and catheter ablation/surgical ablation are reasonable alternatives (see DGPK guideline: http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/13%20LL%20Tachykardie%20HerzrhythmusstoerungenAS.pdf)
- Indications for ICD therapy are summarized in a separate DGPK guideline (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/14%20LL%20Tachykardie%20Herzrhythmusstoerungen%20Indikationen%20zur%20ICD%20TherapieAS.pdf)

Bradycardia-induced tachyarrhythmias

Cardiac pacing in patients with long QT syndrome combined with β -blockers does not prevent sudden cardiac death without ICD back-up

Cardioinhibitory syncope

Permanent cardiac pacing should be avoided in pediatric patients unless unresponsive to any other treatment.

Indications for pacing therapy in children

Level of evidence

Class I

Conditions for which there is evidence and/or general agreement that a given procedure or treatment plan is beneficial, useful, and effective.

Class II

Conditions for which there is conflicting evidence and/or divergence of opinion about the usefulness/efficacy of a procedure or treatment.

- Class IIa: Weight of evidence/opinion is in favor of usefulness/efficacy
- Class IIb: Usefulness/efficacy is less well established by evidence/opinion

Class III

Conditions for which there is general agreement that a procedure or treatment is not useful/effective.

Class I

- Congenital high grade AV block with symptoms and/or progressive ventricular dilatation and/or reduced exercise tolerance;
- Congenital high grade AV block in asymptomatic patients with ventricular rate <55/min, asystole > threefold the cycle length of underlying rhythm, wide QRS complex escape rhythm, complex ventricular ectopy, ventricular dilatation/dysfunction, prolonged QTc interval
- Symptomatic sinus node dysfunction

- Postoperative/postinterventional high grade AV block lasting > 10 days

Class IIa

- Brady-tachy syndrome after surgery involving the atria for prevention of atrial tachyarrhythmias or with combined antiarrhythmic therapy
- Bradycardia in patients with complex congenital heart defects and ventricular rates <40/min or asystole >3 sec.
- Persistent bifascicular block after transient postoperative complete AV block

Class IIb

- Asymptomatic patients with congenital high degree or complete AV block
- Bradycardia in patients with ventricular rates <40/min. or asystole >3 sec.
- Patients with neuromuscular diseases and any form of AV block (including 1° AV block)

Leads and pacemakers

Indications for epicardial pacing

- Small body size (<15–20 kg)
- Congenital heart defect with right-to-left shunt
- Absence of transvenous access to target chamber
- Pacing of the left or systemic ventricle should be preferred

Endocardial leads and DDD pacemakers are recommended in older children. For cardiac resynchronisation therapy in children see guideline on chronic heart failure (http://www.kinderkardiologie.org/fileadmin/user_upload/Leitlinien/LL_Chronische_Herzinsuffizienz_Nov_2015_Rickers.pdf)

Recommendations for follow-up

Children with bradycardias should be seen by a pediatric cardiologist on a regular basis, while young patients with a pacemaker should be followed at least every 6 months.

Conflicts of interests for this guideline:

Thomas Paul: None

Wolfgang Ruschewski: None

Jan Janousek: None

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Link to all references listed in the original German long version “Bradykarde Herzrhythmusstörungen, Schrittmacher”: www.kinderkardiologie.org/leitlinien/

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Indications for ICD therapy in ventricular tachyarrhythmias

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Abstract In children and adolescents life-threatening ventricular arrhythmias may occur in three situations: 1. in the presence of inherited primary arrhythmia syndromes, 2. in cardiomyopathies, and 3. in patients with congenital heart defects pre- and postoperatively. Implantable cardioverter-defibrillator (ICD) therapy is indicated for either primary or secondary prevention of sudden arrhythmic death.

ICD therapy in children covers specific technical challenges due to patient's size and anatomy. Accordingly, a non-transvenous approach is common in infants and young children. Complications are more frequent than in the adult population and are mainly lead-related and/or caused by inappropriate shocks. Psychological problems may necessitate respective support.

Keywords: ICD, ventricular tachyarrhythmias, children, guide line, congenital heart disease

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Ventricular tachyarrhythmia is a non-sustained (<30 sec) or sustained rhythm disturbance arising from the ventricles associated with a temporary or permanent high heart rate. The following types can be differentiated:

- Monomorphic ventricular tachycardia
- Polymorphic ventricular tachycardia
- Ventricular fibrillation

ICD therapy is indicated in patients with a life-threatening ventricular tachyarrhythmia for prevention of sudden cardiac death.

Etiology

In children life-threatening ventricular arrhythmias occur mainly in the following settings:

- Inherited primary arrhythmia syndromes
- Cardiomyopathies
- Congenital heart defects pre- and postoperatively

Clinical presentation

Main symptoms include:

- Palpitations
- Syncope
- Cardiac arrest

Diagnostic work-up

ECG documentation of baseline and pathologic rhythms (including search for electrocardiographic markers of primary arrhythmia syndromes and cardiomyopathies), imaging, hemodynamic evaluation and genetics:

- 12-lead ECG including drug provocation tests and signal-averaged electrocardiogram, treadmill test, Holter monitoring, event recorder
- Echocardiography
- Chest X-ray, cardiac magnetic resonance imaging and CT scans
- Cardiac catheterization, endomyocardial biopsy, intracardiac electrophysiological study
- Genetics and search for metabolic disease if suspected

Sophisticated work-up should be performed in specialized centres experienced in paediatric cardiology, electrophysiology and paediatric cardiac surgery.

Indications for treatment

ICD therapy is indicated for either primary or secondary prevention of sudden arrhythmic death. Primary preventive ICD implantation is a class IIa or IIb indication and follows mainly accepted risk stratification algorithms concerning risk of life-threatening ventricular arrhythmias in adults. Class I indications for secondary preventive implantation include survival of cardiac arrest due to ventricular arrhythmia or spontaneous sustained ventricular tachycardia if no causal treatment can be offered.

Level of evidence

Class I

Conditions for which there is evidence and/or general agreement that a given procedure or treatment plan is beneficial, useful, and effective.

Class II

Conditions for which there is conflicting evidence and/or divergence of opinion about the usefulness/efficacy of a procedure or treatment.

- Class IIa: Weight of evidence/opinion is in favor of usefulness/efficacy
- Class IIb: Usefulness/efficacy is less well established by evidence/opinion

Class III

Conditions for which there is general agreement that a procedure or treatment is not useful/effective.

- Class I:
 1. Cardiac arrest due to ventricular tachycardia or fibrillation if reversible causes have been excluded

2. Spontaneous symptomatic sustained ventricular tachycardia in patients with congenital heart defects; in selected patients catheter ablation may avoid ICD implantation
3. Cardiac syncope in the presence of congenital heart disease and inducible sustained hemodynamically unstable ventricular tachycardia not treatable by catheter ablation or surgery
4. Sustained ventricular tachycardia or syncope refractory to other therapy in the presence of an inherited primary arrhythmia syndrome

- Class IIa:

1. Patients with congenital heart disease and recurrent syncope of unknown origin in the presence of either systemic ventricular dysfunction or inducible ventricular tachyarrhythmia
2. Patients with syncope of unknown origin and non-ischemic dilated cardiomyopathy with significant systemic ventricular dysfunction (ejection fraction $\leq 35\%$)
3. Patients with hypertrophic cardiomyopathy in the presence of one or more risk factors for sudden cardiac death (family history of sudden cardiac death, spontaneous non-sustained ventricular tachycardia, syncope, left ventricular wall thickness ≥ 30 mm and paradoxical blood pressure response during exercise)
4. Patients with arrhythmogenic right ventricular cardiomyopathy and one or more risk factors for sudden cardiac death (ventricular tachycardia inducibility, spontaneous non-sustained ventricular tachycardia, male gender, severe right ventricular dilatation, age at presentation < 5 years, left ventricular involvement, syncope, apical right ventricular aneurysm in association with a genetic locus on chromosome 1q42-43)
5. Patients with congenital long QT syndrome and syncope or ventricular tachycardia despite beta-blocker therapy
6. Ambulatory heart transplant candidates carrying the risk of sudden cardiac death (as bridge to heart transplantation)
7. Patients with Brugada syndrome and syncope and/or documented ventricular tachycardia without cardiac arrest
8. Patients with catecholaminergic ventricular tachycardia and syncope or documented sustained ventricular tachycardia despite beta-blocker therapy
9. Patients with cardiac sarcoidosis, giant cell myocarditis or Chagas disease

- Class IIb:

1. Patients with congenital heart disease and recurrent syncope of unknown origin despite comprehensive non-invasive and invasive diagnostic work-up
2. Asymptomatic patients with congenital long QT syndrome and risk factors for sudden cardiac death (e.g. QTc > 500 ms in males)
3. Familial cardiomyopathy associated with sudden cardiac death (known family history)
4. Patients with left ventricular non-compaction cardiomyopathy

- Class III:

1. Patients with permanent ventricular tachycardia or fibrillation
2. Patients with a significant psychiatric disease potentially aggravated by ICD implantation and/or preventing proper follow-up

3. Patients with drug-refractory heart failure (New York Heart Association class IV) who are no candidates for heart transplantation or cardiac resynchronization therapy
4. Patients with syncope of unclear origin without inducible ventricular tachyarrhythmia in the absence of structural heart disease
5. Ventricular tachycardia or fibrillation amenable to catheter or surgical ablation
6. Patient with ventricular tachyarrhythmias from a reversible cause in the absence of structural heart disease

Management

Choice of ICD system depends on individual hemodynamics and/or anti-bradycardia pacing indication:

- Single-chamber, dual-chamber or CRTD device Implantation technique depends on patient's size and cardiovascular anatomy:
- Endocardial, epicardial, pericardial, pleural, subcutaneous or combined.

Prognosis

While ICD system efficacy is high, the following complications may occur:

- Lead failure/fracture
- Inappropriate shocks
- Infection
- Changes in sensing and defibrillation threshold (mainly in pleural and subcutaneous arrays)
- Cardiac perforation (endocardial leads)
- Heart strangulation (epicardial and pericardial leads)
- Psychological problems.

Recommendations for follow-up

Children with an ICD should be followed in a specialized pace-maker/ICD outpatient clinic with specific focus on:

- Stored tachycardia episodes
- Growth related lead problems and shock vector changes
- Changes in R wave amplitude
- T wave oversensing
- Regular defibrillation threshold testing in devices likely to be affected by growth (e.g. pleural and subcutaneous arrays)
- Heart strangulation (epicardial or pericardial leads)
- Psychological support

Follow-up intervals should not exceed 6 months. Remote follow-up may replace a part of in-office visits and is desired. Participation in competitive sports is in general discouraged but may be discussed on an individual basis. Driving recommendations and choice of occupation follow current adult ICD guidelines.

Conflict of interests for this guideline

Jan Janoušek: None
Wolfgang Ruschewski: None
Thomas Paul: None

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Link to all references listed in the original German long version "ICD-Therapie, tachykarde Herzrhythmusstörungen": www.kinderkardiologie.org/leitlinien/

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Arterial Hypertension

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Abstract Arterial hypertension in infancy and early childhood is a typical symptom of renal, heart or endocrine disease. In later childhood or adolescence secondary hypertension is rare, usually only a

positive family history and/or obesity are present. Therefore this guideline is not only a diagnostic and therapeutic recommendation how to manage patients with arterial hypertension, it should also trigger primary prevention, that is regular blood pressure measurements from three years onwards, as well as a healthy life style.

Ambulatory blood pressure measurements are advocated to exclude white-coat or masked hypertension. The younger, the slimmer and the more hypertensive the patients are, the more extensively a secondary hypertension has to be ruled out. A basic diagnostic program is provided for every hypertensive patient. It has to be individually extended according to findings.

Secondary forms of hypertension have to be treated according to the underlying disease. Otherwise, a high-normal or grade 1 hypertensive patient should be counselled for a healthy life style. If hypertension persists for six months, drug treatment has to be initiated with ACE inhibitors, AT₁ receptor antagonists, calcium channel blockers (amlodipin) or beta blockers, in combinations also with diuretics (furosemide, torasemide). Normotension should be achieved, in patients with renal failure even lower values are recommended.

Long-term surveillance is needed for blood pressure and to evaluate for end organ damage (microalbuminuria, renal failure, myocardial hypertrophy, retinal fundus hypertonicus) or adjunct disease (diabetes mellitus).

Keywords: Arterial Hypertension, children, guideline

The purpose and the mode of generation of these guidelines are described in the Introduction of this supplement.

Definition

Arterial hypertension in childhood and adolescence is classified according to percentiles (table 1). For manual blood pressure (BP) the US norms (2002, online supplement table 1) are recommended, for automated oscillatory BP the German KIGGS data (Neuhauser 2011, online supplement table 2) and for the ambulatory BP the German multicenter study data (Wühl 2002, online supplement table 3).

Relationships to consider

Risk factors for arterial hypertension

- obesity
- prematurity, intrauterine growth retardation, perinatal asphyxia
- intracranial pressure increase

Table 1. Definition and classification of arterial hypertension. If an adolescent surmounts the thresholds for adults, the lower thresholds for adults should be used.

	children and adolescents ≥ 3 single measurements or 24-h-measurement systolic or diastolic	adults ≥ 3 auscultatory measurements systolic diastolic	adults 24-h-measurement (daytime) systolic diastolic	adults 24-h-measurement (nighttime) systolic diastolic
optimal	not jet defined	<120 and <80	not jet defined	not jet defined
normal	<90 th percentile	<130 and <85	<130 and <80	not jet defined
high normal	P90–P94	130–139 or 85–89	130–134 or 80–84	<120 and <70
hypertension ¹	P95 - (P99 + 5 mmHg)	140–159 or 90–99	≥ 135 or ≥ 85	≥ 120 or ≥ 70
hypertension ²	$> P99 + 5$ mmHg	160–179 or 100–109	not jet defined	not jet defined
hypertension ³	not jet defined	≥ 180 or ≥ 110	not jet defined	not jet defined

- cardiovascular diseases (coarctation, vascular inflammatory disease, haemangioma, fistula)
- renal disease
- after solid organ transplantation
- systemic diseases (diabetes, tuberous sclerosis, neurofibromatosis, Turner's syndrome, Williams Beuren syndrome, ...)
- drugs: steroids, nonsteroidal anti-inflammatory drugs, beta mimetics, calciumneurin inhibitors (cyclosporine, tacrolimus), erythropoetin, central stimulants (methylphenidate, atomexin), tricyclic antidepressant, antipsychotics, laxatives, anovulants
- excessive glycyrrhizin consumption (liquorice)
- drug addiction (cocaine, ecstasy, amphetamines, ...)
- sleep apnea

Associated lesions (acute end organ damage)

- left ventricular hypertrophy, heart failure
- microalbuminuria, proteinuria, renal failure
- hypertensive encephalopathy
- retinal changes, blindness

Long-term lesions

- atherosclerosis (coronary artery disease, ischaemic stroke, peripheral artery disease)
- aortic aneurysm, aortic dilatation
- endothelial dysfunction
- hypertensive heart failure (with preserved ejection fraction, HFpEF)
- renal failure, hypertensive nephropathy

Risk factors for cardiovascular disease

- arterial hypertension
- hyperlipidaemia
- smoking
- obesity
- diabetes mellitus
- renal disease
- sedentary life style
- male
- genetic factors
- prenatal factors (prematurity, small for gestational age, gestational hypertension)
- psychosocial factors (low income, isolation, stress, depression, anxiety, hostility, introversion)
- sleep apnea

Differential diagnosis (secondary hypertension)

- renal disease (renoparenchymal or renovascular)
- endocrinological disease
- oncologic disease (hypercalcaemia)
- sleep apnoea syndrome

Clinical presentation

In general, arterial hypertension has no symptoms. However, it can cause headache, dizziness, nose bleeding, attention or concentration deficits, or tinnitus.

Diagnostic work-up

Diagnostic evaluation aims to detect secondary forms of hypertension, risk factors for hypertension, signs of end organ damage, signs of long-term lesions including cardiovascular disease, and other risk factor for cardiovascular disease.

Basic evaluation must include:

- **History:** family history, perinatal history, medical and illegal drugs, smoking, food consumption, dietary supplements, activity profile, sleep history
- **Physical examination:** height, body mass, syndrome-typical signs, peripheral pulses, blood pressure, abdominal masses, cutaneous signs of secondary forms of hypertension, neurologic examination
- **Ambulatory blood pressure measurement:** exclusion of masked or white coat hypertension
- **Blood tests:** blood cells, creatinine, urea, sodium, potassium, calcium, thyroid stimulating hormone, triglycerides, cholesterol (LDL and HDL), glucose
- **Urine tests:** red and white blood cells, protein, glucose, microalbumin
- **Abdominal sonography:** kidneys, urinary tract, renal arterial Doppler, abdominal masses
- **Echocardiography:** left ventricular hypertrophy, exclusion of coarctation
- **Eye examination:** fundus hypertonicus, exclusion of retinal hamartoma
- For young, slim or severely hypertensive patients this basic diagnostic program has to be extended individually with detailed nephrologic and endocrinologic tests (metanephries, cortisol, renin, aldosterone, ...), as well as imaging (renal angiography, cerebral imaging) and sleep diagnostic procedures.

Management

The management algorithm is outlined in figure 1. After confirmation of the elevated office BP by ambulatory BP measurement, the basic and probably some advanced diagnostic programs have to exclude secondary hypertension that has to be treated specifically according to the underlying disease.

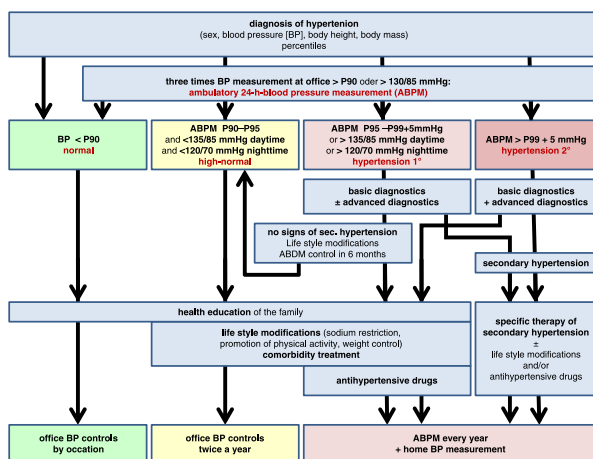


Figure 1.
Management algorithm for arterial hypertension in children and adolescents

Life style interventions

For children and adolescents with hypertension grade 1 and without secondary form of hypertension, sole life style interventions for the first 6 months are warranted. These include:

- weight reduction
- smoking cessation
- physical activity
- high potassium and low sodium diet (caveat renal failure!)
- DASH diet (dietary approach to stop hypertension): low meat, low fat, low cholesterol, low saturated fat, little soft drinks, low sodium, high protein, high fish, high poultry, high whole wheat, high fiber, high potassium, high calcium, high magnesium.

Pharmacologic treatment

If life style modifications fail to achieve normal or high-normal values within six months, drug treatment has to be initiated in addition. Antihypertensive treatment is similar in children and adults, except for beta-adrenergic receptor antagonists that have a higher value in children and should be considered as one of the first line therapies. Otherwise diuretics have less importance as a single first-line drug.

- Licenced drugs with paediatric dosages (cave: age limits for certain drugs) are:
 - First choice (*US-only, + EU-only):
 - angiotensin converting enzyme inhibitor (ACEI: benazepril*, captopril, enalapril, fosinopril*, lisinopril, quinapril*)
 - angiotensin II type 1 receptor antagonist (ARB: candesartan, irbesartan*, losartan, olmesartan*, valsartan)
 - calcium channel antagonist (CCB: amlodipine, felodipine*, isradipine*, extended release nifedipine*)
 - beta-adrenergic receptor antagonist (BB: atenolol*, labetalol*, metoprololsuccinate)
 - Second choice/combination partners
 - diuretics (amiloride*, chlorthalidone*furosemide/frusemide, hydrochlorothiazide*, spironolactone*, torasemide/torsemide +, triamterene*)
 - alpha-1-adrenergic receptor antagonist (doxazosin*, prazosine, terazosin*)
 - central alpha-2-adrenergic receptor agonist (clonidine)
 - vasodilator (hydralazine*, minoxidil)

If single drug therapy fails to lower BP adequately, combination therapy might be necessary. All first choice drug groups can be combined with each other and diuretics, except ACEI and ARB (and probably potassium sparing diuretics). Two substances from the same group should also be omitted except for diuretics, where potassium sparing diuretics (amiloride, spironolactone, triamterene) should *only* be given with potassium wasting ones (chlorthalidone, furosemide/frusemide, hydrochlorothiazide, torasemide/torsemide).

A frequently used strategy is to start with a long acting ACEI (enalapril, lisinopril) and if necessary to combine with CCB (amlodipine), BB (metoprololsuccinate) and/or diuretics.

For certain indications fixed first line drugs are recommended:

- renal failure: ACEI or ARB (prevent proteinuria) with/-out diuretics (oedema treatment)
- obesity, metabolic syndrome, diabetes mellitus: ACEI or ARB (considered metabolically positive) and CCB (neutral); BB and thiazides should be omitted (metabolically negative)
- heart failure: ACEI or ARB in combination with BB with/-out diuretics

- Aortic coarctation: consider surgery or intervention. If after surgery no restenosis can be found, no certain drug group is preferred
- Migraine: BB and/or CCB
- obstruction of the upper airways: treatment of the underlying causes

Recommendations for follow-up

Blood pressure should be lowered below the 90th percentile. In certain conditions (renal failure) lower values should be achieved (e.g. below 75th percentile in non-proteinuric, below the 50th percentile in proteinuric kidney disease). If the BP cannot be lowered adequately with single drug treatment, increase of dosage and/or combination therapy is necessary.

Patients on antihypertensive treatment should be taught to record their BP themselves at home. Nevertheless, annual ambulatory BP measurements and evaluation for end organ damage (blood test for creatinine, urea; urine test for microalbumin, protein, glucose red and white blood cells; eye examination for fundus hypertonicus; echocardiography for left ventricular hypertrophy) is recommended. Medical surveillance is life-long.

Conflict of interests for this guideline for this guideline

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References (DGPK): www.kinderkardiologie.org/leitlinien/

Supplementary materials

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Hypertensive emergency

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Abstract The hypertensive crisis (very high blood pressure values) can present as a hypertensive urgency (without end organ damage) or

hypertensive emergency (with end organ damage). Patients with hypertensive urgency have to be evaluated according to the guidelines for arterial hypertension and oral drug treatment should be initiated urgently. Patients with hypertensive emergency have to be admitted to a paediatric intensive care unit, vital signs including blood pressure must be monitored continuously. Diagnostic evaluation to define the underlying disease and drug treatment to lower blood pressure must be initiated immediately and in parallel. Antihypertensive drugs should be given as a continuous i.v. infusion and titrated to achieve a blood pressure reduction of 15–30% within the first 6–8 hours. Internationally, labetalol, nicardipine and sodium-nitroprussid are the most evaluated drugs. If locally not available off-label nifedipine or off-label urapidil can be considered.

Keywords: *arterial hypertension, hypertensive emergency, children, guideline*

The purpose and the mode of generation of these guidelines are described in the Introduction of this supplement.

Definition

The **hypertensive crisis** is a sudden increase in blood pressure. The threshold is depending on the underlying condition. A hypertensive crisis can present as hypertensive urgency or hypertensive emergency.

A **hypertensive emergency** is a hypertensive crisis with existing or threatening organ damage. Involved end organs can be the central nervous system, kidneys and cardiovascular system. Therefore, the definition of a hypertensive emergency depends less on the absolute level of the blood pressure, than on the clinical condition, like coexisting encephalopathy, haemolytic uraemic syndrome, status post renal or heart transplantation, aortic aneurysm or dissection and pregnancy.

A **hypertensive urgency** is a hypertensive crisis without end organ damage or threat for end organs. It should be managed rapidly according the guidelines for arterial hypertension. Secondary forms of hypertension are very likely and must be evaluated thoroughly. Oral drug treatment has to be started or enhanced after diagnostic (re-) evaluation.

Relationships to consider

In childhood, a hypertensive emergency is rarely seen in primary hypertension. It can present with many forms of secondary hypertension. Typical conditions are

- haemolytic uremic syndrome
- acute renal failure
- exacerbation of congenital kidney disease
- renovascular disease
- endocrine tumor (phaeochromocytoma, Conn's disease, Cushing's disease, ...)
- (iatrogenic) volume overload
- intoxication, drug abuse
- pregnancy, eclampsia
- stroke
- elevated intracranial pressure
- paradoxical hypertension shortly after coarctation repair.

Involved end organs:

- central nervous system
- kidneys
- heart

Clinical presentation

In general, arterial hypertension has no symptoms. Typical signs of a hypertensive emergency are:

- Stroke
- Loss of consciousness
- Memory loss
- disorientation
- Damage to the eyes (visual impairment, blindness)
- Loss of kidney function
- Angina (unstable chest pain)
- Pulmonary edema
- Aortic dissection
- Eclampsia

Diagnostic work-up

Diagnostic evaluation aims to detect rapidly the most probably secondary forms of hypertension similar to the basic and extended diagnostic work-up described in the guideline for arterial hypertension.

Management

Patients with a hypertensive emergency have to be admitted on a paediatric intensive care unit. In addition to the diagnostic procedures, continuous surveillance of vital signs and blood pressure has to be established.

In parallel to these measures, antihypertensive drug treatment has to be started. A continuous intravenous drug application is necessary to titrate the drugs according to the blood pressure. Preferred substances are:

- labetalol
- nitrendipine
- sodium-nitroprusside (cave: cyanide intoxication)

If these drugs are not licensed or are not available, off-label use of nifedipine (cave: alcoholic solution) or off-label use of urapidil has to be considered.

Glyceroltrinitrate, a common drug for hypertensive emergency in adults is not effective in children.

Antihypertensive bolus injections or fast-acting buccal solutions are not recommended because of rapid blood pressure drop.

When the primary diagnosis is established, drug treatment has to be modified.

The aim of the initial treatment is to lower blood pressure about 25–30% of the peak pressure within the first 6–8 hours except in hypertensive encephalopathy, when blood pressure should be lowered only about 15%. In the next 24–48 hours normotension should be achieved. After normalizing the blood pressure the parenteral drugs have to be replaced by an oral antihypertensive drug regime according to the guideline for arterial hypertension.

Recommendations for follow-up

Patients after a hypertensive emergency have to be kept under blood pressure surveillance life long, unless the underlying disease could be completely resolved.

Conflict of interests for this guideline

Alfred Hager: A.H. received compensations for advisory board meetings from Actelion; writer's honoraria from Actelion;

speaker's honoraria from Actelion, OMT, AOP Orphan, Pfizer, GlaxoSmithKline, Medtronic, Schiller, Abbott, and Encysive; as well as travel grants from Pfizer, GlaxoSmithKline, AOP Orphan, Lilly, Actelion, Medtronic, Arrows, and Guidant. A.H. is shareholder of Celgen, Gilead, Vertex, Abbvie, Pfizer, Novartis, Johnson&Johnson, Amgen, Cerner, Lilly, Merck, Biogen and several other biotechnological companies. A.H.'s institution contributed to studies sponsored by Actelion, Medtronic, Edwards, Occlutech, Novartis, and Lilly; as well as it got unrestricted scientific grants from Medtronic, Pfizer, GlaxoSmithKline, Abbott, and Actelion.

Elke Wühl: None

Martin Hulpke-Wette: None

Stephanie Lär: S.L. None

Jochen Weil: J.W. none

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Link to all references listed in the original German long version "Arterielle hypertension": www.kinderkardiologie.org/leitlinien/shmtl

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Paediatric pulmonary arterial hypertension (PAH)

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Abstract Idiopathic, heritable, and congenital heart disease associated pulmonary arterial hypertension are the most frequent forms of pulmonary hypertension in infancy and childhood. This guideline focusses on these forms and highlights diagnosis and management, investigations and therapeutic options according to current knowledge.

Keywords: *pulmonary arterial hypertension (PAH), guideline, children, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- Mean pulmonary arterial pressure ≥ 25 millimetres of mercury at rest, end-expiratory pulmonary artery wedge pressure below 15 millimetres of mercury and pulmonary vascular resistance index of more than $3 \text{ Wood units} \times \text{m}^2$
- Estimated prevalence: 2.1 cases per million children, ratio PAH associated with congenital heart disease to idiopathic PAH 1:0.9. Incidence of idiopathic PAH/heritable PAH 0.48 cases per million children per year.

Aetiology

Most frequent causes:

- Congenital heart defects (predominantly left-to-right shunt lesions)
- Idiopathic and heritable PAH

Differential diagnosis

- Parenchymatous lung disease
- Persistent pulmonary hypertension of the Newborn
- Left heart lesions
- Other rare forms of pulmonary hypertension.

Clinical presentation

In the initial stages of PAH symptoms are often mild and non-specific. In advanced stages symptoms occur even at rest.

- Infants: failure to thrive
- Toddlers: developmental retardation, fatigue, pulmonary symptoms
- Children and adolescents: neurological symptoms as headache, dizziness and blurred vision, shortness of breath on exertion or syncope and chest pain
- Clinical findings:
 - cardiac murmurs, accentuated pulmonary component of the second heart sound
 - occasionally signs of bronchial obstruction (wheezing)
 - exercise intolerance and/or cyanosis
 - signs of right heart failure

Eisenmenger patients: clubbing.

Caveats: diagnosis of PAH may be delayed by misinterpretation of symptoms and clinical findings (e.g. syncope – epilepsy; dyspnoea – obstructive pulmonary disease, bronchial asthma).

Diagnostic work up

Aim: comprehensive diagnostic work-up in PAH to establish diagnosis, to assess severity of PAH, evaluation of right ventricular function and pulmonary vasoreactivity.

Assessment of several diagnostic tools

- **ECG:** right heart pressure overload and arrhythmias. Caveat: normal ECG does not exclude PAH.
- **ECHO:** main diagnostic tool. Parameters for diagnosis and follow-up in PAH include e.g. velocity of the tricuspid regurgitant jet, mean pulmonary artery pressure (peak PI velocity), D-sign, RV eccentricity index. Exclusion of structural heart disease and postcapillary PH.
- **Chest X-ray:** basic diagnostic tool.
- **Pulmonary function test and exercise test (cardiopulmonary exercise test, 6-minute-walk-test):** assessment of physical ability. Useful for intraindividual monitoring in school-aged children and adolescents.
- **Laboratory tests (diagnostic and prognostic marker):** blood gas analysis, haemogram, cardiac enzymes and biomarkers, parameters of liver/kidney/thyroid function, coagulation status, immunologic markers, iron status, genetic screening.
- **Cardiac catheterization:** gold standard with testing of acute pulmonary vasoreactivity before operation/intervention/medical treatment. Including basic hemodynamics, testing of acute

pulmonary vascular reactivity with inhaled nitric oxide, oxygen, aerosolized iloprost or combinations of these substances. Preferably under spontaneous breathing, normoventilation and normocapnia.

- **Cardiac MRI:** additional to ECHO for detailed analysis of cardiac function, and for follow-up monitoring.
- **CT:** assessment of lung parenchyma, high sensitivity of CT angiography of pulmonary arteries and exclusion of embolism.

Indications for treatment

In patients with Idiopathic Pulmonary Arterial Hypertension, Heritable Pulmonary Arterial Hypertension and in patients with congenital heart disease, not suitable for shunt closure, PAH remains still a fatal non-curable disease, although prognosis has been improved by recent progress in specific treatment strategies. Therapeutic goals are

- delay of the progression of PAH
- improvement of clinical symptoms and
- improvement of quality of life.

Specific drugs with vasodilator effects aim to reduce pulmonary vascular resistance and to improve concomitant right ventricular dysfunction by decreasing right ventricular afterload.

Management

Causal treatment options

- Idiopathic PAH/heritable PAH: no causal therapeutic options.
- Congenital heart disease: timely surgery of cardiac defects. Indication and operability are determined by pressure values, shunts and pulmonary to systemic resistance ratio.

General recommendations

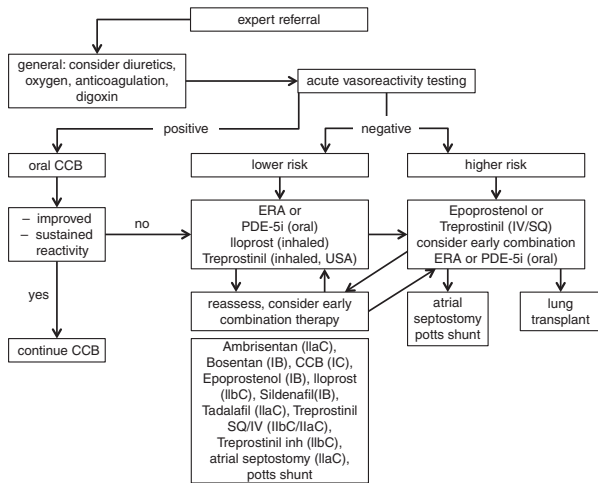
- Complete vaccinations to avoid infections, including RSV prophylaxis, influenza and pneumococcal vaccination.
- Avoid circumstances which can aggravate PH (e.g. fever, increased intrathoracic pressure as pressing by obstipation, blowing trumpets, diving, weight lifting, etc.).
- Physical activity at patients' discretion is useful after careful cardiopulmonary exercise testing. Static exercise and competitive sports should be avoided.
- Non cardiac surgical procedures represent a significant risk and should be performed only in experienced centers.
- Phlebotomy should be avoided except in symptomatic erythrocytosis (Caveats: risk of iron deficiency anaemia), deficiency of folic acid- and vitamin B12 should also be taken into consideration.
- Females in fertile age: ensure appropriate contraception due to the life-threatening risk of severe complications during pregnancy, delivery and postpartal period.
- Long-term oxygen therapy: discussed controversial. Some patients with other forms than PAH associated with congenital heart disease, cyanosis and hypoxaemia might benefit from oxygen supply.
- Diuretics: depending on clinical indication in right ventricular failure and oedema.
- In the absence of contraindications, systemic anticoagulation is recommended in idiopathic PAH/heritable PAH and thromboembolic Pulmonary Hypertension (grade of recommendation IIa, level of evidence C, target INR: 2.0–3.0). In PAH associated with congenital heart disease and Eisenmenger's syndrome, anticoagulation is not recommended routinely and may only be

considered upon strict indications (e.g. rhythm disturbances, systemic or pulmonary thromboembolism).

Specific Medical treatment

As there are no large, controlled studies or guidelines for medical treatment of each subtype of paediatric pulmonary hypertension, PAH associated with congenital heart disease is treated similarly as Idiopathic PAH.

Paediatric consensus treatment algorithm for idiopathic PAH/ heritable PAH from the World Symposium on Pulmonary Hypertension 2013 (mod. Ivy, Pediatric pulmonary hypertension, JACC 2013):



Use of all agents is considered off-label in children aside from sildenafil and bosentan in Europe. CCB: calcium channel blocker; ERA: endothelin receptor antagonist; Heritable PAH: heritable pulmonary arterial hypertension; inh: inhalation; Idiopathic PAH: idiopathic pulmonary arterial hypertension; IV: intravenous; PDE-5i: phosphodiesterase 5 inhibitor; SQ: subcutaneous.

- **Catheter interventions** Atrial septostomy may be a palliative therapeutic option for patients with right heart failure unresponsive to medical treatment. Even in experienced hands, it remains a high risk procedure.

• Surgery

- Potts Shunt

In drug-refractory courses, surgical creation of a Potts Shunt has been performed in a limited number of patients.

- Transplantation

Single lung- or combined heart- and lung-transplantation. Besides of limitations of donor organs long-term prognosis remains unfavourable. Idiopathic PAH/heritable PAH patients have a better outcome than Eisenmenger patients.

Recommendations for follow-up

Regular outpatient follow-up visits in close collaboration with an expert centre for PAH are recommended. Depending on the clinical course, following parameters should be monitored:

- treatment effects by means of evaluation of clinical status
- objective parameters (6 minute-walk-test, pulmonary function test, cardiopulmonary exercise test, transcutaneous oxygen saturation)

- organ specific function parameters (echocardiography: prognostic relevant parameters as right ventricular dysfunction, pericardial effusion, right atrial/ventricular enlargement; laboratory tests: biochemical markers, blood gas analysis, hemogram, liver/kidney parameters).

In some clinical constellations repeated cardiac catheterization for treatment monitoring (e.g. progression of PAH and need of therapeutic escalation) may be useful.

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Link to all references listed in the original German long version "Pulmonal-arterielle Hypertonie (PAH)": www.kinderkardiologie.org/leitlinien/

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Infective endocarditis in childhood and adolescence

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Abstract Infective endocarditis is an infectious disease of the valvular and/or mural endocardium or endothelium of great

arteries due to bacteria, fungi or other pathogens. In childhood congenital heart defects have become the main predisposing risk factor for infective endocarditis in developed countries. The clinical course of infective endocarditis can be *acute* as a life-threatening disease or *subacute* with an indolent course. The most frequent pathogens of infective endocarditis are streptococci and staphylococci. Diagnosis in children is based on the modified Duke criteria including clinical signs, transthoracic or transesophageal echocardiography, and positive blood cultures. Treatment consists of antimicrobial and surgical therapy depending on course and type of infective endocarditis. The overall mortality of treated infective endocarditis is below 10%. Prevention is recommended for the most complex types of congenital heart disease.

Keywords: *infective endocarditis, prophylaxis, children, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definitions

- Infective endocarditis is defined as a microbial infection of the valvular and/or mural endocardium or endothelium of the heart or the great vessels.
- Epidemiology: 0.34 to 0.64 per 100 000 children per year in developed countries.
- Predisposing factors: In childhood infective endocarditis is often associated with congenital heart disease, implanted prosthetic material after previous cardiac surgery or catheter interventions, and previous episodes of infective endocarditis.
- Pathogenesis: Pre-existing damage of valvular or mural endocardium leads to non-bacterial thrombotic endocarditis followed by significant bacteremia with consecutive microbial (bacterial) adherence, colonization and replication within microbial thrombotic endocarditis (= vegetations), with the subsequent risk of septic embolism.
- Classification of infective endocarditis:
 - Native valve endocarditis
 - Prosthetic valve endocarditis
 - Health-care associated endocarditis
 - Endocarditis in drug-addicted persons

Clinical presentation

- Subacute infective endocarditis (commonly caused *Streptococcus viridans* group): Prolonged clinical course with mild or moderate symptoms such as fever, malaise, fatigue, weight loss, failure to thrive, myalgia, arthralgia, chills, headache, nausea, and vomiting. Less frequently symptoms can be associated with extra-cardiac disease due to haematogenous spread by septic emboli.
- Acute infective endocarditis (often due to *Staphylococcus aureus*): Fulminant clinical course with sepsis, myocardial depression, and cardiogenic shock often associated with haematogenous spread and extra-cardiac disease.
- Specific signs for infective endocarditis are a new or changing heart murmur, *immunologic phenomena* such as Roth spots of the retina, Osler nodes of the distal fingers, and glomerulonephritis, and *vascular phenomena* such as Janeway lesions on the palms and soles, Splinter haemorrhages of the nails, conjunctival

Table 1. Modified Duke criteria for diagnosis of infective endocarditis (adapted)

Major criteria

Blood cultures positive for infective endocarditis (IE):

- Typical microorganisms consistent with IE within two blood cultures: *Viridans Streptococci*, *Streptococcus bovis*, *Staphylococcus aureus*, enterococci or HACEK group, in the absence of a primary focus; or
- Microorganisms consistent with IE from persistently positive blood cultures: At least two positive blood cultures of blood samples drawn >12h apart; or All of three or a majority of ≥4 separate blood cultures (with first and last sample drawn at least 1 h apart); or
- Single positive blood culture for *Coxiella burnetii* or phase I IgG antibody titer >1:800.

Evidence of endocardial involvement

- Echocardiography positive for IE: Vegetation, abscess, new partial dehiscence of prosthetic valve
- New valvular regurgitation

Minor criteria

- Predisposition: predisposing congenital heart defect, previous IE, drug addiction
- Fever (>38°C)
- Vascular phenomena: major arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial haemorrhages, conjunctival haemorrhages, Janeway lesions
- Immunologic phenomena: glomerulonephritis, Osler's nodes, Roth spots, rheumatoid factor
- Microbiological evidence: positive blood culture but does not meet a major criterion or serological evidence of active infection with organisms consistent with IE

Diagnosis of IE is possible in the presence of 2 major criteria, or 1 major and 3 minor criteria, or 5 minor criteria	Diagnosis of IE is possible in the presence of 1 major and 1 minor criteria, or 3 minor criteria
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haemorrhages and other findings (splenomegaly, microhaematuria, anaemia).

- Complications of infective endocarditis are congestive heart failure, arrhythmia (AV block), peri-valvular abscess, septic embolism such as cerebral stroke, cerebral abscess, pulmonary embolism, kidney and splenic infarctions.
- The most frequent pathogens of infective endocarditis are *Streptococcus viridans* group and *Enterococcus* spp., and *Staphylococcus* spp. (*Staphylococcus aureus*, coagulase-negative staphylococci), while pathogens of the HACEK group (*Haemophilus/Aggregatibacter*, *Actinobacillus/Aggregatibacter*, *Cardiobacterium*, *Eikenella*, and *Kingella*) and fungi are rare.
- Risk factors for a poor prognosis at admission include delayed diagnosis of infective endocarditis, age less than 3 years, vegetations larger than 10 mm, and specific pathogens such as fungi (*Candida* spp.), gram-negative bacteria (*Pseudomonas* spp.), multiresistant pathogens (methicillin-resistant *Staphylococcus aureus* or vancomycin-resistant *Enterococcus*).

Diagnostic work up

Modified Duke criteria are used for the diagnosis of infective endocarditis with a sensitivity of nearly 90% (Table 1). They are

based on echocardiography, microbiology (blood cultures), and clinical signs:

Transthoracic and transesophageal echocardiography

- Valvular and mural vegetations represent the anatomic correlate of infective endocarditis.
- Size and localisation of vegetations and the consecutive functional damage of valvular and mural structure (abscess, valvular dysfunction, peri-valvular dehiscence), must be determined.
- If present, haemodynamic compromise of myocardial function due to infective endocarditis must be shown.

Transthoracic echocardiography is used as first-line tool, and must be followed by transesophageal echocardiography in the event of diagnostic uncertainty.

Blood cultures

- At least three blood cultures before onset of antimicrobial treatment or at least three days after stopping antimicrobial therapy should be analysed (recommended blood volume per bottle: in infants 1 to 3 ml, in toddlers and pre-school-aged children 3 to 5 ml, and school-aged children 10 ml). Blood cultures should be incubated for at least 7 to 10 days to detect slowly growing microorganisms (HACEK group and fungi). At least two blood cultures more than 12 hours apart should be positive for typical microorganism. After blood cultures are obtained an initial empirical antimicrobial treatment is started. For specific treatment antibiogram and minimal inhibitory concentration are needed. In case of blood culture negative endocarditis serologic tests must be performed for *Brucella* spp., *Coxiella burnetii*, *Bartonella* spp., *Tropheryma whippelii*, *Mycoplasma* spp. and *Legionella* spp.
- Molecular diagnosis of infective endocarditis using eubacterial and pathogen specific polymerase chain reaction of valve tissue and blood culture is recommended.
- Laboratory tests include inflammatory parameters (blood sedimentation rate, C-reactive protein, pro-calcitonin, interleukin 6), blood cell count, coagulation parameter, urea, creatinine, urinary status

Other diagnostic evaluation

- ECG: arrhythmia
- Abdominal and cerebral ultrasound: emboli
- Chest X-ray: heart size, infiltration, emboli
- MR/CT imaging: emboli

Management

- Infective endocarditis is optimally treated in a department of paediatric cardiology with on-site paediatric cardiac surgery by a multidisciplinary team.
- Treatment of infective endocarditis aims at complete eradication of pathogens within the vegetation. Therefore, intravenous microbicidal antibiotics for at least 4 weeks are recommended (Table 2–5). Most often beta-lactamase antibiotics are used (Figure 1). Microbiological diagnosis includes determination of the minimal inhibitory concentration. Intravenous application of antibiotics is needed to reach appropriate tissue levels of antibiotics. Empirical

Table 2. Initial empirical antimicrobial treatment of infective endocarditis and culture negative endocarditis.

Empirical initial treatment and treatment of culture-negative endocarditis			
Antibiotic	Dose	Max. dose	Duration (Wks.)
Native valve (and prosthetic valve > 12 months post-implantation)			
Ampicillin/ β -Lactamase-Inhibitor plus Gentamicin	300 mg/kg/d in (4-) 6 equal doses 3 mg/kg/d in 1 or 3 equal doses	12 g/d dependent on serum levels ^a	4-6 4-6
Patients with β -Lactam allergy			
Vancomycin plus Gentamicin	40 mg/kg/d in 2-3 equal doses 3 mg/kg/d in 1 or 3 equal doses	dependent on serum levels ^b dependent on serum levels	4-6 4-6
plus Ciprofloxacin	30 mg/kg/d in 3 equal doses	400 mg/Dosis	4-6
Prosthetic valve (<12 months post-implantation)			
Vancomycin plus Gentamicin	40 mg/kg/d in 2-3 equal doses 3 mg/kg/d in 1 or 3 equal doses	dependent on serum levels dependent on serum levels	4-6 2
plus Rifampicin	20 mg/kg/d in 3 equal doses	1200 mg/d	4-6

^aGentamicin trough level < 2,0 mg/L

^bVancomycin trough level between 15 und 20 mg/L

Table 3. Antimicrobial treatment of infective endocarditis for oral streptococci and streptococci group D.

Oral streptococci and group D streptococci			
Antibiotic	Dose ^a	Max. dose	Duration (Wks.) ^b
Penicillin-sensitive strains (MIC < 0,125 mg/L)			
Standard therapy			
Penicillin G	200 000 IU/kg/d in (4-) 6 equal doses	12-18 Mio E/d	4
or Ampicillin	300 mg/kg/d in (4-) 6 equal doses	12 g/d	4
or Ceftriaxone	100 mg/kg/d as a single dose	2 g/d	4
Two-week therapy ^c			
Above antibiotics plus Gentamicin	as above 3 mg/kg/d in 1 or 3 equal doses	dependent on serum levels ^d	2
Patients with β -Lactam allergy			
Vancomycin	40 mg/kg/d in 2-3 equal doses	dependent on serum levels ^e (30 mg/kg/T)	4
Streptococci with raised Penicillin-MIC (MIC 0,125-2 mg/L)			
Standard therapy			
Penicillin G	200 000 IU/kg/d in (4-) 6 equal doses	12-18 Mio IU/d	4
or Ampicillin	300 mg/kg/d in (4-) 6 equal doses	12 g/d	4
plus Gentamicin	3 mg/kg/d als equal doses	dependent on serum levels	2

Table 3. *Continued*

Patients with β -Lactam allergy			
Vancomycin	40 mg/kg/d in 2-3 equal doses	dependent on serum levels (30 mg/kg/d)	4
plus Gentamicin	3 mg/kg/d in 3 equal doses	dependent on serum levels	2

^aall antibiotics administered intravenously

^bfor all prosthetic valve endocarditis a therapy of at least 6 weeks is recommended

^conly uncomplicated native valve endocarditis

^dgentamicin trough level < 2,0 mg/L

^evancomycin trough level between 15 and 20 mg/L

Table 4. Antimicrobial treatment of infective endocarditis for staphylococci.

Methicillin-susceptible staphylococci			
Antibiotic	Dose	Max. dose	Duration (Wks.)
Native valve			
Methicillin-susceptible staphylococci			
Oxacillin	200 mg/kg/d in (4-) 6 equal doses	12 g/d	4-6
plus Gentamicin ^a	3 mg/kg/d in 3 equal doses	dependent on serum levels ^b	3-5 Tage
Prosthetic valve			
Methicillin-susceptible staphylococci			
Oxacillin	200 mg/kg/d in (4-) 6 equal doses	12 g/d	≥ 6
plus Rifampicin	20 mg/kg/d in 3 equal doses	1200 mg/d	≥ 6
plus Gentamicin	3 mg/kg/d in 3 equal doses	dependent on serum levels	2
Methicillin-resistant staphylococci and in patients with β -Lactam allergy			
Native and prosthetic valve			
Vancomycin	40 mg/kg/d in 2 bis 3 equal doses	dependent on serum levels ^c (30 mg/kg/d)	≥ 6
plus Rifampicin	20 mg/kg/d in 3 equal doses	1200 mg/d	≥ 6
plus Gentamicin	3 mg/kg/d in 3 equal doses	dependent on serum levels	2

^ano demonstrated advantage for the addition of gentamicin and thus optional.

^bGentamicin trough level < 2,0 mg/L

^cVancomycin trough level between 15 and 20 mg/L

antimicrobial treatment should be adapted to the microorganism detected.

- Cardiovascular surgery is indicated during acute infective endocarditis to control and avoid severe congestive heart failure, irreversible valvular damage, uncontrolled infection (sepsis), and embolic complications. Surgery aims at removal of infected tissue and tissue reconstruction or, if necessary, valve replacement. The use of foreign material should be avoided during cardiovascular surgery, as far as possible.
- Perioperative complications, mortality and time of hospitalisation time can be reduced by early cardiac surgery and by reconstruction of the affected cardiac valve without foreign material. An individual patient-related risk-benefit profile is needed.

Table 5. Antimicrobial treatment of infective endocarditis for enterococci.

Enterococcus spp			
Antibiotic	Dose	Max. dose	Duration (Wks.) ^a
β -Lactam-susceptible strains			
Ampicillin	300 mg/kg/d in (4-) 6 equal doses	12 g/d	4-6
plus Gentamicin ^b	3 mg/kg/T in 3 equal doses	dependent on serum levels ^c	4-6
β -Lactam-resistant strains			
Vancomycin	40 mg/kg/d in 2 bis 3 equal doses	dependent on serum levels ^d	6
plus Gentamicin	3 mg/kg/d in 2 bis 3 equal doses	dependent on serum levels	6

^ain cases of prosthetic valve or symptoms longer than 3 months a duration of therapy of 6 weeks is recommended.

^bin the presence of high-level Gentamicin-Resistance (MIC > 500 mg/L), but Streptomycin-susceptibility gentamicin can be replaced by streptomycin.

^cGentamicin trough level < 2,0 mg/L

^dVancomycin trough level between 15 and 20 mg/L

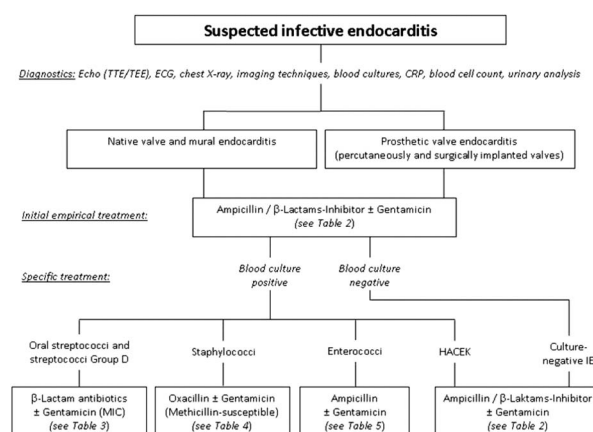


Figure 1.
Management of infective endocarditis.

- During the first two weeks of treatment clinical course must be monitored by laboratory tests, ECG, and echocardiography in a paediatric heart centre.
- Symptomatic treatment of infective endocarditis includes treatment of congestive heart failure, arrhythmia, and renal insufficiency. Anticoagulation is not per se indicated other than for mechanical valves or another independent indication. Use of heparin is preferred. In case of thromboembolic complications an individual patient-related risk-benefit analysis is needed.
- A minimum duration of antimicrobial treatment of four to six weeks is recommended, because slow growing pathogens are hidden in vegetation and biofilm and host defence is limited. As an exception native valve endocarditis with susceptible streptococci spp. (Minimal Inhibitory Concentration < 0.125 mg/l) can be treated with penicillin G and gentamicin for two weeks.

- After successful treatment of infective endocarditis around 50% of the patients will later need elective cardiovascular surgery.

Recommendations for follow-up

- After completion of antimicrobial treatment clinical examination, ECG and transthoracic echocardiography are recommended. If recurrence is suspected, laboratory tests and blood cultures are indicated.
- Children with a previous episode of infective endocarditis remain at high risk for infective endocarditis and should therefore receive antibiotic prevention in clearly defined situations including dental procedures involving manipulation of the gingival or periapical region of teeth or perforation of the oral mucosa.
- Good dental hygiene is of extreme importance and should be encouraged.

Conflict of interests for this guideline

Walter Knirsch: None
 Colin R. Mackenzie: None
 Hans-Joachim Schäfers: None
 Helmut Baumgartner: None
 Hans-Heiner Kramer: None

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Link to all references listed in the original German long version “Endokarditis, Endokarditisprophylaxe”: www.kinderkardio.logie.org/leitlinien/

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Myocarditis

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Abstract Myocarditis is defined as inflammatory disease of the myocardium with or without associated myocyte necrosis. Majority of cases of myocarditis in children are secondary to viral infection resulting in myocyte destruction caused by the infectious agent as well as the host immune response. Acute myocarditis may progress to chronic dilated cardiomyopathy. Myocarditis is often unrecognized, but it is one major cause for morbidity and mortality in children and young adults.

Diagnosis can be established by combining laboratory testing, echocardiography, and cardiac MRI. Endomyocardial biopsy with the use of PCR and in-situ hybridization helps in rapid detection of viral genome, while immunohistochemistry allows characterization and localisation of mononuclear cell infiltrates.

Treatment is guided by severity of heart failure. Results of endomyocardial biopsy may guide novel therapeutic interventions such as immunoglobulin therapy, immunosuppressive therapy, and treatment with interferon.

As true incidence of myocarditis in children is unknown, data on outcome remain speculative ranging from complete recovery, progression to dilated cardiomyopathy and death.

Keywords: *myocarditis, children, guideline*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

- Inflammatory disease of the myocardium with or without associated myocyte necrosis; no ischemic damage associated with coronary artery disease
- Inflammation may involve cardiomyocytes, interstitial and perivascular connective tissue as well as intramural coronary vessels and occasionally epicardial vessels
- May result from infections caused by viruses, bacteria, and parasites; or from drugs as anthracyclines
- Majority of cases of myocarditis in children secondary to viral infection; main viruses involved include enterovirus (coxsackie A and B, echovirus, poliovirus), adenovirus, parvovirus B19, human herpes virus 6, Epstein Barr virus, cytomegalovirus, influenza A and B
- Non-viral causes include mycoplasma, rickettsiae, borrelia burgdorferi (Lyme carditis), protozoae and other parasites, fungi, and yeasts; various drugs and toxins, including antimicrobial medications; hypersensitivity, autoimmune, or collagen-vascular diseases such as systemic lupus erythematoses, mixed connective tissue disease, rheumatic fever, rheumatoid arthritis, and scleroderma; toxic reactions to infectious agents, Kawasaki disease and sarcoidosis
- Often involvement of pericardium – perimyocarditis
- Myocyte destruction is caused by direct cellular damage by the infectious agent as well as the innate and adaptive host immune response
- With persistence of viremia and the accompanying immune response, acute myocarditis may progress to chronic dilated cardiomyopathy.

- Myocarditis evolves in three phases: 1) *acute phase* – inflammatory cell invasion with myocardial necrosis and apoptosis; 2) *subacute phase* – initiation of host's immune cascade with natural killer cells, macrophages and T cells with most destructive impact approximately 7–14 days after infection; 3) *healing phase* – fibrosis of involved myocardium.
- With continued inflammation and persistent infection, left ventricular dysfunction and dilatation may ensue
- Focal cardiac involvement in the majority of cases, only occasionally diffuse disease with ventricular dilatation and frank heart failure as in dilated cardiomyopathy
- Myocarditis is often unrecognized, incidence varies between 4 to 21% in autopsy studies from young accident and sudden death victims; acute myocarditis is one major cause for morbidity and mortality in children and young adults.

Clinical presentation

- Clinical presentation varies from asymptomatic cases to those with complete cardiovascular collapse (acute fulminant myocarditis), occasionally bradycardia (AV block) or supra-ventricular and/or ventricular tachycardia
- Symptoms dependent on age – infants: nonspecific symptoms suggestive of infectious disease including poor appetite, vomiting, fever, irritability, pallor, diaphoresis and dyspnea. In newborns and infants, it is very likely that intrauterine myocarditis is expressed as chronic disease (PVB19, CMV); – older children and adolescents: often nonspecific respiratory or gastrointestinal complaints, chest pain.

Diagnosis

Often difficult to establish as symptoms may vary and are nonspecific.

Echocardiography

- Assesses cardiac function and associated valvar insufficiency as well as wall motion abnormalities, presence of pericardial effusion
- Typical findings: systolic and/or diastolic dysfunction and/or left ventricular dilatation, mitral valve regurgitation

Electrocardiogram

- Abnormalities present in >90% of cases of myocarditis: pathologic findings include sinus tachycardia, repolarisation abnormalities, axis deviation, ventricular hypertrophy, infarction patterns, and decreased ventricular voltages

Chest radiograph

- Cardiomegaly most often reported finding

Laboratory investigation

- No current serologic biomarkers available that clearly differentiate myocarditis from other causes of acute myocardial dysfunction such as ischemia
- CK, CKMB and troponin may be elevated; BNP/proBNP allow assessment of cardiovascular function
- Impact of virus serology in patients with suspected myocarditis unproven
- Detection of virus genome in blood and/or stool is recommended

Cardiac MRI (CMR)

- Recommended in symptomatic patients and/or suspected myocarditis
- CMR allows detection of myocarditis and extent of damage by tissue characterization techniques
- CMR lacks information on causative agent of myocarditis.

Endomyocardial biopsy (EMB)

- Historical gold standard for the diagnosis of myocarditis
- Usually biopsies from right ventricular septum
- Invasive procedure with potential significant complications in children (<1% in experienced centers): pneumothorax, hemothorax, arrhythmia, heart block, perforation and death

Histology

- Dallas criteria: definition of *acute myocarditis* by lymphocytic infiltrates in association with myocyte necrosis
- Depending on amount of inflammatory infiltration and myocardial necrosis, diagnosis on initial EMB is established as: *acute myocarditis*, *borderline myocarditis*, *no myocarditis*
- *Borderline myocarditis*: inflammatory infiltrates without evidence of myocyte necrosis
- On repeat EMB, histology may be classified as *ongoing myocarditis*, *resolving myocarditis*, or *resolved myocarditis*
- PCR and in-situ hybridization allows detection of viral genome, resulting in increased diagnostic accuracy
- Immunohistochemistry allows characterization and localisation of mononuclear cell infiltrates resulting in increased detection rate of myocarditis, particularly helpful in resolving vs. chronic myocarditis
- *Chronic myocarditis*: focal or diffuse mononuclear infiltrates (T lymphocytes and macrophages) with >14 cells/mm² and HLA class expression in professional antigen-presenting immune cells

Differential diagnosis

Any disease with impaired left ventricular function, particularly dilated cardiomyopathy, abnormal left coronary artery from pulmonary artery in newborns, and ischemia as in Kawasaki disease

Management

- Data on specific treatment of EMB proven acute myocarditis in children and adults is limited (i.e. eosinophilic myocarditis)
- Appropriate medical treatment depends on clinical presentation of heart failure (see guidelines DGPK on acute and chronic heart failure)
- In children with decompensated heart failure despite optimal medical treatment left ventricular assist device implantation is recommended for bridging as left ventricular dysfunction may resolve spontaneously within weeks.
- Results of heart transplantation in adults with fulminant myocarditis have not been inferior to those with other underlying conditions.
- Arrhythmias should be treated appropriately and aggressively. In life-threatening arrhythmias patients/families may be provided with a “life-vest” or an automated external defibrillator (AED) instead of early ICD implantation.

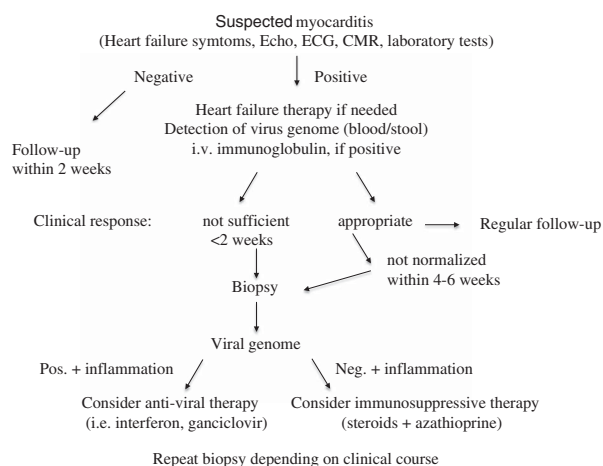


Figure 1.

Diagnosis and treatment of myocarditis in children.

- In acute myocarditis with proven viremia, intravenous immunoglobulin (1–2 g/kg over 48 h) has been reported to be beneficial. Immunosuppressive therapy using steroids is not recommended in acute viral myocarditis as it may enhance virus replication and inhibit the endogenous interferon system.
- Immunosuppressive therapy with steroids should be restricted to patients with postviral chronic myocarditis after ongoing myocardial viral infection has been ruled out by EMB.
- The use of interferon in viral myocarditis (enterovirus) has been proposed as potential treatment option with promising early results in animal studies and human pilot studies. Ganciclovir may be used in herpes virus myocarditis.
- Figure 1 depicts diagnosis and treatment of myocarditis in children.

Outcome

As true incidence of myocarditis in children is unknown, data on outcome remain speculative ranging from complete recovery, progression to dilated cardiomyopathy and death.

Conflicts of interests for this guideline:

Thomas Paul: none
Carsten Tschöpe: none
Reinhard Kandolf: none

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Link to all references listed in the original German long version “Myokarditis”: www.kinderkardiologie.org/leitlinien/

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Acute rheumatic fever

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Abstract Acute rheumatic fever occurs after a Lancefield group A β -hemolytic streptococcal infection. Western Europe belongs to the low risk population defined as ≤ 2 per 100 000 patients in school-aged children. Cardiac involvement is characterized by a diffuse inflammatory process. Further manifestations include the joints, the central nervous system, the vasculature and the subcutaneous tissue. Diagnosis is established according to the Jones criteria, as modified in 2015 by the American Heart Association. Objective of therapy is eradication of the antigen, treatment of the systemic inflammation, prevention of recurrence and environmental prophylaxis. In case of solely joints involvement without cardiac manifestation, the disease is called post-streptococcal arthritis.

Keywords: acute rheumatic fever, children, guideline, heart disease

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Acute rheumatic fever and post-streptococcal arthritis are defined as diseases following Lancefield group A β -hemolytic streptococcal infection. Diagnosis is established according to the Jones criteria, as modified in 2015 by the American Heart Association.

Chronic valvar disease may develop during long term follow-up.

Acute rheumatic fever is diagnosed if 2 major manifestations or 1 major plus 2 minor manifestations are present. According to the revised criteria subclinical findings by applying echocardiographic criteria can now be used to establish cardiac involvement (carditis). If joint involvement is the only major criterion and Jones criteria are not fulfilled, post-streptococcal arthritis is present.

Major Jones Criteria

- Arthritis
- Carditis
- Chorea
- Erythema marginatum
- Subcutaneous nodules

Minor Jones Criteria

- Arthralgia
- Fever
- High ESR and/or CRP
- Prolonged PR interval (unless carditis is a major criterion).

Epidemiology (Incidence)

≤2/100.000 in school-aged children (low-risk population in western Europe)

>2/100.000 up to 400/ 100.000 in school-aged children (high-risk population, as reported for indigenous Australian or New-Zealand population).

According to these two groups different criteria were published.

1. Pathological mitral regurgitation (all 4 criteria met)
 - Seen in at least 2 views
 - Jet length ≥2 cm in at least 1 view
 - Peak velocity >3 m/s
 - Pansystolic jet in at least 1 envelope
2. Pathological aortic regurgitation (all 4 criteria met)
 - Seen in at least 2 views
 - Jet length ≥1 cm in at least 1 view
 - Peak velocity >3 m/s
 - Pansystolic jet in at least 1 envelope

Differential diagnosis

- Cardiac involvement: all forms of endocarditis, myocarditis, congenital heart defects with valvar involvement
- Arthritis: juvenile idiopathic arthritis, reactive arthritis, peri- or parainfectious arthritis, septic arthritis
- Chorea: systemic lupus erythematosus, M. Wilson, CNS vasculitis, anti-Phospholipid syndrome, post pump chorea.

Clinical presentation

- Secondary disease following Lancefield group A β-hemolytic streptococcal infection (tonsillopharyngitis), ARF after 2-3 weeks, PSA after 1-2 weeks
- Carditis as pancarditis with preference of endocardium with mitral and aortic valve involvement
- Migrating arthritis for mostly few days
- Chorea minor
- Skin involvement is rare : subcutaneous nodules, erythema marginatum

Diagnostic work-up

According to the Jones criteria

Table 1. Diagnostic criteria for low and high risk populations

Low risk Population (Western-Europe)	High risk Population
Major criteria	Major criteria
Carditis (clinical and/or subclinical*)	Carditis (clinical and/or subclinical*)
Polyarthritis only (>= 4 joints)	Arthritis (Monoarthritis or polyarthritis, polyarthralgia)
Chorea minor	Chorea minor
Erythema marginatum	Erythema marginatum
Subcutaneous nodules	Subcutaneous nodules
Minor criteria	Minor criteria
Polyarthralgia	Monoarthralgia
Fever ≥ 38.5°C	Fever ≥ 38°C
ESR ≥ 60 mm/h and/or CrP ≥ 3 mg/dl	ESR ≥ 30 mm/h and/or CrP ≥ 3 mg/dl
Prolonged PR-Interval (unless carditis is a major criterion)	Prolonged PR-Interval (unless carditis is a major criterion)

*Doppler findings in rheumatic valvulitis:

Table 2. Eradication of streptococci

Agent	Dose	Dose interval	Duration
Penicillin V	100 000 U/kg body weight/day p.o.	3 times daily, max. 3 x 1 200 000 U	10 days
Clarithromycine	15 mg/kg body weight/day p.o.	twice daily, max. 2 x 250 mg	10 days
Cephalosporines	individual dose, p.o.		10 days

Table 3. Duration and dose of antibiotic prophylaxis in ARF

Category	Duration
Rheumatic fever with carditis and persisting valvular disease	10 years or until 40 years of age (depending on which period is longer), lifelong in severe valvular disease and high potential for exposure to streptococci
Rheumatic fever with carditis no residual heart disease	10 years or until 21 years of age (depending on which period is longer)
Rheumatic fever without carditis	5 years or until age 21 (depending on which period is longer)

Agent	Dose	Mode
Benzathine penicillin G	600 000 U <u>< 27 kg</u> 1 200 000 U <u>> 27 kg</u>	Intramuscular every 4 weeks in high-risk situation every 3 weeks
Penicillin V	2 x 250 mg 2 x 200 000 U	Daily, oral
Erythromycin	2 x 250 mg	Daily, oral

Echocardiography

Echocardiography is the key diagnostic tool to detect carditis (morphologic and functional valvar lesions resulting from valvulitis, aortic and/or mitral regurgitation).

ECG

- prolonged PR interval (minor criterium)
- repolarization abnormalities,
- tachyarrhythmias.
- Holter monitor in cardiac involvement.

Chest X-ray

Cardiomegaly, pulmonary congestion.

Cardiac catheterization, magnet resonance imaging, computed tomography
Usually not indicated.

Arthrosonography in post streptococcal arthritis

Arthrosonography is not necessary to establish diagnosis, but helps to differentiate between arthralgia and arthritis. In arthritis it shows an articular effusion and/or synovial thickening.

Indications for treatment

Objective of therapy is eradication of the antigen, treatment of the systemic inflammation, prevention of recurrence and environmental prophylaxis.

Management*Eradication of streptococci*

- Immediate start of antibiotic treatment after throat swab and antigen testing for streptococci (independent from the result). Treatment of first choice is penicillin V, in case of a penicillin allergy macrolides or cephalosporines (Fig. 1).

Acute treatment of systemic inflammation and carditis

- Treatment of inflammation with NSAID (i.e. Ibuprofen 30 mg/kg/day TID)
- Treatment of carditis with prednisolone (2 mg/kg/day, maximum 80 mg, in 2–3 doses)

Surgical therapy

- pericardial drainage
- acute surgery for severe mitral or aortic insufficiency, if possible after inflammation
- in case of elective indication at least one year after the acute phase of inflammation.

Prognosis

After acute cardiac involvement, chronic valvar disease may develop

- aortic and mitral valve insufficiency and/or stenosis
- recurrence of ARF and worsening of valvar disease.

Recommendations for follow-up

Antibiotic prophylaxis is indicated in all cases of ARF (Fig 2). Regular follow-up examinations are necessary usually at least once

a year in case of cardiac involvement and residual lesions. Examinations should include echocardiographic evaluation.

Conflicts of interest for this guideline

Ralf Trauzeddel: None

Ulrich Neudorf: None

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Link to all references listed in the original German long version “Rheumatisches Fieber”: www.kinderkardiologie.org/leitlinien/

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Acute Heart Failure and Ventricular Assist Device/ Extracorporeal Life Support

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Abstract Acute heart failure can be caused by various cardiac and non-cardiac disorders leading to severe circulatory compromise. It requires prompt diagnosis to guide therapy and frequent re-evaluations to check effectiveness of treatment. Preload, afterload, heart rhythm and contractility have to be evaluated and treated specifically. If medical treatment fails, mechanical circulatory support should be considered.

Keywords: *Acute Heart Failure, Ventricular Assist Device, Extracorporeal Membrane Oxygenation, Guideline, Congenital Heart Diseases*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Acute heart failure occurs if cardiac output does not match oxygen requirement of the body. Myocardial dysfunction is accompanied by multiple pathophysiological changes (e.g. dysregulation of calcium homeostasis, altered architecture of the heart, biochemical and hormonal abnormalities, disordered fluid balance, inflammation). As a consequence, secondary organ failure may develop (e.g. of the lung, kidney, liver, central nervous system). After surgery for congenital heart diseases, time on cardiopulmonary bypass, cross-clamp time, ischaemia, capillary leak, inflammation, acute unloading or loading of the ventricles, and changes in vascular resistance may all contribute to the extent of myocardial and circulatory impairment.

Epidemiology

- Incidence due to cardiomyopathy or myocarditis in children <16 years 0.87/100,000 children/year
- Hospital admissions (excluding acute postoperative heart failure) 15.2–17.9/100,000 children/year
- After cardiac surgery (mainly transient): in newborns 42%, in children <7 years of age 11.7–25.9%.

Aetiology

The aetiology of acute heart failure is multifactorial (see table 1). Postoperative contributing factors such as arrhythmias, cardiac tamponade, anaemia, systemic inflammatory response syndrome etc. should be considered.

Clinical Presentation

- Congestion: tachypnea, dyspnea, orthopnea, cough; hepatomegaly, ascites, peripheral oedema, distension of jugular veins
- Impaired perfusion: cold, pale, greyish skin, prolonged capillary refill time, flat pulses, peripheral cyanosis, tachycardia, low arterial blood pressure, low pulse pressure, cardiac arrest, vomiting, oliguria, anuria, agitation, somnolence, coma.

Diagnostic work-up

Describe haemodynamics especially in respect to preload and afterload. Define aetiology. Procedures must be outweighed against the need for prompt circulatory stabilization. Establish monitoring; repeat evaluation according to patient's condition. Besides a comprehensive physical examination the following diagnostic measures are recommended:

Echocardiography:

Describe function of left and right ventricle, valves, and anatomy, look for residual defects after surgery, effusions, or thrombus formation.

Laboratory:

Blood gases including lactate. Simultaneous arterial and central venous blood gases to estimate oxygen extraction and shunts (see table 2). Full blood count, creatinine, urea, liver function tests, bilirubin, coagulation studies, troponin I, creatinine kinase/CK-MB, (NT-)pro-BNP.

Chest X-ray:

Identify pulmonary venous congestion, oedema, pleural effusion. Rule out treatable pulmonary complications or alternative diseases which may contribute to dyspnea.

Table 1. Aetiology of acute heart failure

In normal anatomy	
Myocarditis	
Cardiomyopathy	<i>Dilated, hypertrophic, restrictive</i> <i>Primary and secondary cardiomyopathies</i> <i>(post-infectious, genetic, metabolic, ischaemic, in muscular disease, post chemotherapy)</i> <i>e.g. idiopathic/hereditary, lung disease, pulmonary arterial embolism</i>
Pulmonary hypertension	
Arrhythmia	<i>AV block, supraventricular or ventricular tachycardia</i>
Pericardial tamponade	
Hypertensive crisis	
Valvular disease	<i>Stenosis or regurgitation; endocarditis, rheumatic fever</i>
Non cardiac disease	<i>Septic shock, anaemia, tension pneumothorax, trauma, intoxication etc.</i>
In congenital heart disease	
Volume overload	<i>Shunt lesions, valvular regurgitation, arteriovenous fistula</i>
Obstruction	<i>of ventricular outflow, valves, or vessels</i>
Hypoxia	<i>Cyanotic heart disease</i>
Ischaemia	<i>Coronary anomaly</i>
Complex anatomy	<i>Univentricular heart, before and after palliation</i>

Table 2. Haemodynamic parameters

Calculation of important haemodynamic parameters	
Arterio-venous oxygen extraction ^(x)	$(1.34 [\text{mL/g}] \times \text{Hb} [\text{g/dL}]) \times (\text{SaO}_2 - \text{SvO}_2)$ simplified: $(\text{SaO}_2 - \text{SvO}_2) [\%]$
Ratio of pulmonary to systemic blood flow (Qp/Qs)	$(\text{SaO}_2 - \text{SvO}_2) / (\text{SpvO}_2 - \text{SpaO}_2)$

12-lead-ECG:

Helpful in identifying underlying cardiac disease or precipitants (arrhythmia, ischaemia).

Cardiac catheterization:

If suspected coronary anomaly is not ruled out otherwise, to obtain endomyocardial biopsies, if intervention is intended (critical aortic or pulmonary stenosis, coarctation, restrictive interatrial septum). Caveat: catheterization may trigger fatal circulatory compromise.

Monitoring

Continuous ECG, pulse oximetry, balancing of fluids, central/peripheral temperature. In children on intravenous vasoactive medication invasive arterial and central venous lines are indicated.

Other

Pulmonary artery pressure line, Swan-Ganz-catheter, pulse contour analysis and near-infrared spectroscopy are helpful in selected patients.

Table 3. Parameters indicating adequate cardiac output

Parameter	Target range
CVS	Without cyanosis: >60%
AVDO ₂	4–6 mL/dL or 30–40%
Blood gases	Base excess balanced, normocapnic
Lactate	Falling/normalizing (<20 mg/dL or <2 mmol/L)
Diuresis	2–3 mL/kg/h
Central to peripheral temperature difference	<4 °C (depends on patient age, medication and ambient temperature)
Capillary refill	<2 sec

Abbreviations: SaO₂, systemic arterial oxygen saturation; SvO₂, central venous oxygen saturation; SpvO₂, pulmonary venous oxygen saturation; SpaO₂, pulmonary arterial oxygen saturation; CVS, central venous oxygen saturation; AVDO₂: arterio-venous oxygen extraction.

^(x) Saturations are expressed as fraction i.e. 100% as 1.0 or 60% as 0.6

Management

Treatment should be goal oriented and the effect reassessed frequently (see table 3).

General measures

- Adjust therapy according to target parameters (see table 4)
- In circulatory collapse cardiopulmonary resuscitation
- Apply oxygen application according to cardiac physiology:
 - In non-cyanotic heart diseases or normal anatomy: achieve more than 90%
 - In cyanotic heart defects consider oxygen supply, if saturation is <75%
 - Caveat: oxygen may aggravate pulmonary recirculation
- Correct anaemia (in cyanotic patients haemoglobin 14–16 g/dL)
- Avoid agitation and fever to reduce oxygen consumption. Consider mechanical ventilation. Caveat: Sedation may provoke acute decompensation.
- Eliminate existing metabolic acidosis.

Specific therapy (see table 4)

- Administer prostaglandin in newborns unless duct-dependent circulation is ruled out
- Preload

Assess preload by echocardiography, atrial pressures, or volume challenge (head-down position, liver pressure).

 - In volume depletion administer 10–30 mL/kg isotonic fluids.
 - In cases of severe congestion administer diuretics.
 - Drugs to reduce afterload or to improve contractility may be indicated.
 - Caveat: In restrictive physiology (restrictive cardiomyopathy, cardiac tamponade, hypertrophic myocardium) reduction of preload or afterload can be deleterious.
- Afterload
 - As long as the coronary perfusion pressure is sufficient an inodilator or dobutamine can be recommended to reduce afterload.

Table 4. Pharmacotherapy in acute heart failure in children

Drug	Dosage (i.v.)	Half-life	SVR	PVR	Comments
Inotropes					
Increase cardiac output, heart rate, and myocardial oxygen consumption					
Epinephrine	0.01–0.1 (–1) µg/kg/min	3–10 min	↑	↑	
Dobutamine	5–10 (–20) µg/kg/min	2–3 min	↓	↓	
Dopamine	1–10 (–15) µg/kg/min	5–10 min	↑	↑	Vasoconstriction with high dosages. Not indicated for prophylaxis or treatment of renal failure.
Inodilators					
Increase cardiac output, little effect on heart rate. No effect on myocardial oxygen consumption					
Milrinone	0.375–1 µg/kg/min	2 hrs	↓	↓	Bolus of 50–75 µg/kg over 15 min may be considered. Thrombocytopenia. Prolongs closure of patent ductus arteriosus. Half-life depends on renal function: up to 10 hrs in premature infants.
Enoximone	2–10 µg/kg/min	Same as milrinone	↓	↓	Bolus of 0.2–0.5 mg/kg over 10 min may be considered. Thrombocytopenia. Should be used short-term (max. 24 hrs).
Levosimendan	(0.05–) 0.1 (–0.2) µg/kg/min over 24 h	~80 hrs	↓	↓	Bolus of 6–12 µg/kg over 15 min may be considered. Metabolites active up to 1 week. Repetitive application possible. Calcium sensitizer and inhibition of phosphodiesterase-III.
Vasopressors					
Pronounced effect on systemic vascular resistance. Increase myocardial oxygen consumption. May worsen systolic heart failure due to increased afterload.					
Norepinephrine	0.01–0.1–1 µg/kg/min	2 min	↑	↑	
Vasopressin	0.0003–0.002 U/kg/min	1 min	↑	→	In refractory low SVR. Effect on PVR controversial.
Terlipressin	5–10 ng/kg/min	50–80 min	↑	↓→	Alternatively: bolus of 10–20 µg/kg every 4–6 hrs. In refractory low SVR. Effect on PVR controversial.
Vasodilators (predominantly systemic)					
Sodium nitroprusside	0.5–10 µg/kg/min	3–4 min	↓	↓	Add sodium thiosulphate to infusion. Combination with sildenafil contraindicated.
Urapidil	0.5–2 mg/kg/h	3 hrs	↓	↓	Alternatively: 1–4 mg/kg as bolus. Little reflex tachycardia due to blockade of 5-HT _{1A} -receptors.
Phentolamine	(0.1–) 2 (–10) µg/kg/min	19 min	↓	↓	Consider bolus of 250 µg/kg over 5 min. Blockade of post-(α ₁) and presynaptic-(α ₂) receptors, pronounced reflex tachycardia.
Clonidine	0.5–3 µg/kg/h	6–24 hrs	↓	↓	Lowers blood pressure as a central sympatholytic drug (agonist of presynaptic α ₂ -receptors). Initial rise in blood pressure by stimulation of peripheral α ₂ -receptors. Half-life in renal failure up to 40 hrs.
Nitroglycerin	0.5–3–20 µg/kg/min	2 min	↓	↓	Lowers preload only with low dosages (<3 µg/kg/min), in higher dosages additional effect on afterload.
Vasodilators (predominantly pulmonary)					
iNO	Inhaled		→	↓	In mechanically ventilated patients.
Sildenafil	2–40 ppm				Phosphodiesterase-V inhibitor. Consider bolus of 0.4 mg/kg over 3 hrs. Do not exceed 30 mg/d iv.
	0.06–0.1 mg/kg/h	4 hrs	↓	↓	Oral: 8–20 kg: 3 × 10 mg; >20 kg: 3 × 20 mg
Epoprostenol	1–20 (–60) ng/kg/min	6 min	↓	↓	Start with 1–3 ng/kg/min, increase gradually. Inhibits platelet aggregation. Half-life varies individually. Start with low dosage, increase gradually.
Iloprost	1–5 ng/kg/min	30 min	↓	↓	Inhibits platelet aggregation. Inhaled: 0.25 µg/kg, max. 5 µg, every 3 hrs.
β-blocking agents					
Negative inotropic and negative chronotropic. Contraindicated for acute heart failure due to systolic dysfunction. Useful in tachycardia with hypertrophic myocardium.					
Metoprolol	0.5–2.5 µg/kg/min	3–5 hrs			β ₁ -selective. Half-life may be longer in genetic predisposition.
Esmolol	Start: 500 µg/kg/min, maintenance: 50–200 µg/kg/min	7 min			β ₁ -selective, for short-term therapy (24 hrs).
Diuretics					
Furosemide	3–12 mg/kg/d (max. 0.5 mg/kg/h)	1 hr			

- If hypotensive use positive inotropic drugs (epinephrine), and consider vasopressors to increase coronary perfusion pressure.
- Vasopressors are also used to raise afterload in hypertrophic myocardium.
- Treatment of pulmonary vascular resistance is crucial in right ventricular failure, pulmonary hypertension or Fontan physiology. Inhaled nitric oxide, prostacyclins (intravenously, inhaled), oxygen and normoventilation are useful to lower the pulmonary vascular resistance.
- Systemic hypertensive crisis is rare in children. Normalization of afterload is achieved with vasodilators (see table 4).
- Arrhythmias
 - Sinustachycardia is a compensatory mechanism of the failing heart and in most cases not treated specifically. Arrhythmias reducing cardiac output should be treated.
 - Bradycardia causing acute heart failure can be treated with atropine (AV block) inotropes, or pacemaker.
- After correction of preload and afterload inotropic drugs and/or inodilators are indicated for temporary support in acute heart failure with poor end-organ perfusion/cardiogenic shock, or hypotension.
- If inotropic therapy fails, mechanical circulatory support should be considered.

Surgical and interventional therapy

- Pericardial effusion, tamponade, bleeding or residual defects after cardiac surgery may lead to acute heart failure and require immediate therapy. In children with postoperative left ventricular asynchrony, biventricular stimulation should be considered.

Mechanical circulatory support

Children with increasing dosages of inotropes, sustained metabolic acidosis, rising lactate, elevated oxygen extraction, or end-organ-failure should be evaluated for mechanical circulatory support. Treatment goals are bridge to recovery or bridge to transplantation. Contraindications (e.g. severe neurologic injury) must be ruled out, and informed consent of parents or legal guardian has to be obtained before implantation.

Extracorporeal life support (ECLS)

ECLS is indicated if weaning from cardiopulmonary bypass is not possible, in early postoperative low cardiac output refractory to medical therapy, or as support during cardiopulmonary resuscitation (E-CPR). After implantation of the extracorporeal circuit diagnosis of the aetiology of the circulatory collapse is mandatory to guide further therapy. Between 40% and 50% of children can be discharged home after ECLS therapy. If long-term support is required and lung function has recovered, a ventricular assist system may be indicated. In pulmonary hypertension and right ventricular failure ECLS can be used as bridge to transplantation.

Ventricular assist devices (VADs)

Paracorporeal pulsatile VADs are suitable for all patient sizes (infants from a body weight of 2.5 kg) and can be implanted as left, right or biventricular VADs. Alternatively in children with a body surface area over 1 m², continuous flow VADs can be used. Support of children with univentricular hearts is possible. Survival rates are significantly higher with VAD than with ECMO when

used as a bridge to heart transplantation. Timing of implantation of mechanical support is dependent on circulatory compromise. Generally accepted are cardiogenic shock and progressive decline on inotropic support. Impairment of liver or kidney function before implantation increases mortality on VAD. Overall survival in children depends on diagnosis (in cardiomyopathy 70–80%, in congenital heart disease 40%), on patient selection and waiting time. Main complications are thromboembolic events.

Recommendations for follow-up

Life-long follow-up examinations are mandatory in children with residual chronic heart failure or after recovery from mechanical circulatory support. Long-term neurological follow-up is recommended.

Conflicts of interest for this guideline

Oliver Miera: is on an advisory board and receiving consultancy fees from Cardiorentis, and received speaker fees from Berlin Heart, and Actelion.

Ingo Dähnert reports receiving consultancy fees from Medtronic, Amplatzer, Saint Jude Medical, Boston Scientific.

Nikolaus Haas reports receiving consultancy and speaker fees from pfm medical, Occlutech, Medtronic, and AGA medical and consultancy fees from Osypka, NMT medical, AndraMed Edwards.

Marc N. Hirt: None

Josef Thul: None

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Chronic Heart Failure

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Abstract Heart failure in the paediatric population is an important cause of morbidity and mortality and can be caused by multiple aetiologies. It requires urgent assessment to establish the diagnosis, determine the hemodynamic status and to initiate appropriate therapy. This paper reviews current evidence and expert consensus on diagnosis and management of chronic heart failure in childhood.

Keywords: *chronic heartfailure, children, guideline, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Chronic heart failure (CHF) is characterized by the inability of the heart to pump sufficiently to meet the organs oxygen demands. This leads to activation of the sympathetic nervous system and the renin-angiotensin-aldosterone-system (RAAS), aggravating heart failure symptoms such as shortness of breath, tachycardia, limited ability to exercise, tiredness, systemic vasoconstriction and fluid retention in the pulmonary and systemic circulation. Historically, heart failure, often referred to as congestive heart failure, has been commonly used synonymously for reduced systolic left ventricular (LV) ejection fraction (HFrEF) but in recent years it became obvious that patients with preserved left ventricular function (HFpEF) can present with heart failure symptoms due to impaired ventricular relaxation (diastolic heart failure). This guideline is focused on heart failure due to impaired left ventricular systolic function.

Epidemiology

Chronic heart failure occurs less often in children than in adults; it is three times more frequent in infants as compared to older children. A population based study found 4 children <1 year and 1.3 children aged 1 to 10 years with chronic heart failure in 1000 person years (USA). In a German study from a tertiary paediatric cardiology referral center the incidence of heart failure was 23 in 1000 children admitted.

Aetiology

In approximately 80% of children affected, heart failure is associated with congenital cardiac malformations. Other reasons include cardiomyopathies (most common in structurally normal hearts), myocarditis, rhythm disturbances or rare diseases.

A) Structural heart disease

- Volume loaded ventricle:

- Left-to-right shunt lesions (e.g. ventricular septal defect, persistent ductus arteriosus)
- Valve insufficiencies (e.g. aortic regurgitation)
- Pressure loaded ventricle:
 - Left sided obstructions (e.g. aortic stenosis)
 - Right sided obstructions (e.g. pulmonary stenosis)
- Complex congenital heart defects:
 - Univentricular hearts (e.g. hypoplastic left heart syndrome)
 - Systemic right ventricle (e.g. congenital corrected transposition of the great arteries)

B) Structurally normal heart

- Cardiomyopathies (CMP):
 - Dilated CMP
 - Hypertrophic CMP
 - Restrictive CMP
 - Non-compaction CMP
 - Arrhythmogenic right ventricular dysplasia/CMP
- Myocarditis
- Rhythm disturbances (tachycardia and bradycardia)
- Rare causes (e.g. thalassemia, thyroid dysfunction)

Clinical presentation

See Tables 1 and 2

Heart failure has been classified to describe severity of symptoms. The two most common heart failure classifications used in paediatrics are from the New York Heart Association (NYHA) and the Ross score for infants and small children. The latter was modified by Läer et al. in 2002.

Assessment of severity of heart failure:

- none: 0 – 2 points
- mild: 3 – 6 points
- moderate: 7 – 9 points
- severe: 10 – 12 points

Diagnostic work-up

Aims: confirm diagnosis, assess severity of heart failure, evaluate differential diagnosis, perform risk stratification and evaluate therapeutic options.

Table 1. Clinical Presentation

	Commonly encountered	Less commonly
Infants and young children	Tachypnea Tachycardia Feeding difficulties Diaphoresis Pallor	Cyanosis Palpitations Syncope Edema Ascites
Older children and adolescents	Fatigue Effort intolerance Dyspnea Orthopnea Abdominal pain Nausea Vomiting	Palpitations Chest pain Leg edema Ascites

Table 2. Heart failure severity score (Ross score modified by L  er et al.)

	Clinical Score		
	0	1	2
History			
1. <i>Diaphoresis</i>	head only	head and body during exercise	head and body at rest
2. <i>Tachypnoe</i>	normal	retractions	dyspnea
Physical exam			
3. <i>Breathing</i>	normal	retractions	dyspnea
4. <i>Respiratory rate/min</i>			
0-1 years	<50	50-60	>60
2-6 years	<35	35-45	>45
7-10 years	<25	25-35	>35
11-14 years	<18	18-28	>28
5. <i>Heart rate (beats/min)</i>			
0-1 years	<150	150-170	>170
2-6 years	<105	105-115	>115
7-10 years	<90	90-100	>100
11-14 years	<80	80-90	>90
6. <i>Hepatomegaly</i> (liver edge from right costal margin; cm)	<2	2-3	>3

First line investigation

- Physical examination (heart failure score)
- Echocardiography is an indispensable part of the initial evaluation. It should be performed immediately by an experienced investigator to describe function of the left and right ventricle separately (FS, FAC and/or EF) including long-axis function (MAPSE, TAPSE), valve function and anatomy, effusions or thrombi. Systolic heart failure in children is currently defined as SF < 25% and/or EF < 50%.
Additional function tests: Tei-index (global systolic and diastolic function), global and regional strain and strain rate (longitudinal deformation and dyssynchrony), stress echocardiography (for regional wall motion impairment) and diastolic function parameters (e.g. E/E', mitral valve inflow, LA size, pulmonary venous flow).
- Chest X-ray (cardiomegaly, CTR >50%)
- 12-lead ECG (nonspecific but frequently abnormal in heart failure)
- Laboratory tests: NT-pro-BNP (significant relationship with symptoms and LV function), troponin I, creatinine kinase/CK-MB, full blood count, creatinine, urea, liver function tests, bilirubin, albumin, total serum protein, glucose, CRP, coagulation studies.

Secondary line investigation

- Cardiopulmonary exercise testing (objective assessment of exercise capacity, differentiation of pulmonary from cardiac origin of dyspnea)
- Magnetic resonance imaging (gold standard for volumes, function and mass; tissue characterization, shunt quantification, valve function, anatomy)
- Holter ECG (heart rate level, arrhythmia detection)
- Catheter examination: No general indication in paediatric heart failure. Indicated if non-invasive methods are not sufficient to assess haemodynamics and anatomy in complex congenital heart disease, to obtain endomyocardial biopsies, to perform an intervention, to assess pulmonary pressures and vascular resistance, to assess coronary artery abnormalities and prior to heart transplantation.

Management

General measures

- Physical activity and recreational sports are recommended, competitive sports should in general be avoided, but may be discussed on an individual basis.
- Healthy life style and diet.
- In severe HF, sufficient calorie supply is essential to avoid cachexia.
- Vaccinations according to current guidelines: in addition RSV prophylaxis may be considered in the first year of life (October to April, second year optional, see guideline); influenza immunization after 6 months of age annually; first pneumococcal vaccinations during infancy, completion within first 2 years of life.
- Correct anaemia, obtain higher haemoglobin levels in cyanotic patients (>14 g/dl)

Pharmacologic treatment

- General considerations and principles:
 - Pharmacokinetics and pharmacodynamics are age dependent.
 - Few randomized controlled trials (RCT) exist with low evidence in the paediatric population (heterogenous study groups, often under-powered).
 - Since 2006 EU legislation requires paediatric studies before approval of new drugs. Therefore, currently many traditional heart failure drugs are used *off-label* in paediatrics.
 - Structural normal heart: pharmacotherapy according to recent adult guidelines on chronic heart failure (see Figure 1)
 - First line therapy for systolic heart failure: neurohumoral antagonists (ACE/AT1-Inhibitors, mineralocorticoid receptor antagonists and β -blockers) have shown survival benefit in large adult studies.
 - Dosing: in general, maximal dose for chronic heart failure drugs comparable to pharmacotherapy in arterial hypertension; but lower starting dose (1/4 or 1/2 as used in arterial hypertension; see Table 3), doubling after 1 to 3 weeks, further increase until maximal dose or side effects occur.
 - Congenital heart disease: in left-to-right shunt lesions, β -blocker and diuretics are used as transient pharmacological therapy until operation or catheter intervention. Systemic RV, heart failure therapy according to failing LV but no evidence!
- Specific pharmacotherapy:
 - Angiotensin-converting enzyme inhibitors and angiotensin I receptor blockers:*
Major adult studies (>7,000 pts) have shown improved survival and symptoms with the use of ACE inhibitors in chronic HF. AT1-receptor antagonists can be considered as an alternative in patients intolerant of an ACE inhibitor (limited experience in children). Therefore, ACE/AT1-Inhibitors should be started in all heart failure patients (NYHA I-IV) with structurally normal hearts. In left-to-right shunt lesions, single ventricle physiology and in patients with a systemic right ventricle, studies have failed to show improvement of heart failure symptoms or exercise capacity. Caveat: ACE/AT1-inhibitors may cause fetopathy in the second and third trimester of pregnancy.
 - β -adrenergic antagonists:*
Key randomized controlled trials in adult patients with heart failure showed reduced mortality and hospitalization rate. β -blockers should only be applied in hemodynamically

Table 3. Dosages of oral heart failure medication in childhood in various age groups according to the European Medicines Agency (EMA) and expert consensus (off-label use) of the guideline committee of the German Society of Paediatric Cardiology (DGPK).

Category	Drug	Age group/body weight	Starting dose (oral)
ACE inhibitor	Captopril*	PT, NB, IN 1-18 years	0.15 mg/kg/d q8 h 0.15 mg/kg/d q8 h max. 18.75 mg/d
	Enalapril*	PT, NB IN, 1-18 years	no data 0.05 mg/kg/d in 1 dose or q12 h max. 2.5 mg/d
	Lisinopril*	PT, NB IN, 1-18 years	no data 0.05 mg/kg/d in 1 dose resp. max. 2.5 mg/d
	Ramipril*	PT, NB, IN 1-18 years	no data 0.05 mg/kg/d in 1 dose resp. max. 2.5 mg/d
AT1-Receptor antagonists	Candesartan*	PT, NB, IN 1-18 years	no data 0.05 mg/kg/d in 1 dose resp. max. 4 mg/d
	Losartan*	PT, NB, IN 1-18 years	no data 0.2 mg/kg/d in 1 dose resp. max. 12.5 mg/d
	Valsartan*	PT, NB, IN 1-18 years	no data 0.5 mg/kg/d in 1 dose resp. max. 40 mg
β-Adrenergic antagonists	Metoprolol (succinate)*	PT, NB, IN 1-18 years	0.2 mg/kg/d q12 h 0.25 mg/kg/d in q12 h max. 11.88 mg in q12 h
	Bisoprolol*	PT, NB IN, 1-18 years	no data 0.05 mg/kg/d in 1 dose max. 1.25 mg/d
	Propranolol*	PT, NB, IN, 1-6 years 6-18 Jahre	0.5 mg/kg/d q8h no data
	Carvedilol*	PT, NB IN, 1-18 years	no reports 0.1 mg/kg/d q12 h max. 6.25 mg/d
Aldosterone antagonists	Spironolactone**	PT, NB, IN, 1-18 years	0.5 mg/kg/d in 1 dose max. 25 mg/d
	Eplerenone**	PT, NB, IN, 1-3 years 4-18 years	no data 0.5 mg/kg/d in 1 dose max. 25 mg/d
Diuretics	Furosemide	PT, NB, IN, 1-18 years	0.5 mg/kg/d in 1 dose 2 mg/kg/d q6-8 h max. 60 mg/d
	Torsemide	PT, NB IN, 1-18 years	no data 0.1 mg/kg/d in 1 dose max. 5 mg in 1 dose
Digitalis glycosides	Digoxin	PT, NB, IN, 1-18 years	1 mg/kg/d in 1-2 doses max. 25 mg/d start with maintenance dose: 8-10 mcg/kg/d in 1 dose max 200 mcg/d
	Ivabradine*	PT, NB, IN, 1-5 years 6-18 years	no data 0.1 mg/kg q12 h max. q12 h
Calcium-Sensitiser	Levosimendan		Initial i.v. bolus: 6-12 mcg/kg cont' infusion 0.1-0.2 mcg/kg/min for 24 h
	Drug	Target dose	Comments
ACE inhibitor	Captopril*	2-3 mg/kg/d q8 h 2-3 mg/kg/d in q8 h max. 150 mg/d	Age >1 year: consider longer acting ACE inhibitors (Enalapril or Lisinopril).
	Enalapril*	0.5 mg/kg/d in (1dose) q12h bzw. max. 40 mg/d	caveat: - reduced dose in impaired renal function
	Lisinopril*	0.5 mg/kg in 1st dose resp. max. 40-(80) mg/d	- monitor serum K ⁺ and Na ⁺ levels - potential fetotoxicity in 2nd and 3rd trimester
	Ramipril*	0.5 mg/kg in 1 dose resp. max. 10 mg/d	- potential hypotensive effects after starting dose (monitor blood pressure for 2 hours)
AT1-Receptor antagonists	Candesartan*	0.3 mg/kg/d in 1 dose bzw. max. 16 mg/d	Age >1 year: consider longer acting ACE inhibitors (Enalapril or Lisinopril).
	Losartan*	1 mg/kg/d in 1 dose bzw. max. 50 mg/d	caveat: - in pts with impaired renal or liver function
	Valsartan*	3 mg/kg/d q12 h resp. max. 320 mg/d	- serum K ⁺ and Na ⁺ levels - potential fetotoxicity in 2nd and 3rd trimester
β-Adrenergic antagonists	Metoprolol (succinate)*	2 mg/kg/d q12 h 2 mg/kg/d q12 h max. 190 mg/d	β1-receptor blocker
	Bisoprolol*	0.4 mg/kg in 1 dose max. 20 mg/d	β1-receptor blocker, long half-life
	Propranolol*	3 mg/kg/d q8 h	nonspecific β-blocker caveat: hypoglycemia!
	Carvedilol*	1 mg/kg/d q12 h max. 50 mg/d	nonspecific β-blocker

Table 3. (Continued).

Category	Drug	Target dose	Comments
Aldosterone antagonists	Spironolactone**		caveat: hyperkalemia
	Eplerenone**		caveat: hyperkalemia limited experience in children
Diuretics	Furosemide	Seek lowest effective maintenance dose. Increase according to clinical requirements.	Indicated in pulmonary or systemic congestion but only to achieve euvoemia with the lowest effective dose. Reduction of diuretics should be followed by increase of ACE/AT1- inhibitors, β-blockers, and mineralocorticoid-receptor- blocker. Caveat: diuretics may cause nephrocalcinosis in preterms.
	Torsemide	Seek lowest effective maintenance dose. Increase according to clinical requirements.	
Digitalis glycosides	Digoxin		USA: FDA approval for children >2 years Target plasma level: 0.5 – 0.9 ng/ml, Impaired renal function: 50% dosage Limited data in children
	Ivabradine*	0.3 mg/kg/d q12 h max. 15 mg/d	
If-Channel-blocker	Levosimendan	Interval therapy according to individual requirements	Intravenous interval therapy may be considered in severe heart failure (NYHA III/IV).

PT = preterm, NB = newborn, IN = infant, d = day

*double starting dose every two weeks at the beginning, subsequently increase by smaller steps until target dose reached or side effects occur.

**starting dose equals target dose.

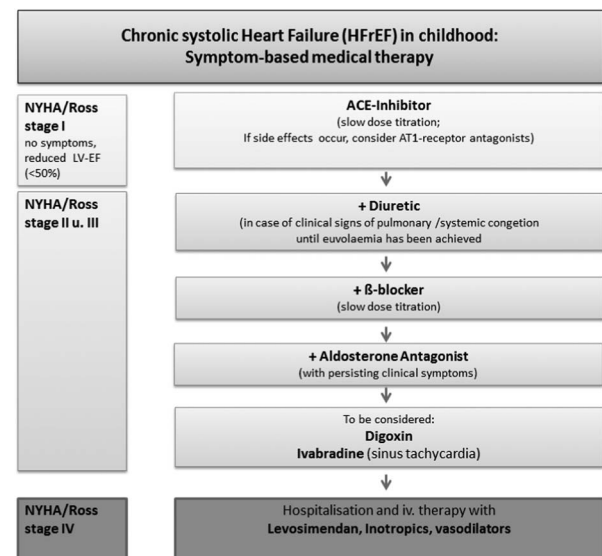


Figure 1.

Flow sheet of pharmacotherapy according to clinical symptoms in children with structural normal hearts. In patients with left-to-right shunts medical therapy with β-blockers and diuretics is recommended.

stable patients. β-1 selective blockers might be advantageous due to different adrenergic adaptation in paediatric heart failure but there is no evidence from clinical studies. Propranolol has shown improvement in heart failure

symptoms (Ross score) and reduction of neurohormonal activation in left-to-right shunt lesions. Carvedilol did not result in any significant improvement in clinical heart failure outcomes in a heterogenous group of paediatric heart failure patients, but subgroup analysis showed a beneficial trend for patients with systemic LV.

- **Aldosterone antagonist therapy:**
Well established in adult heart failure therapy (survival benefit). Experience in paediatric heart failure is limited (no RCTs). Low doses in heart failure therapy are recommended to manage secondary hyperaldosteronism. Caveat: Hyperkalemia may result in patients with impaired renal function and/or concomitant ACEi. Male gynecomastia may occur.
- **Diuretics:**
Have shown to improve clinical symptoms in congestive heart failure and should therefore be considered as additional therapy. The effects of diuretics on morbidity and mortality have not been studied in patients with heart failure (no RCTs in adults and children). Caveat: Chronic diuretic therapy may aggravate RAAS activation. Therefore, it is only indicated to achieve euvoemia with the lowest effective dose.
- **Digoxin:**
Did not alter all-cause mortality in adult HF but can improve symptoms and prevent deterioration. May be considered for rate control in atrial fibrillation. No RCTs in the paediatric population are available. At the present time, no clear benefit in left-to-right shunt lesions with regard to heart failure symptoms. Target plasma level: 0.5 – 0.9 ng/ml.
- **Ivabradine:**
May be considered in patients with sinus tachycardia in addition to adequate treatment with β -blockers. Showed significant risk reduction of hospitalizations due to heart failure in adults.
- **Levosimendan:**
Calcium sensitizer used in acutely decompensated heart failure. Should be reserved for patients with severe reduction of cardiac output. Only few reports of outpatient therapy in paediatric HF (intermittent intravenous therapy).
- **Inotropic agents and i.v. vasodilators:**
See DGPK guideline on acute heart failure
- **Phosphodiesterase-5 inhibitors and Endothelin-Receptor antagonists:**
Use is based on the hypothesis that vasoactive substances have a beneficial effect in Fontan patients by lowering pulmonary artery pressure and resistance. Two studies have shown improved exercise capacity in Fontan patients, but long-term benefit is unclear.
- **Anticoagulation:**
There is an increased risk for thromboembolic events in children with dilated cardiomyopathy/normal hearts. In patients with left-to-right shunts medical therapy with β -blockers and diuretics is recommended. No RCTs exist in the paediatric population, experience is based on small studies. Vitamin K antagonists or heparin may be used in patients with EF < 25%, atrial fibrillation or with a history of thromboembolic events (expert consensus). Aspirin may be considered in patients with EF > 25%.
- **Antiarrhythmic agents:**
For pharmacological therapy of tachyarrhythmias see recent guideline of the German Society of Paediatric Cardiology (DGPK).

Table 4. Disease staging according to ISHLT

Class	Description	Examples
A	At an increased risk of developing heart failure but with normal cardiac function and chamber size	Exposure to cardiotoxic agents, family history of heritable cardiomyopathy, and single ventricle anatomy
B	Abnormal cardiac morphology or function with no past or current symptoms of heart failure	Aortic insufficiency with left ventricular dilation, history of anthracycline exposure with left ventricular dysfunction
C	Past or current heart failure symptoms and structural or functional heart disease	Cardiomyopathy or congenital heart disease patients with symptomatic systolic ventricular dysfunction
D	Marked symptoms at rest despite maximal medical management	Requiring specialized interventions e.g. continuous infusion of inotropic agents, mechanical circulatory support, heart transplantation or hospice care

Non-pharmacologic treatment

- **Cardiac resynchronization therapy (CRT):**
In dyssynchronous contraction pattern, multisite pacing offers the possibility to improve systolic ventricular function. CRT is recommended in symptomatic patients with pacemaker induced dyssynchrony/cardiomyopathy and in symptomatic patients (NYHA II-IV) with systemic LV, sinus rhythm, QRS $\geq$ 150 ms and left bundle branch block pattern. CRT should be considered in all patients with significantly impaired function (EF < 35%) of the systemic ventricle (RV or LV or biventricular circulation) and dyssynchrony despite optimized pharmacological therapy according to an expert consensus statement from 2014 (Khairy et al.) on adults with congenital heart disease.
- **Implantable cardioverter defibrillator (ICD):**
For prevention of sudden cardiac death and electrical therapy of ventricular tachyarrhythmias in children with chronic heart failure see recent guideline of the German Society of Paediatric Cardiology (DGPK).
- **Mechanical support and heart transplantation:**
Ventricular Assist Devices (VADs)/Extracorporeal Membrane Oxygenation (ECMO) and heart transplantation should be considered for selected patients according to the recent guidelines of the German Society of Paediatric Cardiology (DGPK) and the International Society of Heart and Lung Transplantation (ISHLT).
- **PA banding:**
Pulmonary artery banding in infants and young children with cardiomyopathy has recently been described as a new therapeutic strategy. Therefore, no recommendation can be made at this early stage.

Prognosis

Heart failure based on myocardial dysfunction is a progressive disease. Therefore a task force of the American College of Cardiology (ACC) and the American Heart Association (AHA) developed a staging system (Class A-D) which was modified by the ISHLT (Table 4). Stages of HF emphasize development and progression of disease and can be used to describe individuals and populations, whereas the NYHA classification focuses on exercise capacity and symptomatic status of the patient.

Recommendations for follow-up

Life-long follow-up is mandatory in children with chronic heart failure even after recovery.

Conflict of interests for this guideline

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Link to all references listed in the original German long version “Chronische Herzinsuffizienz”: www.kinderkardiologie.org/leitlinien/

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Primary Cardiomyopathies

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Abstract The diagnosis cardiomyopathy comprises a broad spectrum of different myocardial diseases. This guideline covers exclusively primary cardiomyopathies and focusses on specific diagnostic and therapeutic aspects of dilated cardiomyopathy including left ventricular non-compaction cardiomyopathy, hypertrophic cardiomyopathy, arrhythmogenic right ventricular cardiomyopathy/dysplasia and restrictive cardiomyopathy. Up to now, cardiomyopathies cannot be cured. Medical, interventional and surgical approaches are available to treat several specific symptoms and to lower risks of definite forms of cardiomyopathy. In the care of patients with congestive heart failure due to cardiomyopathy, mechanical ventricular assist devices and heart transplantation are currently final treatment options for the patient. Special attention should be given to the development of elevated pulmonary vascular resistance.

Keywords: *Cardiomyopathy, children, guideline*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Cardiomyopathies are defined as exclusive primary disease of the myocardium (primary cardiomyopathy) compared to myocardial disease as part of a systemic affection (secondary cardiomyopathy). The American Heart Association has proposed a diagnostic classification of primary cardiomyopathies which discriminates between the underlying pathophysiologic causes:

- genetic primary cardiomyopathies:
 - hypertrophic cardiomyopathy
 - arrhythmogenic right ventricular cardiomyopathy /dysplasia
 - left ventricular non-compaction
 - conduction system disease (e.g. Lenegre disease, Wolff-Parkinson-White syndrome)
 - ion channelopathies
- acquired primary cardiomyopathies:
 - inflammatory (myocarditis)
 - stress provoked
 - peripartal
 - tachycardia induced
 - in infants of insulin-dependent diabetic mothers
- genetic or acquired caused:
 - dilated cardiomyopathy
 - restrictive cardiomyopathy

Secondary cardiomyopathies in infants and children's are most frequently associated with storage diseases, Noonan-syndrome, muscular dystrophy, and autoimmune/collagen diseases or with chemotherapy.

Epidemiology

The prevalence of primary cardiomyopathies is given:

- 1.13/100.000 children and adolescents <19 years of age

- 8.34/ 100.000 in infancy
- allocation of different types of cardiomyopathy in infants and children:
 - 51% dilative congestive cardiomyopathy
 - 42% hypertrophic cardiomyopathy
 - 7% all other types

Relationships to consider

Associated lesions

- Mitral valve regurgitation is frequent in patients with left ventricular dysfunction
- Pulmonary and tricuspid regurgitation is frequent in patients with pulmonary artery hypertension

Differential diagnosis

- in dilative cardiomyopathy:
 - acute myocarditis
 - anomalous left coronary artery from the pulmonary artery
 - secondary dilative cardiomyopathy
 - tachycardia induced cardiomyopathy
- in hypertrophic cardiomyopathy:
 - left heart obstructive lesions
 - myocardial hypertrophy in athletes
 - secondary hypertrophic cardiomyopathy (storage disease, Noonan and other syndromes)
- in restrictive cardiomyopathies:
 - constrictive pericarditis

Haemodynamics

Haemodynamic consequences follow the extent of systolic and diastolic dysfunction of the left and right ventricle and to the extent of valvar regurgitation. Arrhythmias may have acute or chronic impact on haemodynamics. Typical haemodynamic implications of cardiomyopathy are:

- reduced cardiac output
- low arterial blood pressure
- left atrial/pulmonary venous congestion
- pulmonary hypertension
- elevated central venous pressure
- tachycardia

Clinical presentation

Clinical presentation varies substantially according to the age of the patient and the extent of myocardial dysfunction. Typical findings are:

- in dilative cardiomyopathy:
 - pulmonary oedema may be misinterpreted as pulmonary infection
 - children may present diaphoretic, tachycardic and tachypnoeic
 - expiratory grunting in younger or wheezing in older children is common
 - auscultation and palpation:
 - systolic murmur of mitral regurgitation
 - gallop rhythm
 - palpable precordial pulse

- in left ventricular non-compaction cardiomyopathy
 - symptoms of heart failure
 - arrhythmias
 - less frequent thromboembolic events
- in hypertrophic cardiomyopathy:
 - exertional limitation
 - dyspnea and/or fatigue
 - palpitations
 - syncope or near syncope
 - heart failure symptoms are less frequent
 - rarely arterial thromboembolic events
 - character of systolic heart murmur depends mostly on the presence and degree of left ventricular outflow tract obstruction
- in restrictive cardiomyopathy
 - symptoms of congestive heart failure
 - arrhythmias
 - syncope or near syncope
 - dyspnea on exertion
 - cough
 - oedema, ascites, hepatomegaly
 - auscultation:
 - loud second heart sound
 - systolic heart murmur
 - gallop rhythm
- in arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - ventricular arrhythmias
 - syncope or near syncope on exertion
 - less frequent symptoms of congestive right or biventricular heart failure

Diagnostic work up

Echocardiography

Is usually diagnostic in all types of cardiomyopathy. Examination should particularly focus on the assessment of:

- in dilated cardiomyopathy:
 - ventricular systolic and diastolic diameters and function
 - mitral regurgitation
 - identification of coronary artery abnormalities
 - assessment of pulmonary artery pressure by Doppler-quantification of pulmonary and/or tricuspid regurgitation
- in left ventricular non-compaction cardiomyopathy:
 - double layer left ventricular wall with compacted and non-compacted myocardium
 - intertrabecular fissures with visualisable blood flow
 - ratio of non-compacted to compacted sub endocardial myocardium >2
- in hypertrophic cardiomyopathy
 - left ventricular systolic and diastolic diameters and function
 - diastolic wall diameter septal and inferior lateral
 - maximum wall diameter at any segment
 - symmetric (concentric) or asymmetric pattern of hypertrophy
 - assessment of the left ventricular apex (apical hypertrophic cardiomyopathy)
 - assessment of mitral valve movement (systolic anterior movement?)

- assessment of right ventricular wall movement (from subcostal or parasternal view)
- assessment of left ventricular obstruction (in patients <30 mmHg gradient at rest, stress-echocardiogram or Valsalva-manoeuve may be diagnostic)
- in arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - depiction of right ventricular akinesia, dyskinesia, or aneurysm
 - depiction of right ventricular outflow tract dilation in parasternal long or short axis view
- in restrictive cardiomyopathy
 - dilated atria
 - normal or nearly normal ventricular sizes and function
 - transmitral Doppler: elevated E/A ratio > 2.9
 - reduced mitral deceleration time of early diastolic inflow
 - tissue-Doppler: reduced Ea and Aa waves
 - assessment of mitral annular velocity and different additional parameters are described in the specific literature to be helpful to distinguish between constructive pericarditis and restrictive cardiomyopathy
 - assessment of pulmonary artery pressure by Doppler-quantification of pulmonary and/or tricuspid regurgitation.
- hypertrophic cardiomyopathy
 - not required for diagnostic reasons
 - endomyocardial biopsy may be helpful in individual patients with uncertain phenotype or concentric hypertrophy and suspicion of secondary cardiomyopathy
 - left ventricular and coronary angiography may be indicated to prearrange surgical treatment in patients with outflow tract obstruction
 - different techniques of septal ablation in selected adolescents
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - electrophysiological study
 - endomyocardial biopsy showing pathognomonic lipofibrotic degeneration in individual patients can confirm the diagnosis if echocardiography and MRI are not conclusive
- restrictive cardiomyopathy
 - not required for diagnostic reasons
 - assessment of pulmonary pressure and resistance
 - endomyocardial biopsy may expose rare forms of endocardial fibroelastosis or storage diseases

ECG

- dilated cardiomyopathy:
 - nonspecific, may show signs of left ventricular strain or arrhythmias
- left ventricular non-compaction cardiomyopathy:
 - nonspecific, may exhibit repolarization disturbances or branch bundle blocks
- hypertrophic cardiomyopathy
 - increased voltage and signs of left ventricular hypertrophy
 - repolarization changes and T-wave inversions
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - following ECG findings may be diagnostic:
 - QRS duration > 110 ms
 - right bundle branch block
 - T-wave abnormalities in right precordial leads V₁-V₃, epsilon wave
 - Premature ventricular beats or tachyarrhythmia
- restrictive cardiomyopathy
 - right and/or left atrial enlargement
 - ST-segment depression in advanced cases

Cardiac catheterization

- dilated cardiomyopathy:
 - recommended for exclusion of coronary anomalies (alternatively CT-scan)
 - assessment of pulmonary pressure and resistance
 - endomyocardial biopsy including histological, immunohistological and molecular work-up and quantitative analysis of viral load is recommended in patients with newly diagnosed and severe congestive heart failure to differentiate primary from secondary dilated cardiomyopathy and acute myocarditis
- left ventricular non-compaction cardiomyopathy:
 - not required for diagnostic purpose
 - assessment of pulmonary pressure and resistance

Cardiac MRI/CT

- dilated cardiomyopathy:
 - detection of myocardial oedema by MRI in newly diagnosed patients can help to differentiate dilated cardiomyopathy from acute myocarditis by specific techniques
- left ventricular non-compaction cardiomyopathy:
 - diagnostic MRI criteria are available. MRI is recommended to clarify uncertain findings
 - late gadolinium enhancement and analysis of fibrotic areas may have a prognostic impact
- hypertrophic cardiomyopathy
 - MRI is recommended if echocardiography depicts not all segments of the left ventricle
 - late gadolinium enhancement and analysis of fibrotic areas may have prognostic impact
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - MRI is more sensitive compared to echocardiography to depict regional right ventricular dys-/akinesia and/or fibrofatty degeneration of the myocardium
- restrictive cardiomyopathy
 - MRI can be indicated to differentiate constrictive pericarditis

CT-scan is an alternative for patients with contraindication for MRI.

Specific laboratory tests/genetic testing

Today genetic testing is of diagnostic value for many forms of cardiomyopathies. In addition it may be useful for therapeutic and preventive decisions and better family counselling. Detailed recommendations for genetic testing are available from the German Society of Paediatric Cardiology and the European Society of Cardiology.

- dilated cardiomyopathy:
 - cardiac biomarkers (natriuretic peptides) are helpful to monitor the course of disease and therapeutic effects

- specific tests to clear up suspicion and differential diagnosis of secondary cardiomyopathy
- left ventricular non-compaction cardiomyopathy:
 - cardiac biomarkers are helpful to monitor the course of disease and therapeutic effects
 - specific tests to clear up suspicion and differential diagnosis of secondary cardiomyopathy
- hypertrophic cardiomyopathy:
 - specific tests for differential diagnosis of secondary cardiomyopathy
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - genetic testing should be performed

Holter ECG and exercise test

Not diagnostic. As supraventricular and ventricular arrhythmia and tachycardia are frequently associated with any type of cardiomyopathy, regular Holter ECGs should be inherent part in the care of cardiomyopathies. Regular exercise testing is recommended in particular for patients with hypertrophic cardiomyopathy and arrhythmogenic right ventricular cardiomyopathy/dysplasia as part of primary diagnostics and follow-up examinations.

Chest X-ray

Not diagnostic in all types of cardiomyopathy. It shows left or right ventricular or atrial enlargement and pulmonary congestion.

Evaluation of patient's relatives

- Precise medical history covering 3 generations is recommended in all families with one patient with primary cardiomyopathy.
- First degree relatives and siblings should receive a complete cardiac investigation.
- Any conspicuous finding in relatives – regardless from the results of genetic testing – should be tracked annually.
- Brothers and sisters of patients are recommended for cardiac investigation every 3 years.
- Brothers and sisters of patients with hypertrophic cardiomyopathy > 12 years of age should receive cardiac testing annually.

Indications for treatment

- dilated cardiomyopathy:
 - treatment of congestive heart failure is recommended according to the parameters listed in the DGPK guideline “chronic congestive heart failure”
 - prevention of complications like thromboembolism and arrhythmia/sudden cardiac death should be considered according to patient's age, left ventricle diameter and function
- left ventricular non-compaction cardiomyopathy:
 - treatment indications like in other forms of dilative cardiomyopathy (see above)
 - anticoagulation should be considered in reduced left ventricular function (compared to left ventricular ejection fraction $\leq 35\%$ in adults)
- hypertrophic cardiomyopathy
 - medical treatment aims at improvement of patient's symptoms and lowering the risk for sudden cardiac death

- no prospective randomized evidence is given in the literature concerning medical treatment and lowering the risk of sudden cardiac death
- strongest risk factors for sudden cardiac death in adults are:
 - cardiac arrest/sustained ventricular arrhythmia
 - familial sudden death
 - multiple-repetitive non-sustained ventricular arrhythmia (>3 beats, heart rate >120/min.)
 - massive thickness of the septum >30 mm (under debate)
 - insufficient rise of blood pressure on exertion (systolic blood pressure rise ≤ 20 mmHg)
 - rapid progression
- relief of left ventricular outflow tract obstruction is recommended in symptomatic patients or high degree gradient
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - primary ICD prevention is indicated in pts with increased risk for SCD: syncope, symptoms in young age, familial SCD, sportive activities, high QRS- and QT dispersion, severe RV (and LV-) dysfunction, induced VTs during EP study
 - arrhythmia
- restrictive cardiomyopathy
 - only symptomatic therapy available
 - listing for heart transplantation should be considered timely

Management

- dilated cardiomyopathy:
 - treatment of congestive heart failure and initiation of anticoagulation is recommended according to the DGPK guideline “chronic congestive heart failure”
 - cardiac resynchronisation therapy should be considered according to the DGPK guideline “chronic congestive heart failure”
 - an implantable cardioverter should be considered according to patient's age, left ventricle diameter and function, see also DGPK guideline “tachycardia”
 - heart transplantation should be timely considered in progressive disease or treatment failure
 - implantation of ventricular assist device may be indicated in acute congestive heart failure (see DGPK guideline “acute congestive heart failure”) or in patient with elevated pulmonary resistance as bridge to transplant
- left ventricular non-compaction cardiomyopathy:
 - treatment like in other forms of dilative cardiomyopathy (see above)
 - anticoagulation should be considered in reduced left ventricular function (compared to left ventricular ejection fraction $\leq 35\%$ in adults)
- hypertrophic cardiomyopathy
 - symptomatic patients should receive lipophilic beta-blockers like Propranolol, Metoprolol or Bisoprolol. A Propranolol dosage of 2 mg/kg is recommended by the American Consensus group; higher dosage up to 5 mg/kg may be effective
 - Verapamil and other calcium-antagonists should not be given in infants and in patients with severe hypertrophy because of negative inotropy and elevated risk for cardiac death
 - long-term prophylactic treatment with amiodarone should not be established because the likelihood of side effects

- an implantable cardioverter is the only safe prophylaxis for sudden cardiac death and should be considered as primary or secondary prophylaxis, balancing the specific problems of ICD implantation in children and the patient's individual risk profile
- surgical septal myectomy is indicated in symptomatic drug-refractory patients
- successful transcatheter intervention with alcohol or high frequency septal ablation in selected adolescents has been reported in small series of selected patients
- need for heart transplantation is uncommon
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - arrhythmia may be treatable by transcatheter ablation or should be treated medically
 - implantable cardioverter
 - in rare treatment-refractory patients surgical exclusion of the right ventricle and total cavo-pulmonary connection may be considered as an alternative to heart transplantation
- restrictive cardiomyopathy
 - symptomatic medical treatment of arrhythmias
 - an implantable cardioverter may be discussed in individual patients
 - timely heart transplantation with respect to pulmonary vascular resistance
 - (bi-) ventricular assist device as bridge to transplant

Prognosis

- Dilated cardiomyopathy and left ventricular non-compaction cardiomyopathy with symptoms of congestive heart failure are frequently progressive and the most common cause for paediatric heart transplantation.
- Hypertrophic cardiomyopathy, arrhythmogenic right ventricular cardiomyopathy/dysplasia and ion channelopathies are the most frequent findings in sudden death or cardiac resuscitation in young athletes <35 years of age. Thus, investigation of patient's history, ECG and echocardiogram is reasonable before adolescents start competitive sports (consensus of the German guideline committee).
- Restrictive cardiomyopathy is frequently progressive and early at risk for pulmonary congestion with elevated pulmonary vascular resistance

Recommendations for follow-up

All children and adolescents with primary cardiomyopathy should receive annually cardiac investigation including patient's history, ECG and echocardiogram. Beyond that the following recommendations are given by the German guideline board:

- dilated cardiomyopathy and left ventricular non-compaction cardiomyopathy:
 - annually measurements of cardiac biomarkers (e.g. NT-pro BNP), Holter-ECG at least every 2 years and stress-ECG at least every 3 years are recommended in clinically stable patients
 - timely referral to a heart transplant center is reasonable
 - suspicion of pulmonary congestion/pulmonary artery hypertension from echocardiography should be verified by (repeated) right heart catheterization
- hypertrophic cardiomyopathy
 - participation in sports activities should be considered according to DGPK guideline "sports with congenital heart disease"

- Holter-ECG and exercise test should be performed annually in all symptomatic patients and patients with high risk profile for sudden cardiac death
- arrhythmogenic right ventricular cardiomyopathy/dysplasia
 - patient's should go strictly without competitive sports and physical exertion (see also DGPK guideline "sports with congenital heart disease")
- restrictive cardiomyopathy
 - timely referral to a heart transplant center is reasonable
 - in suspected pulmonary congestion/pulmonary artery hypertension from echocardiography should be verified by right heart catheterization.

Conflict of interest for this guideline

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Syncope

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Abstract: Syncope has to be differentiated from loss of consciousness of different aetiologies. Reflex syncope is frequent in adolescents and should be diagnosed adequately in the majority of patients by basic diagnostics. Advanced diagnostic management is necessary to rule out or to confirm suspected cardiac disease. Treatment of reflex syncope is based on reassurance and education while the impact of drug treatment is controversial. Selective use of basic and advanced diagnostics should increase the specificity for detection of cardiac syncope and avoid overdiagnostics in reflex syncope.

Keywords: *syncope, children, guideline*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Syncope is defined as temporary loss of consciousness and postural tone due to transient inadequate global cerebral perfusion.

Differential diagnosis

Syncope must be discriminated from loss of consciousness due to cerebral seizures, neurological, metabolic, psychogenic and various other diseases.

Clinical presentation

Syncope may be a singular event or happen frequently in some individuals. Loss of consciousness in syncope characteristically shows

- sudden onset
- short duration
- spontaneous complete recovery

An aura may be present, sudden loss of postural tone may cause injury and symmetric or asymmetric convulsions may occur. Typical age of manifestation is 12–19 years, preferentially in females. At least one syncope occurs in about 15% of all children. A clinical classification of syncope is given in table 1. Breath holdings spells represent a special type of syncope in 6 months to 5 years old children (prevalence about 2–5% of all children). Loss of consciousness in children aged 5–11 years per se may be a hint not to be a reflex syncope. Though relapses after the first reflex syncope occur frequently, the prognosis for freedom of symptoms in the long-term outcome is excellent.

Basic diagnostics:

Diagnostic work-up

Basic diagnostics

- case, medical and family history
- physical examination
- 12-lead-ECG
- blood pressure measurement, if applicable orthostatic intolerance test

Diagnostic targets

A checklist for basic diagnostics is given in table 2 and should be used to

- differentiate a syncope from other diseases with loss of consciousness
- classify the syncope
- specify the risk for a possible cardiac syncope

Table 1. Clinical classification of syncope in children and adolescents

- 1) Reflex (neurally mediated) syncope
 1. Toddler's breath holding spells (6 months – 5 years)
 - cyanotic spells (expiratory apnoea -> cyanosis -> sinus tachycardia)
 - pallid spells (pallor -> agitation -> asystole due to abnormal sympathetic-vagal regulation)
 2. Adolescent reflex syncope (12–19 years)
 - Vasodepressive syncope (decline of systemic vascular resistance and blood pressure)
 - cardioinhibitory syncope (decline of heart rate up to asystole)
 - mixed forms (vasodepressive, cardioinhibitory)
 - postural tachycardia (progressive inadequate increase of heart rate -> vagal back-regulation -> decline of heart rate up to asystole)
- 2) Orthostatic hypotension (typical syncope of the elderly)
 - to be discriminated from vasodepressive reflex syncope, which does primarily not show arterial hypotension and which is much more frequent in children
- 3) Cardiac syncope
 - arrhythmias
 - congenital functional and structural heart diseases

Table 2. Basic diagnostics to attain the diagnostic targets – checklist

Was it syncope? Exclude other diseases if loss of consciousness

- was not of sudden onset
- was not short-lasting
- did not recover complete and spontaneously
- happened in presence of circumstances which may hint on one of the various differential diagnoses

Reflex or orthostatic reflex syncope at young age is adequately defined (in absence of other pathologic findings) if

- syncope was preceded by typical prodromal stages like dizziness, light headedness, visual snow, sudden sweating
- syncope was mediated by emotional distress because of unpleasant circumstances (smell, fear, pain, medical instrumentation, blood phobia)
- syncope was mediated by orthostatic stress (e.g. long standing in a warm and crowded environment)
- syncope happened in typical trigger situations like cough, sneeze, gastrointestinal stimulation (swallow, defaecation, visceral pain), post-prandial (often in context with nausea or vomiting), micturition, post-exercise, in context with laughing, brass instrument playing
- syncope happened after standing up into vertical position (morning wake-up) or while standing upright (not immediately) after exertion
- orthostatic intolerance test leads to syncope within 3 minutes

“red flags” possibly indicating a cardiac syncope:

- atypical age of manifestation (5-11 years)
- syncope during exercise (incl. swimming)
- scaring and emotional stress as trigger situation (i.e. loud noise, cold water)
- syncope without prodromi while being in horizontal position
- syncope in context with chest pain, tachycardia, palpitations
- history of congenital heart disease or cardiac surgery, Kawasaki disease or intake of drugs known to cause QT-prolongation
- family history of unexplained death in young age

ECG-findings indicating cardiac syncope:

- sustained ventricular tachycardia or rapid supraventricular tachycardia
- QRS-prolongation (≥ 120 ms in adolescents), bundle branch block
- High grade AV-block
- repolarisation changes or signs of ventricular hypertrophy (myocarditis, cardiomyopathy)
- prolongation or shortening of the QTc-interval (long/short QT syndrome)
- right bundle branch block + ST segment elevation in leads V1-V3 (Brugada syndrome)
- repolarisation changes in right precordial leads (arrhythmogenic right ventricular cardiomyopathy)
- J-wave in the inferior leads and/or V3-V5 (syndrome of early repolarization)
- delta-wave (WPW syndrome)
- deep Q waves in leads I and AVL (ALCAPA, Kawasaki)
- low voltage, ST segment elevations, flattened T waves (pericarditis, pericardial effusion)

- **Echocardiography:** outflow tract obstructions, pericardial effusion, pulmonary arterial hypertension, coronary anomalies, impaired ventricular function, cardiomyopathy
- **Advanced ECG monitoring** including **24h-Holter-ECG** (suspected arrhythmia in case of frequent relapse of syncope), **external event recorder** (only meaningful if patient during earlier syncopes did feel announcements and is compliant) and **implantable loop recorder** (highly sensitive and specific, recommended betimes in cases of reasonable suspicion)
- **Exercise test** may be sensitive for coronary ischemia or catecholamines induced tachycardia.
- **A tilt table test** is not necessary to diagnose, classify or stratify therapy of reflex syncope. It may be indicated if diagnosis of reflex syncope or orthostatic hypotension is ambiguous. It can be helpful if the patient is in a high risk situation for syncope (risk of injury, education, business) or to increase individual sensitivity for reflex syncope.

Indications for treatment

Treatment of reflex syncope is required in cases with a high frequency of relapses and high degree of suffering. Cardiac syncope should be treated targeted.

Management

Treatment is based on **reassurance and education** of the patient's perception of the benign character of reflex syncope and excellent prognosis. Benefit to prevent relapses have been shown with

- rescue-training
- avoidance of trigger situations
- general conditioning of the circulatory system

Drug treatment is limited to individual and exceptional cases and – due to the benign character of reflex syncope – should always be temporarily limited. While clear evidence is missing, some authors have reported successful treatment with

- alpha-adrenergic agonists
- serotonin-reuptake blockers
- mineralo-corticosteroids.

Due to explicit scientific findings the use of beta-blockers, dihydroergotamin and etilefrin is no longer recommended. Indication for **pacemaker implantation** in reflex syncope is limited to rare cases with recurrent pallid breath holding spells and/or extended asystole. Reflex syncope may reoccur after pacemaker implantation because the vascular component of the pathologic reflex is not addressed.

Prognosis

- In adolescents relapses after a first reflex syncope are not rare
- Prognosis for freedom of recurrent reflex syncopes is excellent in young adulthood

Conflict of interest for this guideline

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Advanced diagnostic measures

Indicated if basic diagnostics provide no evidence for the presence of reflex- or orthostatic reflex syncope or if cardiac syncope is suspected.

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Sport in patients with congenital heart disease

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Abstract Physical activity and sports are essential for human development and social integration and should only be restricted if serious medical risks are present. Restrictions are only necessary if serious arrhythmias, special forms of cardiomyopathies, implanted medical devices, Fontan circulation, cyanosis, pulmonary hypertension, aortic dilatation, severe systemic ventricular dysfunction, or antithrombotic drug treatment are present. The same applies to the early period after surgical and catheterization procedures. For those patients detailed counselling should clarify which kind of sport can be recommended.

Keywords: *sport, guideline, children, congenital heart disease*

The purpose and the mode of generation of these guidelines are described in the Introduction to this supplement.

Definition

Physical activity is any kind of body movement by muscular contractions that leads to an increase in metabolic demand on top of the resting condition.

Exercise or exercise training is any planned, structured physical activity repeatedly performed to increase fitness. It is defined by type of exercise (dynamic, static), intensity, duration, and frequency.

Sport is physical activity that is highly structured in detailed rules. It is divided into leisure sports where the motivation is fun and improvement in fitness, and competitive sport, which is characterised by a much higher time demand and focussing primarily on performance in contests.

Exercise capacity is the maximal work load that can be achieved.

Fitness is the combination of endurance (cardiorespiratory capacity), strength, flexibility and body coordination.

Diagnostic evaluation

In healthy children no eligibility screening is necessary as long as sport is on a leisure level. For competitive sport a detailed medical, family and training history, as well as a structured physical examination are recommended. From a cardiologist's point of view a 12 lead ECG is useful. The “Seattle Criteria” should be applied for adolescents and adults, whereas paediatric modifications are necessary in children. If any suspicious findings arise, further studies like echocardiography or exercise testing are necessary.

Children with congenital heart disease should undergo a sport eligibility screening latest in preschool age by a paediatric cardiologist. It should reveal findings that encompass medical risks at exercise. Sport restriction should not be based on exercise capacity, but only on acute risks during exercise or long-term exercise induced cardiac problems.

Screening should include:

- Family history, medical history, training history
- Physical examination including orthopaedic aspects and motor skills

- Resting ECG, Holter ECG
- Exercise testing, preferably a cardiopulmonary exercise test
- Echocardiography
- Cardiac magnetic resonance tomography depending on the underlying cardiac disease
- Chest X-ray, computed tomography or cardiac catheterization are not warranted for sports eligibility screening unless there is a clear clinical indication.

After eligibility screening, an exercise programme should be established in collaboration with an expert in sports medicine.

Sport recommendations

Physical activity and sports are essential for human development and social integration and should only be restricted if serious medical risks are present.

Healthy children should perform at least 60 minutes daily with moderate or vigorous activity, screen time should be restricted. Exercise should be diversified, age adjusted and enjoyable. In addition, some strength training should be included.

Children with congenital heart disease should perform the same physical activity as recommended to their healthy peers.

There are only few medical restrictions according to the current medical findings:

- Arrhythmias
 - Supraventricular premature beats: no restrictions
 - Ventricular premature beats: no restrictions if monomorphic and suppressed with exercise (otherwise exclude myocarditis, cardiomyopathy, ion channel defects or structural heart disease)
 - supraventricular tachycardia: no restrictions if asymptomatic (except preexcitation) or after successful ablation (otherwise no sport before therapy)
 - asymptomatic preexcitation (Wolff-Parkinson-White pattern): If preexcitation disappears suddenly during exercise, no restriction (otherwise no competitive sports till electrophysiological study)
 - non-sustained idiopathic monomorphic ventricular tachycardia without symptoms: no restrictions
 - Ion channel defects: restriction according to the diagnosis (long-QT syndrome type 1, QTc > 500 msec and catecholaminergic polymorphic ventricular tachycardia have the highest risk)
 - Cardiomyopathies: restrictions according to diagnosis (only non-compaction cardiomyopathy probably has low risk)
- Syncope: sport cannot be recommended until the definite diagnosis has been established
- Implanted medical devices: restrictions according to the type of the device, contact sports should be avoided in most of the devices
- Fontan circulation: no Valsalva manoeuvre (no diving)
- Cyanosis: no restriction as long as asymptomatic

- Pulmonary hypertension: only light sport activities if completely asymptomatic
- Aortic dilatation: restrictions according to diagnosis and individual findings.
- Shunts at atrial level, cross shunts or right-to-left shunts: no scuba diving
- Severe systemic ventricular dysfunction: avoid intense sport activities
- Severe outflow tract obstruction: avoid moderate and heavy sport activities
- Antiplatelets or anticoagulation: no contact sports
- After cardiac catheterisation or cardiac surgery: start exercise after 1 week, increase intensity according to individual haemodynamics/condition. After sternotomy no contact sport or other sport with risk of thorax compression for 6 weeks.

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